Data Analysis

Main Reference Textbook v3

Chapters 1–12 with Math Foundations source maps

Contents

Notation Guide

This guide collects the notation used across the regression portion of the text. It is not a replacement for the chapter explanations; it is a quick reference for keeping the object types straight.

Object Types

Notation	Object type	How to read it
x, y, b, p, n	scalars	Single numerical values or indices.
X, Y, ϵ	random variables	Population-level or model-level random quantities.
$\mathbf{x}$, $\mathbf{y}$, $\mathbf{e}$	vectors	Observed sample-level vectors, usually in $\mathbb{R}^n$.
$\mathbf{X}$, $\mathbf{H}$, $\mathbf{I}_n$	matrices	Data, projection, or identity matrices.
$\boldsymbol{\beta}$, $\boldsymbol{\epsilon}$	parameter/error vectors	Coefficient and error vectors in regression models.
$\widehat{\ }$	estimate or fitted quantity	A hat marks something estimated from data or predicted by a fitted model.

Data and Statistical Learning Notation

For a single observation, $\mathbf{x} = (x_1, \ldots, x_p)^T$ is the vector of p input values and Y is the response. The general supervised-learning model is

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

For a data set with n observations, i indexes observations and j indexes variables:

$$x_{ij} = \text{value of predictor } j \text{ for observation } i.$$

The observed response vector is

$$\mathbf{y} = (y_1, \ldots, y_n)^T.$$

The notation $\mathbf{x}_j$ means the full observed column of predictor j. The notation $\mathbf{x}_i^T$ means the row of predictors for observation i.

Regression Notation

In the regression chapters, the design matrix $\mathbf{X}$ includes an intercept column unless stated otherwise. With p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n\times(p+1)}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{p+1}.$$

The fitted values and residuals are

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}, \qquad \mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}.$$

The true error vector is $\boldsymbol{\epsilon}$; the residual vector is $\mathbf{e}$. Errors are part of the data-generating model and are not observed. Residuals are computed after fitting a model.

The ordinary least squares estimator is

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$

when $\mathbf{X}$ has full column rank. The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T, \qquad \widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}.$$

The residual-maker matrix is $\mathbf{I}_n - \mathbf{H}$.

Later chapters also use $\mathbf{Z}$ for an *engineered* design matrix. Its columns may include transformed predictors, indicator/dummy variables for categories, and interaction columns. Once those columns are built, least squares treats them like any other design-matrix columns; the new work is choosing and interpreting the columns responsibly.

Observed Vector Versus Random Vector

Most chapters use $\mathbf{y}$ for the observed response vector from the data set in hand. Chapter 10 temporarily uses $\mathbf{Y}$ because it studies the fixed-design theoretical model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\mathbf{X}$ is treated as fixed and $\mathbf{Y}$ varies randomly through $\boldsymbol{\epsilon}$. After that theory chapter, the text returns to $\mathbf{y}$ when discussing a realized fitted model.

Fit, Testing, and Diagnostics

Notation	**Meaning**
RSS	Residual sum of squares, $\sum_i (y_i - \widehat{y}_i)^2 = \|\mathbf{e}\|^2$.
TSS	Total sum of squares, $\sum_i (y_i - \bar{y})^2$.
RegSS	Regression or explained sum of squares, often TSS − RSS when an intercept is included.
R^2	Proportion of response variation explained relative to the mean-only baseline.
RSE	Residual standard error, the estimated residual standard deviation.
h_{ii}	Leverage of observation i, the ith diagonal entry of $\mathbf{H}$.

The Big Three Regression Assumptions

The regression chapters use a common three-step assumption ladder:

1. **Mean-zero errors:** the model is centered correctly, e.g. $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$.
2. **Constant variance and uncorrelated errors:** $\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$.
3. **Gaussian errors:** $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

The stronger assumptions support stronger inferential claims, but they are not all needed for every modeling task.

Linear Models

Early chapters use "linear" in the familiar geometric sense: a line, plane, or hyperplane in the displayed predictors. Chapter 11 sharpens the definition: for least squares, the key requirement is that the model be linear in the unknown coefficients. Curved functions of the original predictors, category indicators, and interaction terms can still be linear models if they can be written as a design matrix times a coefficient vector plus error.

Chapter 1

Data Analysis as Statistical Learning

1.1 Why This Course Begins with Statistical Learning

Data analysis is not just the act of computing summaries or fitting models. It is the disciplined process of using data to learn something about a system, make predictions about future observations, and communicate conclusions in a way that can be checked, reproduced, and defended. In this course, we use the phrase *statistical learning* to emphasize that our models are tools for learning from data rather than formulas that are true simply because we wrote them down.

This course serves as a bridge. It connects the probability, statistics, and mathematical tools developed in earlier operations analysis courses to the modeling and machine-learning methods that appear later. The immediate goal is to build the statistical modeling intuition needed for regression, exploratory data analysis, model validation, and diagnostics. The longer-term goal is to prepare you to use more advanced methods without treating them as black boxes.

A recurring theme is the statement, often summarized as "all models are wrong." The point is not that models are useless. The point is that a model is a deliberate simplification. It keeps the structure we think is useful and ignores or absorbs the rest as noise, approximation error, measurement error, or unexplained variation. A good analyst understands both sides: what the model is trying to capture and what it is failing to capture.

1.2 What Is Data?

A data set is a collection of measurements, records, or observations. Each observation corresponds to one unit of analysis: a person, ship, aircraft, market, mission, experiment, time step, or other object being studied. Each variable records one attribute of that observation.

In statistical modeling, we usually distinguish between *input variables* and an *output variable*. Input variables are the measured quantities we use to explain or predict something. They are also called *predictors*, *features*, *covariates*, or *independent variables*. The output variable is the quantity we are trying to explain or predict. It is also called the *response* or *dependent variable*.

For example, suppose we want to model a service member's physical fitness test score. Possible inputs might include age, training frequency, prior score, sleep, and injury status. The output could be the next test score. In a logistics setting, inputs might be demand history, location, distance, and inventory status, while the output could be delivery time or stockout risk. In a classification problem, inputs might be sensor readings and the output might be whether a track is classified as friendly, hostile, or unknown.

We collect the input variables for a single observation into a vector. For an observation with p inputs, write

$$\mathbf{x} = (x_1, x_2, \dots, x_p)^T \in \mathbb{R}^p.$$

Here p is the number of input variables, and x_j is the value of the jth input for this observation. The corresponding output is denoted by Y or, for a realized observed value, by y.

Math Foundations Connection. Math Foundations treats a vector as an ordered collection of quantities that belong to the same object, not merely as an arrow in physical space. That is exactly how we use $\mathbf{x}$ here. The vector $\mathbf{x}$ is one observation's collection of input measurements. Its entries may be physical quantities, demographic attributes, encoded categories, time summaries, or engineered features. The geometric interpretation of vectors will become useful later, but the first interpretation is simpler: a feature vector is a structured record of measured information. See the Math Foundations textbook, Chapter 2.

1.3 A First Statistical Model

The central modeling idea in this chapter is that an output variable Y is related to input variables $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$ through an unknown relationship plus random error:

$$Y = f(\mathbf{X}) + \epsilon. \tag{1.1}$$

The function f represents the systematic part of the relationship between the inputs and the output. The random variable ϵ represents the part of Y not explained by $f(\mathbf{X})$.

A simple example with two predictors is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \tag{1.2}$$

In this model, β_0 is an intercept, β_1 and β_2 are coefficients, and ϵ is the unobserved error. The inputs x_1 and x_2 are observed. The output Y is observed after the fact. The error term ϵ is not directly observed; it is the difference between the observed output and the systematic part of the model.

Equation (1.1) is intentionally general. The function f might be linear, nonlinear, simple, complicated, interpretable, or opaque. In this course, we often begin with linear models because they are mathematically transparent and remain important for inference. But the broader statistical learning framework is not limited to linear models.

We usually impose basic assumptions on the error term. A common starting point is

$$\mathbb{E}[\epsilon] = 0, \tag{1.3}$$
$$\mathrm{Var}(\epsilon) > 0, \tag{1.4}$$
$$\epsilon \text{ is independent of } \mathbf{X}. \tag{1.5}$$

The mean-zero assumption says that the error does not systematically push the output upward or downward once the systematic relationship has been accounted for. The positive-variance assumption says that there is genuine unexplained variability. The independence assumption says that the leftover error is not still carrying predictable information from the inputs. These assumptions are not decorations. They are part of the model's claim about the world, and later chapters will show how violations of assumptions affect inference, prediction, and diagnostics.

Since f is unknown, we estimate it using data. We denote the estimated function by $\hat{f}$. Given a new input vector $\mathbf{x}$, the corresponding predicted output is

$$\widehat{Y} = \hat{f}(\mathbf{x}). \tag{1.6}$$

The hat notation signals that the quantity is estimated or predicted rather than directly observed or known.

Math Foundations Connection. When the model is linear in the input variables, the systematic part can be written compactly as

$$f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = \beta_0 + \boldsymbol{\beta}^T \mathbf{x},$$

where $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T$. The term $\boldsymbol{\beta}^T \mathbf{x}$ is an inner product. Equivalently, it is a linear combination of the entries of $\mathbf{x}$, with coefficients $\beta_1, \ldots, \beta_p$. This is why linear algebra becomes the natural language of regression: the model combines measured inputs by weighting them and adding them together. See the Math Foundations textbook, Chapters 2, 3, and 5.

1.4 Inference and Prediction

There are two major reasons to estimate f: inference and prediction. They are related, but they are not the same task.

1.4.1 Inference

Inference is about understanding the relationship between the inputs and the output. In the linear model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon,$$

inference focuses on questions such as:

- What is the meaning of β_1 and β_2?
- Are these coefficients positive, negative, or plausibly zero?
- Which predictors are most relevant to Y?
- How much does the output change when one input changes and the others are held fixed?
- What assumptions are required before we trust those conclusions?

Inference is about learning structure. If a commander asks why readiness changed, which drivers matter, or whether a particular intervention had an effect, then the analysis is inferential. Accuracy still matters, but the model must also be interpretable enough to support an explanation.

1.4.2 Prediction

Prediction is about estimating the output for new or future inputs. Given a new vector $\mathbf{x}_{\text{new}}$, we use the fitted model to compute

$$\widehat{Y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}}).$$

Prediction focuses on questions such as:

- Given these inputs, what output should we expect?
- How accurate is the prediction likely to be?
- How much uncertainty remains even after we use the model?
- How well will the model perform on cases it has not yet seen?

Prediction is about learning what is likely to happen next. If an analyst is building a model to forecast demand, estimate casualty risk, classify a sensor return, or predict equipment failure, then the model may be judged primarily by its predictive performance.

1.4.3 The Same Model Can Serve Both Goals

The same mathematical model can sometimes support both inference and prediction, but the priorities differ. A highly interpretable model may be preferred for inference even if a more flexible model predicts slightly better. Conversely, in a pure prediction task, a less interpretable model may be acceptable if it performs substantially better on new data. A defensible analysis states which goal is primary and evaluates the model accordingly.

1.5 Regression and Classification

A first way to categorize statistical learning problems is by the type of output variable.

In a *regression* problem, the output is numerical or quantitative. Examples include predicting fuel consumption, time to complete a task, exam score, cost, weight, or sales. A regression model returns a number. A simple linear regression model has the form

$$\widehat{y} = wx + b, \tag{1.7}$$

where w is a slope or weight and b is an intercept or bias term.

In a *classification* problem, the output is a category or class label. Examples include predicting pass/fail, disease/no disease, hostile/friendly/unknown, or low/medium/high risk. A classifier may return a class directly, but many classifiers first estimate probabilities for each class and then choose the most likely class.

For binary classification, where $Y \in \{0, 1\}$, logistic regression models the probability of class 1 as

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma(w_1 x_1 + w_2 x_2 + b), \tag{1.8}$$

where

$$\sigma(t) = \frac{1}{1 + e^{-t}} \tag{1.9}$$

is the logistic or sigmoid function. The expression $w_1x_1 + w_2x_2 + b$ is a score. The sigmoid maps that score to a probability between 0 and 1.

For multiclass classification, a model may compute several scores, often called *logits*. For example, an image classifier might produce a cat logit, dog logit, and ship logit. Those logits can then be transformed into class probabilities. The key idea is that classification models often separate two steps: first produce class-specific scores, then convert those scores into a decision or probability distribution.

1.6 Linear and Nonlinear Models

The previous examples introduce a second distinction: whether the model class is linear or nonlinear. This distinction is separate from the regression/classification distinction. We can have linear regression, nonlinear regression, linear classification, and nonlinear classification. For now, "linear" means the familiar geometric picture: lines, planes, and hyperplanes in the predictors as written. Chapter 11 later sharpens this by showing that a model can still be linear after transforming predictors, as long as it remains linear in the unknown coefficients.

A *linear regression model* predicts a numerical output using a linear function of the inputs:

$$\widehat{y} = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

With one predictor, this is a line. With two predictors, it is a plane. With more predictors, it is a hyperplane.

A *linear classification model* uses a linear score to separate classes. In two dimensions, the decision boundary is a line. Logistic regression is a standard example: the model computes a linear score and then converts that score into a probability.

But data are not always linear. Sometimes the pattern in a regression problem is curved rather than straight. For example, a one-dimensional nonlinear regression model might be written as

$$\widehat{y} = w(x-a)^2 + b. \tag{1.10}$$

This model can represent a curved relationship because the input appears through the transformed term $(x-a)^2$.

Similarly, a nonlinear classifier may use a nonlinear score before applying the sigmoid function. For example,

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma\left(\gamma_0 + \gamma_1(x_1 - h)^2 + \gamma_2(x_2 - k)^2\right). \tag{1.11}$$

This model can create a curved decision boundary in the (x_1, x_2) plane. The boundary is no longer forced to be a straight line.

The important point is that *data shape* and *model shape* must be considered together. If the true relationship is nonlinear but the model is linear, the model may miss essential structure. If the model is extremely flexible, it may capture real structure but also chase random noise. Later chapters develop this tradeoff carefully.

1.6.1 Neural Networks as Repeated Linear and Nonlinear Operations

A fully connected neural network layer is another example of this linear/nonlinear pattern. A basic layer takes an input vector $\mathbf{x}$, forms a vector of linear combinations, and then applies a nonlinear activation function:

$$\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b}). \tag{1.12}$$

Here W is a matrix of weights, $\mathbf{b}$ is a vector of intercepts or biases, and σ is applied component-by-component. Without σ, stacked layers are still only affine transformations (linear maps plus shifts). The nonlinearity is what lets neural networks represent more complicated functions.

This course does not begin with neural networks because the basic modeling questions are easier to see in linear regression. However, the same foundational ideas remain: inputs, outputs, parameters, loss, assumptions, prediction error, and the need to justify the modeling choices.

1.7 Statistical Concepts Used Throughout the Course

Before developing prediction error and model diagnostics, we need a small set of statistical concepts. This section reviews expectation, variance, covariance, correlation, and bias. These ideas will reappear throughout the course.

1.7.1 Expectation

The expectation of a random variable is its probability-weighted average. For a discrete random variable X with possible values $x_1, \ldots, x_m$ and probabilities $p_1, \ldots, p_m$, the expectation is

$$\mathbb{E}[X] = \sum_{i=1}^{m} p_i x_i. \tag{1.13}$$

The probabilities weight the possible outcomes. Outcomes that are more likely contribute more to the average.

Math Foundations Connection. If we collect the possible outcomes into a vector $\mathbf{x} = (x_1, \ldots, x_m)^T$ and the probabilities into a vector $\mathbf{p} = (p_1, \ldots, p_m)^T$, then

$$\mathbb{E}[X] = \mathbf{p}^T\mathbf{x}.$$

Thus expectation can be viewed as an inner product between a probability vector and an outcome vector. The probability vector must have nonnegative entries that sum to one. This connects the statistical idea of a weighted average to the linear algebra idea of an inner product. See the Math Foundations textbook, Chapters 2, 3, and 5.

Expectation has two especially important algebraic properties. First, for constants a and b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b. \tag{1.14}$$

Second, for random variables X and Y,

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]. \tag{1.15}$$

This second property does *not* require X and Y to be independent. This is why we call it linearity of expectation.

For the affine rule, the reasoning is straightforward:

$$\mathbb{E}[aX + b] = \mathbb{E}[aX] + \mathbb{E}[b] = a\mathbb{E}[X] + b,$$

since a constant has expectation equal to itself.

1.7.2 Variance

Variance measures the average squared deviation from the mean. If $\mu_X = \mathbb{E}[X]$, then

$$\operatorname{Var}(X) = \mathbb{E}\left[(X - \mu_X)^2\right]. \tag{1.16}$$

A useful computational identity is

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \tag{1.17}$$

To see why, expand the square:

$$\begin{aligned}
\operatorname{Var}(X) &= \mathbb{E}\left[(X - \mu_X)^2\right] \\
&= \mathbb{E}\left[X^2 - 2\mu_X X + \mu_X^2\right] \\
&= \mathbb{E}[X^2] - 2\mu_X \mathbb{E}[X] + \mu_X^2 \\
&= \mathbb{E}[X^2] - 2\mu_X^2 + \mu_X^2 \\
&= \mathbb{E}[X^2] - \mu_X^2.
\end{aligned}$$

Since $\mu_X = \mathbb{E}[X]$, this gives Equation (1.17).

Variance changes predictably when we scale and shift a random variable. For constants a and b,

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X). \tag{1.18}$$

The constant b does not affect variance because shifting all outcomes by the same amount does not change their spread. The scale factor a multiplies deviations by a, so squared deviations are multiplied by a^2.

1.7.3 Covariance and Correlation

Variance measures how one random variable varies around its mean. Covariance measures how two random variables vary together. If $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$, then

$$\operatorname{Cov}(X, Y) = \mathbb{E}\left[(X - \mu_X)(Y - \mu_Y)\right]. \tag{1.19}$$

If X tends to be above its mean when Y is above its mean, the covariance tends to be positive. If X tends to be above its mean when Y is below its mean, the covariance tends to be negative. If their deviations do not move together in a systematic linear way, the covariance may be near zero.

Covariance appears in the variance of a sum:

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y). \tag{1.20}$$

A direct derivation comes from writing centered variables. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. Then

$$\begin{aligned}
\operatorname{Var}(X + Y) &= \mathbb{E}\left[\left((X + Y) - (\mu_X + \mu_Y)\right)^2\right] \\
&= \mathbb{E}\left[\left((X - \mu_X) + (Y - \mu_Y)\right)^2\right] \\
&= \mathbb{E}\left[(X - \mu_X)^2\right] + \mathbb{E}\left[(Y - \mu_Y)^2\right] + 2\mathbb{E}\left[(X - \mu_X)(Y - \mu_Y)\right] \\
&= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y).
\end{aligned}$$

If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$. The converse is not generally true: zero covariance does not necessarily imply independence. Covariance detects linear co-movement, while dependence can be nonlinear.

Correlation is a scaled version of covariance:

$$\rho_{XY} = \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)}\sqrt{\mathrm{Var}(Y)}}. \tag{1.21}$$

Correlation is unitless and lies between -1 and 1 when both variances are positive. The scaling by the standard deviations makes correlation easier to compare across contexts than covariance.

Math Foundations Connection. For observed data, sample correlation can be interpreted geometrically. If we center two data vectors by subtracting their sample means, then their sample correlation is the cosine of the angle between the centered vectors. This is why correlation measures linear alignment: two centered data vectors pointing in nearly the same direction have correlation near 1; two pointing in opposite directions have correlation near -1; and two nearly orthogonal centered vectors have correlation near 0. See the Math Foundations textbook, Chapters 2, 3, and 5.

1.7.4 Bias

An estimator is a rule for using data to estimate an unknown quantity. If the unknown parameter is θ and the estimator is $\hat{\theta}$, then the bias of the estimator is

$$\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta. \tag{1.22}$$

Bias is systematic error. It asks whether the estimator tends to miss in a particular direction over repeated samples. Bias is not the same thing as random error. An estimator can be unbiased and still vary substantially from sample to sample. Conversely, a biased estimator may have low variability but systematically target the wrong value.

Bias can be induced by measurement choices, data collection procedures, model assumptions, or the deliberate use of a simplified model. This is one reason the phrase “all models are wrong” matters. A model can be useful while still introducing bias. The analyst’s job is not to pretend the model is perfect; it is to understand what kind of error the model introduces and whether that error is acceptable for the task.

1.8 Reproducibility and Defensibility

A data analysis must be more than numerically correct at one moment on one computer. It should also be reproducible and defensible.

1.8.1 Reproducibility

Reproducibility means that the same code applied to the same data produces the same results. A reproducible workflow has several features:

- the data source is clear;

- preprocessing steps are documented;
- transformations are recorded;
- modeling choices are traceable;
- assumptions and exclusions are stated;
- results can be rerun without guessing what happened.

A useful standard is this: your future self, or another analyst, should be able to rerun the analysis and understand the major decisions without needing a private explanation from you.

1.8.2 Defensibility

Defensibility concerns what we can explain to a customer, commander, decision maker, or technical reviewer. A defensible analysis answers questions such as:

- Why did we choose this model?
- What assumptions does the method rely on?
- What alternatives did we consider?
- What are the limitations of the data?
- How much uncertainty remains?
- What conclusion is supported, and what conclusion would overstate the evidence?

Reproducibility is about what we did. Defensibility is about what we can justify. Both are essential. A result can be reproducible but not defensible if it repeatedly applies the wrong method. A result can sound defensible but not be reproducible if the workflow cannot be checked. Good data analysis requires both.

1.9 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 1 use	Math Foundations connection	MF textbook location
Feature vectors for observations	$\mathbb{R}^n$ vector notation, vector entries, subvectors, and vectors as collections of quantities belonging to the same object	Ch02 Secs. 2.1, 2.2, 2.6
Linear models as weighted inputs	Scalar multiplication and linear combinations of vectors	Ch03 Secs. 3.4, 3.5
Inner-product form $\boldsymbol{\beta}^T\mathbf{x}$	Dot product definition, dot product properties, and examples of weighted sums	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3
Neural-network layers as repeated linear operations plus nonlinearities	Matrix-vector multiplication as row inner products and as linear combinations of columns	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3; Ch07 Secs. 7.1, 7.6.1, 7.6.2
Correlation as linear alignment	Dot products, Euclidean norm, distance, angle, and orthogonality	Ch05 Secs. 5.1, 5.3, 5.5, 5.7, 5.8
Reasoning from assumptions to conclusions	Logic, truth tables, implications, biconditionals, and De Morgan's laws	Ch01 Secs. 1.1, 1.7

1.10 Chapter Summary

This chapter introduced data analysis as statistical learning. A data set consists of observations and variables. In supervised learning, we distinguish input variables from an output or response variable. We model their relationship using

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

The two major modeling goals are inference and prediction. Inference asks what the model says about the relationship between inputs and output. Prediction asks how accurately the model can estimate outputs for new inputs. Regression problems involve quantitative outputs, while classification problems involve categorical outputs. Models may be linear or nonlinear, and modern methods such as neural networks build complex functions by combining linear operations with nonlinear activations.

Finally, the chapter reviewed expectation, variance, covariance, correlation, and bias. These concepts provide the mathematical language needed to discuss model error, uncertainty, assumptions, and the limits of prediction. The next chapter uses this foundation to study prediction error, model flexibility, overfitting, and generalization.

Chapter 2

Prediction Error, Flexibility, and Generalization

2.1 Why Prediction Error Matters

The previous chapter introduced the supervised learning model

$$Y = f_0(\mathbf{X}) + \epsilon, \tag{2.1}$$

where $\mathbf{X} = (X_1, \ldots, X_p)^T$ is the vector of input variables, Y is the response, f_0 is the true systematic relationship, and ϵ is the error term. In this chapter we ask a prediction-focused question:

> Given the information in $\mathbf{X}$, how accurately can we predict Y?

This question makes precise the phrase "all models are wrong." A model is not judged by whether it perfectly describes reality. It is judged by how much useful signal it captures, how much error remains, and whether its performance holds up on data it has not already seen.

To avoid a common notation problem, this chapter uses f_0 for the true unknown function and $\hat{f}$ for a fitted model. The true function f_0 is fixed but unknown. The fitted function $\hat{f}$ is random because it depends on the training data used to fit it. If a different training data set were collected, the fitted model would generally be different.

2.2 Mean Squared Error

The most common measure of prediction error for a quantitative response is *mean squared error*. For a new observation with predictors $\mathbf{x}_0$ and response Y_0, the prediction is

$$\widehat{Y}_0 = \hat{f}(\mathbf{x}_0).$$

The squared prediction error is

$$\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2.$$

The mean squared error, or MSE, is the expected squared prediction error:

$$\text{MSE}(\mathbf{x}_0) = \mathbb{E}\left[\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2\right]. \tag{2.2}$$

The expectation in Equation (2.2) averages over the randomness in the training data used to produce $\hat{f}$ and the randomness in the new response Y_0.

For an observed data set, we often compute an empirical version of MSE. Given observations $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$ and predictions $\hat{y}_i = \hat{f}(\mathbf{x}_i)$, the empirical MSE is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2. \tag{2.3}$$

Large errors are penalized heavily because the errors are squared. This makes MSE mathematically convenient, but it also means that a few large mistakes can dominate the score.

The square root of MSE is called *root mean squared error*, or RMSE:

$$\text{RMSE} = \sqrt{\text{MSE}}. \tag{2.4}$$

RMSE is often easier to interpret because it is measured in the same units as the response. For example, if Y is measured in hours, RMSE is also measured in hours; if Y is measured in thousands of dollars, RMSE is measured in thousands of dollars.

Math Foundations Connection. Let $\mathbf{y} = (y_1, \ldots, y_n)^T$ be the vector of observed responses and let $\hat{\mathbf{y}} = (\hat{y}_1, \ldots, \hat{y}_n)^T$ be the vector of predictions. The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

Using the Euclidean norm,

$$\|\mathbf{e}\| = \sqrt{e_1^2 + \cdots + e_n^2},$$

the empirical MSE can be written as

$$\text{MSE} = \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{y}}\|^2. \tag{2.5}$$

Thus MSE is a scaled squared distance between the observed response vector and the fitted response vector. This geometric view will become especially useful when we later interpret least squares regression as a projection problem. See the Math Foundations textbook, Chapters 2, 3, 5, and 12.

2.3 The Bias–Variance Decomposition

The bias–variance decomposition explains why prediction is limited even when we use a good model. It separates expected prediction error into three parts:

$$\text{MSE} = \text{bias}^2 + \text{model variance} + \text{noise}. \tag{2.6}$$

This equation is one of the central ideas in statistical learning.

2.3.1 Set-Up for the Decomposition

Fix a predictor value $\mathbf{x}_0$. Suppose the new response is generated by

$$Y_0 = f_0(\mathbf{x}_0) + \epsilon_0, \tag{2.7}$$

where

$$\mathbb{E}[\epsilon_0] = 0,$$
$$\text{Var}(\epsilon_0) = \sigma^2,$$

with ϵ_0 independent of the training data used to construct $\hat{f}$. We want the expected squared prediction error

$$\mathbb{E}\left[\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2\right].$$

Substitute Equation (2.7):

$$\mathbb{E}\left[\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2\right] = \mathbb{E}\left[\left(f_0(\mathbf{x}_0) + \epsilon_0 - \hat{f}(\mathbf{x}_0)\right)^2\right]. \tag{2.8}$$

Rearrange the expression inside the square:

$$f_0(\mathbf{x}_0) + \epsilon_0 - \hat{f}(\mathbf{x}_0) = \left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right) + \epsilon_0.$$

Now expand the square:

$$\begin{aligned}
&\mathbb{E}\left[\left(\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right) + \epsilon_0\right)^2\right] \\
&\quad = \mathbb{E}\left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right)^2\right] + 2\mathbb{E}\left[\epsilon_0\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right)\right] + \mathbb{E}[\epsilon_0^2].
\end{aligned}$$

The middle term is zero because ϵ_0 has mean zero and is independent of the fitted model. Also, since $\mathbb{E}[\epsilon_0] = 0$, we have $\mathbb{E}[\epsilon_0^2] = \text{Var}(\epsilon_0) = \sigma^2$. Therefore,

$$\mathbb{E}\left[\left(Y_0 - \hat{f}(\mathbf{x}_0)\right)^2\right] = \mathbb{E}\left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right)^2\right] + \sigma^2. \tag{2.9}$$

The final term, σ^2, is the noise variance. It is the variation in Y that remains even if f_0 were known perfectly.

2.3.2 Separating Bias and Variance

We now decompose

$$\mathbb{E}\left[\left(f_0(\mathbf{x}_0) - \hat{f}(\mathbf{x}_0)\right)^2\right].$$

Let

$$m(\mathbf{x}_0) = \mathbb{E}\left[\hat{f}(\mathbf{x}_0)\right]$$

be the average prediction we would get at $\mathbf{x}_0$ over many possible training data sets. Add and subtract $m(\mathbf{x}_0)$ inside the square:

$$\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0) = \left(\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0)\right) + \left(m(\mathbf{x}_0) - f_0(\mathbf{x}_0)\right).$$

Squaring and taking expectations gives

$$\mathbb{E}\left[\left(\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0)\right)^2\right] = \mathbb{E}\left[\left(\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0)\right)^2\right] + (m(\mathbf{x}_0) - f_0(\mathbf{x}_0))^2$$
$$+ 2\,(m(\mathbf{x}_0) - f_0(\mathbf{x}_0))\,\mathbb{E}\left[\hat{f}(\mathbf{x}_0) - m(\mathbf{x}_0)\right].$$

The final expectation is zero by the definition of $m(\mathbf{x}_0)$, so the cross term vanishes. Hence,

$$\mathbb{E}\left[\left(\hat{f}(\mathbf{x}_0) - f_0(\mathbf{x}_0)\right)^2\right] = \mathrm{Var}\left(\hat{f}(\mathbf{x}_0)\right) + \mathrm{Bias}\left(\hat{f}(\mathbf{x}_0)\right)^2, \tag{2.10}$$

where

$$\mathrm{Var}\left(\hat{f}(\mathbf{x}_0)\right) = \mathbb{E}\left[\left(\hat{f}(\mathbf{x}_0) - \mathbb{E}[\hat{f}(\mathbf{x}_0)]\right)^2\right], \tag{2.11}$$

$$\mathrm{Bias}\left(\hat{f}(\mathbf{x}_0)\right) = \mathbb{E}\left[\hat{f}(\mathbf{x}_0)\right] - f_0(\mathbf{x}_0). \tag{2.12}$$

Combining Equations (2.9) and (2.10), we obtain

$$\boxed{\mathrm{MSE}(\mathbf{x}_0) = \mathrm{Bias}\left(\hat{f}(\mathbf{x}_0)\right)^2 + \mathrm{Var}\left(\hat{f}(\mathbf{x}_0)\right) + \mathrm{Var}(\epsilon_0).} \tag{2.13}$$

This is the bias–variance decomposition.

2.3.3 Interpreting the Three Terms

The *squared bias* term measures systematic error. It is large when the average fitted model misses the true function. A model that is too simple can have high bias. For example, if the true relationship is curved but we insist on using a straight line, the fitted line may miss the main shape of the data no matter how much data we collect.

The *model variance* term measures how much the fitted model changes from one training data set to another. A highly flexible model may fit one training sample very closely but produce a noticeably different function if the sample changes. This instability is model variance.

The *noise* term, $\mathrm{Var}(\epsilon_0)$, is irreducible error. It is the part of the response that cannot be predicted from $\mathbf{X}$ using any method, because it comes from unmeasured factors, measurement error, randomness in the system, or other variation not captured by the inputs.

This distinction gives us the language of *reducible* and *irreducible* error. Bias and model variance are reducible in principle: we can try different model classes, collect better data, improve features, or tune the model. The noise term is irreducible with respect to the inputs we have chosen. The best possible prediction error approaches this irreducible error from above; it cannot go below it when the goal is to predict the actual future response Y.

2.4 Training Error and Test Error

Prediction models are trained on data. Let

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_{n_{\text{train}}}, y_{n_{\text{train}}})\}$$

be the training data used to fit $\hat{f}$. The *training MSE* is

$$\mathrm{MSE}_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \tag{2.14}$$

Training MSE measures how well the model performs on data it already used to learn.

The prediction problem, however, is not mainly about the training data. If we train a model using data from existing diabetes patients, we already know which of those patients had diabetes. The real goal is to make accurate predictions for future patients whose outcomes are not yet known. In the same way, a logistics forecast, readiness forecast, or maintenance forecast is useful only if it generalizes beyond the cases already used to build it.

Let

$$\mathcal{D}_{\text{test}} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_{n_{\text{test}}}, y_{n_{\text{test}}})\}$$

be a separate set of observations not used to fit the model. The *test MSE* is

$$\mathrm{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \tag{2.15}$$

Test MSE measures generalization: the model's ability to predict unseen data.

Training MSE is usually optimistic. Many fitting procedures explicitly choose model parameters to reduce training error. Therefore, a low training MSE does not guarantee a low test MSE. The model may simply have learned the idiosyncrasies of the training sample.

2.5 Flexibility

A model's *flexibility* is its capacity to approximate a wide range of functional shapes. A model is more flexible if it can represent a larger set of possible relationships $f(\mathbf{x})$.

For example, a straight-line model in one predictor can represent only lines:

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x.$$

A polynomial model with many terms can represent many more shapes:

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \cdots + \hat{\beta}_d x^d.$$

A smoothing spline, decision tree, random forest, boosted model, support vector machine, or neural network may be more flexible still, depending on how it is tuned.

Flexibility is not automatically good or bad. Its value depends on the relationship between the true function, the amount of noise, and the amount of data available. More flexible methods can reduce bias because they can capture more complex structure. At the same time, more flexible methods tend to increase model variance because they are more sensitive to the particular training sample.

As a general rule:

$$\text{more flexibility} \quad \Longrightarrow \quad \text{lower bias but higher variance.} \tag{2.16}$$

This is only a general rule, not a law that applies identically in every model class. It is nevertheless a useful guide for thinking about prediction error.

2.5.1 Degrees of Freedom as a Flexibility Measure

For some models, flexibility can be summarized by a quantity called *degrees of freedom*. In smoothing splines, for example, a low-degree-of-freedom fit is smoother and less flexible, while a high-degree-of-freedom fit is more wiggly and more flexible. In polynomial regression, the number of estimated parameters provides a simple proxy for flexibility. A model with two parameters is less flexible than a model with thirteen parameters.

The important pattern is this: as flexibility increases, training MSE usually decreases, because the model has more freedom to chase the training observations. Test MSE often decreases at first, but then may increase when the model becomes too flexible.

2.6 Overfitting and Overtraining

Overfitting occurs when a model follows noise or accidental patterns in the training data rather than the underlying relationship. The model looks impressive on the data used to fit it, but performs poorly on new observations. In the lecture slides, this is also described as *overtraining*: an overabundance of flexibility causes test MSE to increase compared with simpler models.

Overfitting is undesirable because prediction is about future or unseen data. A flexible model may fit the training data so well that it learns artifacts specific to that sample. These artifacts do not appear in future data, so test performance suffers.

A useful mental picture is the following. Suppose the true relationship is a smooth curve. A very simple model may underfit: it misses important curvature and has high bias. A moderately flexible model may capture the main trend and generalize well. A very flexible model may twist through individual noisy observations; it has low training error but high variance and worse test error.

Therefore, the goal is not to make training MSE as small as possible. The goal is to choose a model whose predictions generalize.

2.7 Train/Test Splits

A train/test split is a simple way to estimate generalization performance.

1. Randomly divide the available data into a training set and a test set.
2. Fit the model using only the training set.
3. Predict the responses in the test set.
4. Compute test MSE or test RMSE using the test-set predictions.
5. Compare candidate models using test performance, not training performance alone.

The key rule is that the test data must remain unseen during model fitting. If the test data are used to choose transformations, tune flexibility, select predictors, or adjust the model, then they are no longer a clean test of generalization. Later in the course we will use cross-validation to improve on a single train/test split, but the principle is the same: separate the data used to fit the model from the data used to evaluate the model.

2.7.1 A Simulation Example

The lecture simulation uses the model

$$Y = f_0(X) + \epsilon, \tag{2.17}$$

where

$$\begin{aligned} f_0(X) &= X^3, \\ X &\sim \text{Uniform}(-3, 3), \\ \epsilon &\sim \mathcal{N}(0, 9^2). \end{aligned}$$

The true function is cubic, but the observed data are noisy. A train/test split separates the simulated observations into data used to fit the model and data used to evaluate it.

Consider four candidate models with increasing numbers of parameters. The lecture example reports the following pattern:

Model	Number of Parameters	Training MSE	Test MSE
1	2	101.99	111.82
2	4	84.56	91.15
3	9	83.52	90.60
4	13	82.46	92.55

The training MSE decreases as the number of parameters increases. This is expected: more parameters give the fitted model more freedom to reduce training error. The test MSE, however, decreases at first and then increases. The 13-parameter model has the lowest training MSE, but not the lowest test MSE. In this example, the 9-parameter model gives the best test performance among the four candidates.

The lesson is not that nine parameters are always best. The lesson is that model selection is a balance between flexibility and test error. A model that is too rigid underfits; a model that is too flexible overfits. The best predictive model is usually somewhere between these extremes.

2.8 Parametric and Nonparametric Estimators

Another way to describe model classes is to distinguish *parametric* and *nonparametric* methods.

2.8.1 Parametric Methods

A method is parametric if the function f is described using a finite number of parameters. The model may be simple or complicated, linear or nonlinear, but it is governed by a finite parameter vector.

The standard linear model is parametric:

$$f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p = \beta_0 + \boldsymbol{\beta}^T \mathbf{X}, \tag{2.18}$$

where

$$\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^T.$$

Once we assume the form in Equation (2.18), learning f becomes the problem of estimating the finite list of parameters $\beta_0, \beta_1, \ldots, \beta_p$.

Math Foundations Connection. The term $\boldsymbol{\beta}^T\mathbf{X}$ is an inner product. It is also a linear combination of the entries of $\mathbf{X}$ with coefficients $\beta_1, \ldots, \beta_p$. This is why linear models connect naturally to the vector notation from Math Foundations: the model turns a feature vector into a prediction by weighting its components and adding the weighted components together. See the Math Foundations textbook, Chapters 2, 3, and 5.

Parametric does not mean "good" or "simple." For example, one could write a finite-parameter model such as

$$f(\mathbf{X}) = \beta_0 + \beta_{13}X_1 + \beta_2 X_2 + \beta_4 \frac{X_3 X_4}{\beta_3} + \cdots + \cos(\beta_p X_p).$$

This is still parametric because it is controlled by a finite number of parameters, but it is not necessarily sensible, interpretable, or easy to fit. Parametric modeling is powerful because it reduces an infinite function-estimation problem to a finite parameter-estimation problem. Its weakness is that the chosen form may be wrong.

2.8.2 Nonparametric Methods

A method is nonparametric if it does not impose a fixed finite-dimensional form on f in advance. Nonparametric methods attempt to learn the shape of f more directly from the data. They can be deceptively simple or highly complex.

The advantage of a nonparametric method is flexibility. If the true relationship is highly nonlinear, a nonparametric method may capture structure that a linear model misses. The disadvantage is that flexibility requires data. Without enough observations, a flexible method may not be able to distinguish signal from noise. This is another expression of the bias–variance tradeoff: nonparametric methods often reduce bias but may increase variance, especially when the sample size is small.

This course will use nonparametric methods, but the early chapters focus on linear models because linear models make the statistical logic visible. Understanding prediction error, bias, variance, residuals, and validation in simple models prepares us to use more complex models responsibly.

2.9 Model Flexibility and Interpretability

Different models occupy different positions on the flexibility–interpretability spectrum. Least squares linear regression is relatively inflexible but highly interpretable. Subset selection and lasso remain close to the linear-model family but modify which predictors are used or how coefficients are estimated. Generalized additive models and trees are more flexible. Bagging, boosting, support vector machines, and deep learning methods are often more flexible still, but their fitted relationships may be harder to interpret directly.

For prediction, increased flexibility can be valuable. Modern prediction problems often involve nonlinear patterns, interactions, and high-dimensional structure that a basic linear model cannot capture. For pure prediction tasks, linear models are often best treated as baselines rather than as the final word.

For inference, linear models remain extremely important. Their assumptions, estimators, uncertainty measures, and diagnostic tools are well-developed. When the goal is to explain relationships, defend assumptions, and communicate uncertainty to a decision maker, the interpretability of a linear model can be a major advantage.

The practical takeaway is not that one class of models is universally best. The task determines the model. If the goal is prediction, prioritize generalization error. If the goal is inference, prioritize interpretability, assumptions, and uncertainty quantification. When the goal includes both, state the tradeoff explicitly.

2.10 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 2 use	**Math Foundations connection**	**MF textbook location**
Response, prediction, and residual vectors	$\mathbb{R}^n$ vector notation, entries, and vector subtraction	Ch02 Secs. 2.1, 2.2, 2.6; Ch03 Sec. 3.1
Empirical MSE as squared distance	Dot products, $\mathbf{a}^T\mathbf{a} = \sum_i a_i^2$, Euclidean norm, and distance between vectors	Ch05 Secs. 5.1, 5.3, 5.5, 5.7
Residual length and RMSE	Euclidean norm properties and distance as $\|\mathbf{a} - \mathbf{b}\|$	Ch02 Sec. 2.6; Ch05 Secs. 5.5, 5.7
Linear models as finite-parameter estimators	Linear combinations, coefficients of linear combinations, and inner products	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3
Flexibility and degrees-of-freedom intuition	Linear independence, bases, and the maximum size of linearly independent sets in $\mathbb{R}^n$	Ch02 Sec. 2.1; Ch06 Secs. 6.3, 6.4, 6.5
Matrix notation behind fitted predictors	Matrix size, transpose notation, and matrix-vector multiplication	Ch02 Sec. 2.6; Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2

2.11 Chapter Summary

This chapter developed the statistical meaning of prediction error. Mean squared error measures average squared prediction error. Root mean squared error places the score back in the units of the response. The bias–variance decomposition shows that expected prediction error consists of squared bias, model variance, and irreducible noise.

Bias is systematic error caused by the model missing the true relationship on average. Model variance is instability caused by sensitivity to the training data. Noise is irreducible error in the response itself. More flexible models tend to reduce bias but increase variance, which is why more flexibility does not automatically mean better prediction.

Training MSE measures performance on data used to fit the model. Test MSE measures performance on unseen data. Because prediction is about generalization, test error is the more important quantity. Train/test splits provide a first method for estimating generalization performance.

Finally, the chapter distinguished parametric and nonparametric estimators. Parametric methods assume a finite-parameter structure for f, while nonparametric methods avoid imposing a fixed finite-dimensional form. Both can be useful, but both must be evaluated by whether they support the goal of the analysis.

Chapter 3

Data Quality, Cleaning, and Trustworthy Variables

3.1 Why Data Quality Comes Before Modeling

Before we build a statistical model, compute a regression line, or compare prediction errors, we need to know whether the variables in the data set can be trusted. Exploratory data analysis begins with the habit of "listening to the data." That means stepping back from the final question we want to answer and first asking what the data are trying to tell us about themselves.

Data cleaning is not a clerical step that happens before the real analysis. It is part of the analysis. A model can only learn from the variables we give it. If those variables are mislabeled, inconsistently recorded, stored in the wrong type, or contaminated by impossible values, then the model may produce output that looks mathematical but is not analytically defensible.

This chapter focuses on the practical question:

> What must be true about our variables before we trust them enough to summarize, visualize, or model them?

A trustworthy variable is one whose meaning, type, units, allowable values, and missing-value behavior are understood well enough that another analyst could reproduce the cleaning decision and defend it to a customer, commander, or reviewer. The goal is not to make the data look perfect. The goal is to make the workflow repeatable and the decisions defensible.

3.2 Data Tables, Observations, and Variables

A data set is often organized as a rectangular table. The rows are observations, and the columns are variables. If there are n observations and p variables, we can represent the data as a matrix

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times p} \tag{3.1}$$

when all entries are numeric. The entry x_{ij} is the value of variable j for observation i.

The ith row,

$$\mathbf{x}_i^T = (x_{i1}, x_{i2}, \dots, x_{ip}),$$

contains all measured information for observation i. The jth column,

$$\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{nj})^T,$$

contains all observed values of variable j.

This matrix notation is useful, but it can also hide an important fact: not every variable is genuinely numeric. Some columns may contain categories, ordered labels, dates, text, identifiers, or codes. A column made of numbers is not automatically a numerical variable in the statistical sense.

Math Foundations Connection. In linear algebra, a vector is a collection of entries that pertain to the same object or concept. In data analysis, a numeric variable can be viewed as a vector of observed values, and a full numeric data table can be viewed as a matrix whose columns are variable vectors. This perspective helps later when we discuss design matrices, regression, and projections. For cleaning, it helps us remember that operations applied to a column should make sense for the type of object that column represents. See the Math Foundations textbook, Chapters 2, 4, and 7.

3.3 Variable Types

A variable is an attribute that may change its value between observations in a data set. The type of a variable determines which summaries, plots, transformations, and models are meaningful.

Variable type	Meaning	Natural summaries
Continuous numeric	Values can be thought of as varying across a range, rather than only at isolated counts. The observed minimum and maximum are sample bounds, not part of the definition.	mean, median, spread, histogram
Discrete numeric	Values are usually counts or integer-valued measurements.	counts, mean, spread, bar plot or histogram
Ordinal	Categories have an order, but distances between categories may not be meaningful.	counts, proportions, median, ordered bar plot
Categorical	Categories are labels without a natural ordering.	counts, proportions, mode, bar plot

Examples of continuous numeric variables include age, temperature, height, time to complete a task, and distance traveled. Examples of discrete numeric variables include number of missions completed, number of defects found during inspection, and number of officers in a unit. Enlisted rank is an ordinal variable: the order matters, but the numerical distance from one rank to the next is not usually treated as constant. Service branch, blood type, platform type, and property view are categorical variables.

A common trap is treating numeric codes as quantitative measurements. A survey might record service branch as Army = 1, Marine Corps = 2, Navy = 3, Air Force = 4, Space Force = 5, and Coast Guard = 6. Those numbers are convenient labels, not measurements. The average of Army and Navy is not Marine Corps, and the value 6 is not larger than the value 1 in a quantitative sense. Later, categorical variables can be encoded numerically for modeling, but those encodings are modeling tools rather than evidence that the original category has arithmetic meaning.

3.4 Why Data Become Messy

Data can be messy for many reasons. Some problems are obvious, such as impossible values. Others are subtle, such as inconsistent units or category codes that look numeric. Common sources of data problems include data-entry errors, non-standardized labels, missing or incomplete data, inconsistent survey responses, inconsistent units or scales, duplicate records, impossible values, incorrect data types, dates stored as text, categorical variables stored as integers, and numeric variables stored as strings.

The analyst's job is not merely to find these issues. The analyst must decide what to do about them and document the decision well enough that someone else could follow the same logic.

3.5 The Data Cleaning Workflow

Data cleaning is the process of identifying, documenting, and correcting data-quality issues before analysis. A useful first-pass workflow is:

1. identify and handle missing values,
2. fix data types,
3. rename and standardize columns,
4. subset rows and columns as needed,
5. detect outliers,
6. conduct basic sanity checks.

This sequence is not strictly linear. Fixing a data type may reveal new missing values; detecting an outlier may reveal a unit mismatch; a sanity check may send you back to the original data source. The workflow is iterative.

A cleaning decision should satisfy two standards. First, it should be *repeatable*: the same code plus the same raw data should produce the same cleaned data. Second, it should be *defensible*: the analyst should be able to explain why the decision was reasonable for the variable, the data source, and the purpose of the analysis.

A minimal cleaning log should record the variable, the issue, the decision, and enough detail for another analyst to reproduce the choice. For example, "`weight_lb`: values less than or equal to 0 are physically impossible for the study population; recoded as missing." The log does not need to be elaborate, but it should prevent the cleaning process from becoming a set of undocumented manual edits.

3.6 Missing Values and Imputation

Missing values occur when an observation should have a value for a variable but the recorded data do not contain that value. Missingness can be explicit, such as `NA`, `NaN`, blank cells, or `NULL`. Missingness can also be implicit, such as a value of 0 used as a placeholder for an impossible measurement.

For a data matrix, define a missingness indicator

$$m_{ij} = \begin{cases} 1, & \text{if } x_{ij} \text{ is missing,} \\ 0, & \text{if } x_{ij} \text{ is observed.} \end{cases} \tag{3.2}$$

Then the number of missing values in variable j is

$$M_j = \sum_{i=1}^{n} m_{ij}, \tag{3.3}$$

and the missingness rate is

$$r_j = \frac{M_j}{n}. \tag{3.4}$$

These simple counts identify which variables have missingness and how severe it is.

One of the most important missing-data checks is distinguishing *true zeros* from *missing zeros*. A true zero means the value is genuinely zero, such as `sibling_count = 0`. A missing zero means zero was used as a placeholder for a missing or invalid value, such as `weight_lb = 0` for an adult human. Missingness cannot be detected by a universal rule such as "replace all zeros." The rule must depend on the variable definition.

Once missing values are identified, the analyst usually has three choices: drop, impute, or flag.

Action	**When it may be reasonable**	**Main risk**
Drop rows	Missingness is limited, the variables are required, and the complete rows are still representative.	Can remove useful data or bias the analysis.
Drop columns	A variable is unnecessary, too incomplete, unreliable, or poorly defined.	May discard a useful signal for a later analysis.
Impute	A missing value can be replaced using a defensible rule or model.	Can create false precision or distort distributions and relationships.
Flag missingness	Missingness itself may be informative.	Requires careful interpretation in summaries and models.

For complete-case analysis, if J is the set of variables required for a particular analysis, the retained row set is

$$I_{\text{complete}} = \{i : m_{ij} = 0 \text{ for all } j \in J\}. \tag{3.5}$$

The cleaned analysis table is the submatrix containing rows in I_{complete} and columns in J.

Imputation replaces missing values with reasonable substitutes. If x_{ij} is missing, an imputed value is often written as $\widehat{x}_{ij}$, giving

$$x_{ij}^{\text{clean}} = \begin{cases} x_{ij}, & \text{if } x_{ij} \text{ is observed,} \\ \widehat{x}_{ij}, & \text{if } x_{ij} \text{ is missing.} \end{cases} \tag{3.6}$$

Simple imputation uses only the variable with missingness. For a numeric variable X_j, mean imputation replaces missing values with

$$\widehat{x}_{ij} = \bar{x}_{j,\text{obs}} = \frac{1}{n - M_j} \sum_{i:m_{ij}=0} x_{ij}. \tag{3.7}$$

Median, mode, constant, or explicit-category imputation may also be used when appropriate. Conditional imputation uses other variables to make better guesses, such as regression imputation, k-nearest-neighbor imputation, predictive mean matching, or other model-based methods. Multiple imputation creates several plausible completed data sets, analyzes each, and combines the results.

The key caution is that imputation is an assumption, not a fact. Avoid imputing automatically when "did not answer" is itself meaningful, when the variable is not needed, when missingness is too extensive, when imputation would introduce obvious bias, when the missing value represents a choice rather than a failed measurement, or when the data source is too unreliable to support replacement.

3.7 Fixing Data Types and Standardizing Variables

A software data type is how software stores and treats a variable. A statistical variable type is what the variable means analytically. Cleaning often requires aligning both.

Common data-type problems include numeric columns stored as strings, categorical variables stored as integers, dates stored as text, binary variables stored as text labels, identifiers stored as numbers, and numbers with commas, units, or symbols stored as text. The practical cleaning question is:

> Does the software representation support the analysis we intend to perform?

Problem	**Examples**	**Cleaning check**
Numeric stored as text	`1,250`, `72 in`, `unknown`, `--`	Parse valid numbers; decide whether nonnumeric entries are missingness, labels, or errors.
Category stored as integer	Service branch coded 1–6	Ask whether arithmetic on the codes is meaningful. If not, treat as categorical or ordinal.
Date stored as text	`04/03/2026`, `2026-04-03`, `3 Apr 26`	Parse to date format, then validate impossible dates, unexpected future dates, and time-zone issues.
Identifier stored as number	ID, hull number, survey response number	Preserve identity; do not average or model as a quantitative variable unless explicitly meaningful.

Column names are also part of the analysis. A useful naming convention should be consistent, descriptive, short enough to use in code, free of spaces and special characters, and stable across related data sets. Including units in the name can prevent later confusion: `height_in` and `height_cm` are safer than simply `height`.

Subsetting means selecting only part of a data set. Let $I \subseteq \{1, 2, \ldots, n\}$ be a set of row indices and let $J \subseteq \{1, 2, \ldots, p\}$ be a set of column indices. The submatrix

$$\mathbf{X}_{I,J}$$

keeps rows in I and columns in J. For example, if $x_{i,\text{age}}$ is the age for observation i, then

$$I_{\text{adult}} = \{i : x_{i,\text{age}} \geq 18\} \tag{3.8}$$

keeps adult observations. Column subsets are usually defined by the analysis purpose, such as keeping the response, the intended predictors, and identifiers needed for traceability.

Math Foundations Connection. Subsetting is a set operation. The row set I identifies which observations remain. The column set J identifies which variables remain. Thinking in sets helps avoid vague language such as "we removed bad rows." A defensible cleaning rule should be expressible as a clear predicate, such as "remove rows where weight is missing or impossible." See the Math Foundations textbook, Chapters 2 and 4.

3.8 Outliers and Sanity Checks

An outlier is an observation that is unusually far from the rest of the data or does not follow the overall pattern. Outliers matter because they can strongly affect summaries, visualizations, and fitted models. They should be examined before computing final summaries or fitting final models.

Common visual tools include histograms, boxplots, and scatterplots. A histogram can reveal extreme values, unusual gaps, skewness, or multiple modes. A boxplot can reveal observations outside the central bulk of the data. A scatterplot can reveal points that do not follow a bivariate pattern.

Outliers are not automatically errors. An outlier may be a data-entry error, a unit-conversion error, a duplicate or merged-record problem, a rare but valid observation, evidence of subgroups, or evidence that the model or summary is too simple. Removing a valid extreme observation just because it is inconvenient can bias the analysis.

A common numerical rule for outlier flagging uses the interquartile range. Let Q_1 be the first quartile, Q_3 be the third quartile, and

$$\text{IQR} = Q_3 - Q_1. \tag{3.9}$$

Tukey fences flag values outside

$$[Q_1 - 1.5\,\text{IQR},\ Q_3 + 1.5\,\text{IQR}]\,. \tag{3.10}$$

This rule is a flag, not an automatic deletion command.

Sanity checks ask whether the data make sense before we treat them as evidence. One useful warning is:

> Be wary of taking data at face value, especially when you do not know how it was generated.

At minimum, ask:

- Are units consistent, or might a variable mix inches with centimeters, miles with kilometers, or dollars with thousands of dollars?
- Are there impossible values, such as negative ages, probabilities outside $[0, 1]$, counts below zero, or physically impossible measurements?

- Does the number of rows match the expected row unit: one person, one transaction, one unit, one day, or one person-day?
- Are duplicates expected or suspicious?
- Are categorical levels spelled consistently?
- Do minimums, maximums, histograms, and summary statistics agree with subject-matter expectations?

Impossible values are not always easy to repair. Sometimes the correct value can be recovered from documentation or another field. Sometimes the value should be treated as missing. Sometimes the entire record should be excluded from a specific analysis. The decision should be documented.

3.9 A Trustworthy Variable Checklist

Before using a variable in a summary, plot, or model, ask the following questions:

1. **Definition:** What does the variable measure, and what is the row unit associated with the measurement?
2. **Type:** Is the variable continuous, discrete, ordinal, categorical, date/time, text, or an identifier? Does the software type match the statistical meaning?
3. **Units and scale:** Are units consistent? Are values raw, standardized, logged, percentages, or rates?
4. **Missingness:** What values indicate missingness? Are placeholders such as 0, 999, -1, or blanks being used? Is missingness itself meaningful?
5. **Validity:** What values are possible, impossible, or unusual but possible? Are there outliers, duplicates, or inconsistent labels?
6. **Defensibility:** Can the cleaning decision be explained in plain language, implemented in code, and documented in a cleaning log?

This checklist is intentionally practical. A variable is not trustworthy merely because it has numbers in a column. It becomes trustworthy when its meaning and cleaning decisions are clear enough to reproduce and defend.

3.10 Realities of Data Cleaning

Real data-driven studies often involve very large data sets. In an ideal world, errors would be fixed at the data-entry point. In practice, analysts often receive data after many upstream choices have already been made. We fix what we can, document what we cannot fix, and avoid pretending that the data are cleaner than they are.

Some issues are easier to spot than others. Numerical problems may be visible through summary statistics and visual EDA. Categorical problems may require careful inspection of labels and codes. Missingness may require knowledge of how the data were collected. Credibility issues may require comparing the data against external expectations or subject-matter knowledge.

Errors can also be informative. If a toy example contains several obvious errors, those errors may be a warning about the credibility of the data source. The right conclusion may not be "clean harder." It may be "find a better data set."

3.11 Representative Mini-Examples

Two common cleaning traps illustrate the chapter's main ideas.

Categorical codes are not quantities. Suppose a survey records service branch using integer codes. A careless analysis might treat the column as numeric and compute an average branch code. That summary is meaningless. The defensible cleaning decision is to recode the integers to meaningful labels and treat the variable as categorical.

Zero may mean different things in different variables. A value of `sibling_count = 0` may be a valid true zero. A value of `weight_lb = 0` for an adult is likely missing or invalid. The same recorded number has different meanings because the variables have different definitions. Cleaning rules must be variable-specific.

3.12 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 3 use	**Math Foundations connection**	**MF textbook location**
Data rows and variables as structured vectors	$\mathbb{R}^n$ vector notation, entries, subvectors, and vectors as collections of quantities	Ch02 Secs. 2.1, 2.2, 2.6
Numeric, ordinal, and categorical variable domains	Set membership, complements, set difference, subsets, and common number systems	Ch02 Sec. 2.6; Ch04 Secs. 4.1, 4.2, 4.4
Filtering rows and selecting columns	Subvector/slice notation and matrix submatrix extraction	Ch02 Sec. 2.2; Ch07 Secs. 7.1, 7.2
Cleaning by subsetting observations	Set operations, subset notation, quantified statements, and negation of quantified statements	Ch01 Sec. 1.1; Ch02 Sec. 2.6; Ch04 Secs. 4.1, 4.2, 4.4, 4.7, 4.8
Outlier detection as distance from typical pattern	Euclidean norm, distance between vectors, and angle/orthogonality intuition	Ch05 Secs. 5.5, 5.7, 5.8
Design-matrix readiness after cleaning	Matrix dimensions, sparse matrices, transpose notation, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2

3.13 Chapter Summary

Data cleaning is the foundation for trustworthy data analysis. Before modeling, the analyst must understand what each variable means, what type it is, how it is stored, what values are possible, and how missingness is represented.

The main ideas from this chapter are:

- Exploratory data analysis begins by listening to the data before asking targeted modeling questions.
- Variables may be continuous, discrete, ordinal, or categorical.
- Numeric storage does not guarantee numeric meaning.
- Missing values must be detected, counted, interpreted, and handled deliberately.
- True zeros must be distinguished from placeholder zeros.
- Dropping data is simple but can remove information or introduce bias.
- Imputation can preserve rows but introduces assumptions.
- Data types, column names, row subsets, and outliers must be checked systematically.
- Sanity checks protect the analyst from taking unreliable data at face value.
- Cleaning decisions should be repeatable and defensible.

The next chapter builds on this foundation by using numerical summaries and visual structure to describe cleaned data. Once variables are trustworthy enough to inspect, we can ask what their distributions, relationships, and patterns reveal.

Chapter 4

Exploratory Data Analysis: Summaries and Visual Structure

4.1 Why Summaries and Visual Structure Matter

The previous chapter focused on whether variables are trustworthy enough to analyze. Once we have checked definitions, missingness, units, data types, impossible values, and outliers, we can begin describing what the data appear to say.

Exploratory data analysis, or EDA, is the habit of listening to data before forcing it into a model. This chapter develops two complementary tools:

1. **single-number summary statistics**, which compress one feature of a distribution into one number; and
2. **visual EDA**, which uses plots to reveal distributional shape, relationships, subgroups, and unusual structure.

Good EDA combines both. Numbers make comparisons precise; plots make structure visible. The practical question for this chapter is:

> How do we summarize and visualize variables in a way that is accurate, interpretable, and defensible?

4.2 A Numeric Variable as a Vector

Let $x_1, x_2, \ldots, x_n$ be the observed values of a numeric variable X for n observations. We can collect these values in a vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n. \tag{4.1}$$

This notation is useful because most numerical summaries are functions of the entries of $\mathbf{x}$.

For example, the sample mean can be written as

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{4.2}$$

Using the ones vector $\mathbf{1} \in \mathbb{R}^n$, the same quantity is

$$\bar{x} = \frac{1}{n}\mathbf{1}^T\mathbf{x}. \tag{4.3}$$

This linear algebra view is not required for basic EDA, but it helps us see that summaries are not magic. They are operations applied to the observed data vector.

Math Foundations Connection. A vector is an ordered collection of quantities that pertain to the same object or concept. In EDA, a variable column is a vector of observed values. Dot products, sums, centered vectors, and norms give compact ways to express means, variances, covariances, and correlations. See the Math Foundations textbook, Chapters 2, 3, and 5.

4.3 Single-Number Summary Statistics

A single-number summary statistic characterizes one aspect of a variable's distribution. It does not describe everything about the variable. It answers a specific question.

Summary family	Guiding question	Examples
Center	Where is a typical value?	mean, median, mode
Spread	How variable are the values?	range, IQR, variance, standard deviation
Shape	How is the distribution arranged?	skewness, kurtosis
Extremes	What are the edge cases?	minimum, maximum, outlier flags

Table 4.1: Single-number summaries answer different questions about a variable.

The most common EDA mistake is to compute a statistic without saying what question it answers. The same data set may require several summaries: a center for a typical value, a spread for variability, a shape summary for asymmetry or tails, and an extreme-value check for unusual observations.

4.4 Center and Spread

4.4.1 Measures of Center

Measures of center summarize where a distribution is located. The three most common are the mean, median, and mode.

The **mean** is the balance point:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{4.4}$$

It uses every value, which makes it natural for many numerical variables, but also sensitive to outliers.

The **median** is the middle value after sorting the data. If

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

denotes the ordered data, then the median is the middle ordered value for odd n and the average of the two middle ordered values for even n. The median is robust to outliers because changing an extreme value usually does not move the middle of the ordered list.

The **mode** is the most common value. It is especially useful for categorical variables. For continuous numerical variables, exact ties may be rare, so analysts often discuss peaks of a histogram or density estimate rather than an exact modal value.

Situation	Useful center summary	Reason
Symmetric numeric variable	Mean or median	Often similar
Skewed or outlier-prone numerical variable	Median	More robust to tail values
Categorical variable	Mode	Categories do not have arithmetic averages
Ordinal variable	Median or mode	Respects ordering without assuming equal spacing

4.4.2 Measures of Spread

Measures of spread summarize variability. Two variables can have the same center but very different spread.

The **range** is

$$\text{Range}(\mathbf{x}) = \max_i x_i - \min_i x_i. \tag{4.5}$$

It is easy to interpret but depends only on the two most extreme observations.

The **interquartile range** is

$$\text{IQR} = Q_3 - Q_1, \tag{4.6}$$

where Q_1 is the first quartile and Q_3 is the third quartile. Because it measures the spread of the middle half of the data, the IQR is more robust to outliers than the range.

The **sample variance** and **sample standard deviation** are

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2, \tag{4.7}$$

$$s_x = \sqrt{s_x^2}. \tag{4.8}$$

Variance is mathematically powerful but is measured in squared units. Standard deviation is often easier to interpret because it is measured in the original units of the variable.

Define the centered data vector

$$\widetilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}. \tag{4.9}$$

Then

$$s_x^2 = \frac{1}{n-1} \widetilde{\mathbf{x}}^T \widetilde{\mathbf{x}}. \tag{4.10}$$

Thus spread can be interpreted geometrically: large spread means the centered data vector has large squared length.

4.5 Shape and Extremes

Measures of shape describe how a distribution is arranged beyond its center and spread. The two shape ideas emphasized here are **skewness** and **kurtosis**.

For a random variable X with mean μ and standard deviation σ, the population skewness is

$$\gamma_1 = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}. \tag{4.11}$$

Positive skewness indicates a long or heavy right tail; negative skewness indicates a long or heavy left tail. In a right-skewed distribution, the mean is often larger than the median because the right tail pulls the mean upward.

The population kurtosis is

$$\gamma_2 = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4}. \tag{4.12}$$

For EDA, it is safest to interpret kurtosis as tail weight rather than simply as "peakedness." The normal distribution has kurtosis 3, so excess kurtosis is $\gamma_2 - 3$.

Robust analogs. Mean, variance, skewness, and kurtosis are moment-based summaries; they use arithmetic powers of deviations from the mean. Median, IQR, Bowley skewness, and Moors kurtosis are quantile-based analogs. Quantile-based summaries are often more robust when outliers or heavy tails are present. For this course, the main habit is more important than memorizing every robust formula: when outliers or skewness are visible, compare moment-based summaries with quantile-based summaries.

The simplest measures of extremes are the minimum and maximum. Extremes are not automatically bad: they may be rare but valid observations, data-entry errors, unit mistakes, duplicated records, or evidence of subgroups.

A common outlier flag uses Tukey fences. Values are flagged when they fall outside

$$[Q_1 - 1.5\,\mathrm{IQR},\ Q_3 + 1.5\,\mathrm{IQR}]\,. \tag{4.13}$$

This rule is tied to the box plot. It is a screening tool, not an automatic deletion rule. A defensible action is: flag the value, inspect the context, and document the decision.

4.6 Visual EDA

Numerical summaries compress. Plots reveal. Visual EDA uses graphics to show the behavior of observations directly. A plot should match the variable types involved.

4.6.1 Univariate Visual EDA

Univariate EDA examines one variable at a time. For categorical or ordinal variables, a **bar plot** displays counts or proportions by category. It helps reveal common levels, rare levels, unexpected labels, and ordering problems.

For numerical variables, a **histogram** groups observations into bins and displays the number or proportion of observations in each bin. A histogram helps inspect symmetry, skewness, modality, gaps, clusters, outliers, and approximate bell shape. Because histograms depend on bin width, it is good practice to inspect more than one bin choice before making a strong claim about shape.

A histogram may be unimodal, bimodal, or multimodal. Multiple peaks often suggest subgroups, different data sources, or different operating conditions. The next EDA step is usually to color, facet, or group the data by a relevant categorical variable.

4.6.2 Bivariate Visual EDA

Bivariate EDA examines the relationship between two variables. The plot depends on the type of each variable.

Variable 1	Variable 2	Recommended plot
Categorical / ordinal	Categorical / ordinal	Grouped bar plot
Categorical / ordinal	Numerical	Grouped box plot
Numerical	Categorical / ordinal	Grouped box plot
Numerical	Numerical	Scatterplot

Table 4.2: Recommended bivariate EDA plots by variable-type pairing.

Grouped bar plots compare category combinations, especially counts or proportions. Grouped box plots compare numerical distributions across categories, including medians, IQRs, skewness, and outlier flags. Scatterplots compare two numerical variables and are often the first place we notice whether a linear model is a reasonable first approximation. A visible curve, fan shape, cluster pattern, or outlier should be noted before modeling.

4.6.3 Multivariate Visual EDA

Multivariate EDA examines more than two variables. True high-dimensional visualization is difficult, so practical multivariate EDA often uses combinations of pairwise views.

A **scatterplot matrix** displays many pairwise relationships at once. If there are p numerical variables, there are

$$\frac{p(p-1)}{2} \tag{4.14}$$

unique pairwise scatterplots. The diagonal entries are often histograms or density plots for the individual variables.

A **correlation heatmap** displays many pairwise correlations compactly. It can reveal blocks of related variables, strongly positive or negative relationships, possible multicollinearity before regression, and variables that appear weakly related to all others. A heatmap is not truly multivariate in the sense of showing all variables simultaneously; it summarizes many pairwise linear relationships.

4.7 Applications of Standard Deviation

Standard deviation becomes especially useful when the distribution is approximately bell-shaped. If X is approximately normal with mean μ and standard deviation σ, then approximately

$$\mathbb{P}(\mu - \sigma \le X \le \mu + \sigma) \approx 0.68, \tag{4.15}$$

$$\mathbb{P}(\mu - 2\sigma \le X \le \mu + 2\sigma) \approx 0.95, \tag{4.16}$$

$$\mathbb{P}(\mu - 3\sigma \le X \le \mu + 3\sigma) \approx 0.997. \tag{4.17}$$

This is the **empirical rule**. It is an approximation, not a universal law. It should be used only when the histogram is reasonably bell-shaped or when a normal model is otherwise justified.

When the data are not approximately normal, Chebyshev's inequality provides a more general lower bound. For a random variable X with finite mean μ and finite nonzero variance σ^2, for any $k > 0$,

$$\mathbb{P}(|X - \mu| \ge k\sigma) \le \frac{1}{k^2}. \tag{4.18}$$

Equivalently, for $k > 1$,

$$\mathbb{P}(|X - \mu| < k\sigma) \ge 1 - \frac{1}{k^2}. \tag{4.19}$$

Window	Empirical rule, if bell-shaped	Chebyshev lower bound
Within 1σ	about 68%	no useful positive bound
Within 2σ	about 95%	at least 75%
Within 3σ	about 99.7%	at least 88.9%

The practical takeaway is simple: if the histogram looks bell-shaped, the empirical rule is more informative; if the shape is unknown or non-normal, Chebyshev is safer but more conservative.

4.8 Covariance, Correlation, and Heatmaps

For two numeric variables X and Y, covariance measures whether they tend to move together. The population covariance is

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \tag{4.20}$$

The sample covariance is

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}). \tag{4.21}$$

Covariance depends on units, which makes it hard to compare across variable pairs. Correlation rescales covariance into the interval $[-1, 1]$:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}, \tag{4.22}$$

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}. \tag{4.23}$$

Let

$$\widetilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1},$$
$$\widetilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}.$$

Then the sample correlation can be written as

$$r_{xy} = \frac{\widetilde{\mathbf{x}}^T \widetilde{\mathbf{y}}}{\|\widetilde{\mathbf{x}}\| \, \|\widetilde{\mathbf{y}}\|}. \tag{4.24}$$

This is the cosine of the angle between the centered data vectors. If they point in nearly the same direction, correlation is positive and close to 1. If they point in nearly opposite directions, correlation is negative and close to -1. If they are nearly orthogonal, correlation is near 0. See the Math Foundations textbook, Ch05 Sec. 5.8.

Important caution. Correlation measures linear association. A correlation near 0 does not prove that two variables are unrelated. The relationship may be curved, segmented, or driven by subgroups. Always pair correlation with a scatterplot when possible.

If $\mathbf{X} \in \mathbb{R}^{n \times p}$ contains p numerical variables, then the sample correlation matrix is

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix}, \tag{4.25}$$

where r_{jk} is the sample correlation between variables j and k. For nonconstant variables, the diagonal entries satisfy $r_{jj} = 1$; constant-variable correlations are undefined. The matrix is symmetric because $r_{jk} = r_{kj}$.

4.9 A Practical EDA Workflow

A reusable EDA workflow is:

1. Confirm the variable type and check missingness.
2. Compute center, spread, shape, and extreme summaries for key numerical variables.
3. Make a univariate plot and compare the plot to the summaries.
4. For pairs of variables, choose the plot based on the two variable types.
5. For many numerical variables, inspect a scatterplot matrix or correlation heatmap.
6. Record unusual patterns, possible explanations, and follow-up questions.

The goal is not to produce every possible plot. The goal is to learn enough about the data to make the next modeling or analytic step more defensible.

Coding companion reference. Package-specific Python/R commands for EDA summaries, grouped plots, and heatmaps have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual EDA workflow and interpretation guidance.

4.10 Compact EDA Example

Suppose `pft_score` is a numerical variable. A first EDA pass might compute the mean, median, standard deviation, IQR, minimum, and maximum, then create a histogram. If the mean and median are close and the histogram is roughly symmetric, the empirical rule may be a reasonable approximation. If the histogram is skewed or multimodal, the median and IQR may better represent the typical value and spread.

If `score` is numerical and `gender` is categorical, a grouped box plot compares distributions across groups. Do not stop at comparing medians. Inspect IQRs, outlier flags, skewness within each group, group sample sizes, and overlap between groups. A grouped box plot can reveal a pattern, but it does not explain why the pattern exists.

Finally, suppose a medical data set contains variables such as `triceps`, `insulin`, and `bmi`. If zeros are impossible or implausible for those measurements, then a correlation heatmap before cleaning may be misleading because placeholder zeros create artificial relationships. Recode impossible zeros as missing, document the decision, and create summaries and plots after cleaning.

4.11 Reading EDA Output Defensibly

EDA should produce observations that are specific, cautious, and tied to evidence.

Weak statement	Better EDA statement
The data are normal.	The histogram is roughly bell-shaped and symmetric, so the empirical rule may be reasonable.
There are outliers.	Tukey fences flag three high values; check units and entry errors before removal.
These variables are related.	The scatterplot shows a positive, roughly linear association; the sample correlation is positive.
Group A is better than Group B.	Group A has a higher median and smaller IQR in this sample; this plot alone does not establish causation.
The heatmap proves multicollinearity.	The heatmap suggests strong pairwise correlations; formal modeling diagnostics can check multicollinearity later.

A defensible EDA statement distinguishes what was observed from what was inferred.

4.12 Reusable EDA Checklist

Use this compact checklist when exploring a cleaned data set:

- Identify each variable type: continuous, discrete, ordinal, categorical, date, text, or identifier.
- Check missingness, minimum, maximum, impossible values, and suspicious zeros.
- Choose center and spread summaries that match the variable and distribution.

- Inspect shape: symmetry, skewness, modality, tails, gaps, clusters, and outlier flags.
- For pairs of variables, choose the plot by variable types and look for linear, curved, grouped, or outlier-driven patterns.
- For many numerical variables, use scatterplot matrices or heatmaps to decide which relationships deserve closer inspection.
- Document the evidence, the follow-up question, and the limitation of each claim.

4.13 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 4 use	**Math Foundations connection**	**MF textbook location**
A numeric variable as a data vector	$\mathbb{R}^n$ vector notation, entries, and vectors as collections of quantities	Ch02 Secs. 2.1, 2.2, 2.6
Mean as a vector operation	Dot product examples, including $\mathbf{1}^T\mathbf{a} = \sum_i a_i$ and $(1/n)\mathbf{1}^T\mathbf{a}$ as a sample mean	Ch02 Sec. 2.4; Ch05 Secs. 5.1, 5.3
Variance and standard deviation as squared length	Dot products, $\mathbf{a}^T\mathbf{a} = \sum_i a_i^2$, and Euclidean norm $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}}$	Ch05 Secs. 5.1, 5.3, 5.5
Centered-vector view of spread	Vector subtraction, scalar multiplication, norm properties, and distance	Ch03 Secs. 3.1, 3.2, 3.4; Ch05 Secs. 5.5, 5.7
Correlation as geometric alignment	Angle formula, dot products, norms, and the orthogonality case $\mathbf{a}^T\mathbf{b} = 0$	Ch05 Secs. 5.1, 5.3, 5.5, 5.8
Scatterplot matrices and heatmaps	Matrix dimensions, submatrix notation, transpose, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.2, 7.4, 7.6.1, 7.6.2

4.14 Chapter Summary

The main ideas from this chapter are:

- Single-number summaries describe center, spread, shape, and extremes.
- The mean is a balance point but is sensitive to outliers; the median and IQR are more robust.
- Skewness measures asymmetry; kurtosis emphasizes tail weight.
- Tukey fences flag possible outliers but do not justify automatic deletion.
- Histograms show univariate numerical structure, including skewness, modality, and approximate bell shape.
- The empirical rule is useful for approximately Gaussian distributions; Chebyshev's inequality gives a conservative bound when normality is not justified.

- Bivariate plots should be selected based on the two variable types.
- Scatterplot matrices and correlation heatmaps summarize many pairwise relationships.
- Correlation is a scaled covariance and can be interpreted geometrically as the cosine between centered data vectors.
- EDA claims should be specific, cautious, and tied to the actual summary or plot that supports them.

The next chapter begins simple linear regression. The EDA habits from this chapter will matter immediately: before fitting a line, we should understand the response distribution, inspect the predictor, check the scatterplot, and decide whether a linear relationship is a reasonable first approximation.

Chapter 5

Simple Linear Regression: Model, Assumptions, and Least Squares Estimation

5.1 The Big Picture

The previous chapters built the workflow needed before formal modeling: define trustworthy variables, clean the data, summarize distributions, and visualize relationships. This chapter begins the formal regression sequence.

The central question is:

> Given one input variable and one quantitative response variable, what is the equation of the best line through the data?

Simple linear regression, or SLR, is the first model we use to answer that question. It predicts a quantitative response Y using a single predictor X. The word *simple* means one predictor. The word *linear* means that the conditional mean of Y is modeled as a straight-line function of X.

SLR is important for three reasons.

1. It is the cleanest place to see the difference between a true model and an estimated model.
2. It gives explicit formulas for the least squares estimators $\widehat{\beta}_0$ and $\widehat{\beta}_1$.
3. It introduces residuals, normal equations, and model assumptions that appear throughout the rest of regression.

5.2 Review: From Statistical Learning to Linear Regression

In the statistical learning framework, we assume that a response variable Y is related to predictors through

$$Y = f(\mathbf{X}) + \epsilon, \tag{5.1}$$

where f is the systematic part of the relationship and ϵ is random error or noise.

If $\mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] = 0$, then

$$\begin{aligned}\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] &= \mathbb{E}[f(\mathbf{X}) + \epsilon \mid \mathbf{X} = \mathbf{x}] && (5.2)\\ &= f(\mathbf{x}) + \mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] && (5.3)\\ &= f(\mathbf{x}). && (5.4)\end{aligned}$$

Thus, under the mean-zero error condition, $f(\mathbf{x})$ is the conditional mean of Y given the input values. It is the best mean-squared-error predictor of Y given the information in $\mathbf{X}$.

In simple linear regression, there is only one predictor. We replace the general unknown function $f(X)$ with a parametric straight-line model:

$$f(X) = \beta_0 + \beta_1 X. \tag{5.5}$$

The modeling assumption becomes

$$Y = \beta_0 + \beta_1 X + \epsilon. \tag{5.6}$$

The unknown parameters are:

- β_0, the intercept;
- β_1, the slope.

Once we estimate these parameters from data, our fitted regression function is

$$\widehat{f}(X) = \widehat{\beta}_0 + \widehat{\beta}_1 X. \tag{5.7}$$

In data analysis language, we say that we *regress* Y *onto* X.

5.3 Data Notation

Suppose we observe n paired data points:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \tag{5.8}$$

Here:

- x_i is the observed predictor value for observation i;
- y_i is the observed response value for observation i.

It is often useful to collect the observations in vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \tag{5.9}$$

The fitted response vector is

$$\widehat{\mathbf{y}} = \begin{bmatrix} \widehat{y}_1 \\ \widehat{y}_2 \\ \vdots \\ \widehat{y}_n \end{bmatrix}, \qquad \widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i. \tag{5.10}$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}, \qquad e_i = y_i - \widehat{y}_i. \tag{5.11}$$

Math Foundations Connection. A variable column is a vector in $\mathbb{R}^n$; see the Math Foundations textbook, Chapters 2, 3, 5, and 12. The dot product and Euclidean norm give compact ways to write sums of products and sums of squared deviations. This is why least squares can be written both as a calculus problem and as a geometry problem.

5.4 Predicted Values, Residuals, MSE, and RSS

For any proposed line

$$\widehat{y} = b_0 + b_1 x, \tag{5.12}$$

where b_0 and b_1 are candidate values, the residual for observation i is

$$e_i(b_0, b_1) = y_i - (b_0 + b_1 x_i). \tag{5.13}$$

The empirical mean squared error is

$$\mathrm{MSE}(b_0, b_1) = \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2. \tag{5.14}$$

The residual sum of squares is

$$\mathrm{RSS}(b_0, b_1) = \sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2. \tag{5.15}$$

Because $\mathrm{MSE}(b_0, b_1) = \mathrm{RSS}(b_0, b_1)/n$, minimizing MSE and minimizing RSS give the same fitted line.

The least squares estimators are the values of b_0 and b_1 that minimize the empirical mean squared error:

$$(\widehat{\beta}_0, \widehat{\beta}_1) = \arg\min_{b_0, b_1} \frac{1}{n} \sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2. \tag{5.16}$$

5.5 The Big Three Regression Assumptions

The source lectures present these as three increasingly strong assumptions. We keep that same three-step ladder throughout the regression chapters.

The model

$$Y = \beta_0 + \beta_1 X + \epsilon \tag{5.17}$$

can be paired with three assumption levels. These levels matter because they determine what claims we are allowed to make.

5.5.1 Assumption 1: Mean-zero errors

The first assumption is

$$\mathbb{E}[\epsilon \mid X = x] = 0 \qquad \text{for every } x. \tag{5.18}$$

Then

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x. \tag{5.19}$$

In simple linear regression, this means that the line is the conditional mean of Y given $X = x$. In this setting, the line is a statement about average behavior, not about every individual observation.

5.5.2 Assumption 2: Constant variance and uncorrelated errors

The second assumption is stated after conditioning on the observed predictor values. In other words, we treat the predictor values as fixed and study the remaining randomness in the errors:

$$\mathbb{E}[\epsilon_i \mid X_i = x_i] = 0, \tag{5.20}$$

$$\operatorname{Var}(\epsilon_i \mid X_i = x_i) = \sigma^2 \qquad \text{for all } i, \tag{5.21}$$

$$\operatorname{Cov}(\epsilon_i, \epsilon_j \mid X_1 = x_1, \ldots, X_n = x_n) = 0 \qquad \text{for } i \neq j. \tag{5.22}$$

The second condition is constant error variance, also called homoskedasticity. The third condition says that, after the predictor values are fixed, errors are uncorrelated across observations.

These assumptions support the classical standard error formulas and hypothesis tests that appear later.

5.5.3 Assumption 3: Gaussian errors

The third assumption is the Gaussian version of the same fixed-predictor idea:

$$\boldsymbol{\epsilon} \mid X_1 = x_1, \ldots, X_n = x_n \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \tag{5.23}$$

Equivalently, conditional on the observed predictor values, the errors are independent normal random variables with mean zero and common variance σ^2. This Gaussian error model supports exact finite-sample inference using t and F distributions. This assumption is powerful, but it should not be made casually. Residual diagnostics later in the course help us check whether it is plausible.

Assumption level	What it says	What it supports
Mean-zero errors	$\mathbb{E}[\epsilon \mid X = x] = 0$	Line estimates the conditional mean
Constant variance and uncorrelated errors	Conditional mean zero, constant variance, uncorrelated errors after fixing predictor values	Standard errors and approximate inference
Gaussian errors	Conditional independent normal errors with common variance	Exact classical inference under the model

Table 5.1: The three-step regression assumption ladder for simple linear regression.

5.6 Deriving the Least Squares Estimators

We now derive the formulas for $\widehat{\beta}_0$ and $\widehat{\beta}_1$ by minimizing the empirical MSE. Define the objective function

$$Q(b_0, b_1) = \frac{1}{n}\sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2. \tag{5.24}$$

Let

$$r_i(b_0, b_1) = y_i - (b_0 + b_1 x_i). \tag{5.25}$$

Then

$$Q(b_0, b_1) = \frac{1}{n}\sum_{i=1}^{n} r_i(b_0, b_1)^2. \tag{5.26}$$

5.6.1 Partial Derivative with Respect to b_0

Using the chain rule,

$$\begin{aligned}
\frac{\partial Q}{\partial b_0} &= \frac{1}{n}\sum_{i=1}^{n} 2r_i(b_0, b_1)\frac{\partial r_i}{\partial b_0} && (5.27)\\
&= \frac{1}{n}\sum_{i=1}^{n} 2r_i(b_0, b_1)(-1) && (5.28)\\
&= -\frac{2}{n}\sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]. && (5.29)
\end{aligned}$$

Setting this derivative equal to zero gives

$$\frac{1}{n}\sum_{i=1}^{n} \left[y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i)\right] = 0. \tag{5.30}$$

5.6.2 Partial Derivative with Respect to b_1

Similarly,

$$\begin{aligned}
\frac{\partial Q}{\partial b_1} &= \frac{1}{n}\sum_{i=1}^{n} 2r_i(b_0, b_1)\frac{\partial r_i}{\partial b_1} && (5.31)\\
&= \frac{1}{n}\sum_{i=1}^{n} 2r_i(b_0, b_1)(-x_i) && (5.32)\\
&= -\frac{2}{n}\sum_{i=1}^{n} x_i\left[y_i - (b_0 + b_1 x_i)\right]. && (5.33)
\end{aligned}$$

Setting this derivative equal to zero gives

$$\frac{1}{n}\sum_{i=1}^{n} x_i\left[y_i - (\widehat{\beta}_0 + \widehat{\beta}_1 x_i)\right] = 0. \tag{5.34}$$

5.6.3 The Normal Equations

Equations (5.30) and (5.34) are called the normal equations. They can be written in a compact residual form:

$$\sum_{i=1}^{n} e_i = 0, \tag{5.35}$$

$$\sum_{i=1}^{n} x_i e_i = 0. \tag{5.36}$$

The first equation says that least squares residuals sum to zero. The second says that least squares residuals have zero dot product with the predictor vector.

Using sample averages, define

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \tag{5.37}$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \tag{5.38}$$

$$\overline{x^2} = \frac{1}{n}\sum_{i=1}^{n} x_i^2, \tag{5.39}$$

$$\overline{xy} = \frac{1}{n}\sum_{i=1}^{n} x_i y_i. \tag{5.40}$$

Then the normal equations become

$$\bar{y} - \widehat{\beta}_0 - \widehat{\beta}_1 \bar{x} = 0, \tag{5.41}$$

$$\overline{xy} - \widehat{\beta}_0 \bar{x} - \widehat{\beta}_1 \overline{x^2} = 0. \tag{5.42}$$

5.6.4 Solving for $\widehat{\beta}_1$

From (5.41),

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}. \tag{5.43}$$

Substitute this into (5.42):

$$\overline{xy} - (\bar{y} - \widehat{\beta}_1 \bar{x})\bar{x} - \widehat{\beta}_1 \overline{x^2} = 0 \tag{5.44}$$

$$\overline{xy} - \bar{x}\bar{y} + \widehat{\beta}_1 \bar{x}^2 - \widehat{\beta}_1 \overline{x^2} = 0. \tag{5.45}$$

Rearranging gives

$$\widehat{\beta}_1(\overline{x^2} - \bar{x}^2) = \overline{xy} - \bar{x}\bar{y}. \tag{5.46}$$

Therefore,

$$\boxed{\widehat{\beta}_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2}.} \tag{5.47}$$

Then

$$\boxed{\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}.} \tag{5.48}$$

5.7 Equivalent Centered Form

The slope formula in (5.47) can be written in a more interpretable centered form. First,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}) = \frac{1}{n}\sum_{i=1}^{n}(x_iy_i - x_i\bar{y} - \bar{x}y_i + \bar{x}\bar{y}) \tag{5.49}$$

$$= \overline{xy} - \bar{x}\bar{y} - \bar{x}\bar{y} + \bar{x}\bar{y} \tag{5.50}$$

$$= \overline{xy} - \bar{x}\bar{y}. \tag{5.51}$$

Similarly,

$$\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2 = \overline{x^2} - \bar{x}^2. \tag{5.52}$$

Thus,

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2}. \tag{5.53}$$

This is often the most useful formula for interpreting the slope.

5.8 Slope as Covariance Divided by Variance

The unbiased sample covariance of x and y is

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}), \tag{5.54}$$

while the unbiased sample variance of x is

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})^2. \tag{5.55}$$

Because the same denominator $n-1$ appears in both quantities, it cancels in the ratio. Therefore,

$$\boxed{\widehat{\beta}_1 = \frac{s_{xy}}{s_x^2}.} \tag{5.56}$$

Equivalently, if we use biased denominator-n versions,

$$\widehat{\sigma}_{xy} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y}), \tag{5.57}$$

$$\widehat{\sigma}_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2, \tag{5.58}$$

then

$$\widehat{\beta}_1 = \frac{\widehat{\sigma}_{xy}}{\widehat{\sigma}_x^2}. \tag{5.59}$$

For the slope, the biased-versus-unbiased denominator choice does not change the answer because the denominator cancels.

5.8.1 Interpretation of the Ratio

The numerator measures how x and y move together. The denominator measures how much x itself varies. Thus,

> The least squares slope is the co-movement of x and y, scaled by the variation in x.

If x and y tend to move together, then $s_{xy} > 0$ and the slope is positive. If larger x values tend to pair with smaller y values, then $s_{xy} < 0$ and the slope is negative. If x and y have little linear co-movement, then s_{xy} is near zero and the slope is near zero.

5.9 When the Slope Is Not Identified

The formula

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{5.60}$$

requires

$$\sum_{i=1}^{n}(x_i - \bar{x})^2 > 0. \tag{5.61}$$

This condition means that the observed x values are not all identical.

If every observation has the same predictor value, then there is no information about how Y changes when X changes. The intercept may still describe the mean response at that one X value, but the slope cannot be estimated from the data. This is the simple-regression version of identifiability.

5.10 Why the Critical Point Is a Minimum

The calculus derivation found a critical point. We also need to know that this critical point minimizes the objective. Let

$$Q(b_0, b_1) = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - (b_0 + b_1 x_i)\right]^2. \tag{5.62}$$

This is a sum of squared functions of b_0 and b_1. Therefore, it is a convex quadratic function.

The Hessian matrix is

$$\nabla^2 Q(b_0, b_1) = \frac{2}{n}\begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}. \tag{5.63}$$

This matrix is positive definite when the x_i values are not all identical. Therefore, the critical point is the unique global minimizer.

5.11 Matrix and Geometry View

Math Foundations Connection. The matrix notation below uses the design matrix, transposes, matrix-vector multiplication, dot products, norms, and orthogonality. These supporting concepts are covered in the Math Foundations textbook, Chapters 5 and 7.

Although this chapter derives the estimators using calculus, the same result has a clean linear algebra form. Define the design matrix

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}. \tag{5.64}$$

Then the fitted vector is

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}, \tag{5.65}$$

and the least squares objective can be written as

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \tag{5.66}$$

Using the dot product definition of the Euclidean norm,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{5.67}$$

Taking derivatives in matrix form gives

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{0}. \tag{5.68}$$

Equivalently,

$$\boxed{\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}.} \tag{5.69}$$

These are the normal equations in matrix form.

The vector equation

$$\mathbf{X}^T\mathbf{e} = \mathbf{0} \tag{5.70}$$

says that the residual vector is orthogonal to both columns of $\mathbf{X}$: the ones column and the predictor column. This is the geometric meaning of least squares. The fitted vector is the point in the line-generated model space that is closest to the observed response vector.

Bridge to Later Chapters. In the next regression geometry chapters, we will generalize this same idea: least squares projects $\mathbf{y}$ onto the subspace spanned by the columns of the design matrix.

5.12 Interpreting $\widehat{\beta}_0$ and $\widehat{\beta}_1$

The fitted simple linear regression model is

$$\widehat{\mathbb{E}}[Y \mid X = x] = \widehat{\beta}_0 + \widehat{\beta}_1 x. \tag{5.71}$$

5.12.1 Intercept

The intercept $\widehat{\beta}_0$ estimates the expected value of Y when $X = 0$:

$$\widehat{\beta}_0 \approx \mathbb{E}[Y \mid X = 0]. \tag{5.72}$$

This interpretation is meaningful only when $X = 0$ is within the scope of the data and the model.

For example, if X is years of experience, then $X = 0$ may be meaningful. If X is adult height measured in inches, then $X = 0$ is not meaningful in the observed population. In that case, the intercept is still mathematically necessary for the line, but it should not be given a literal operational interpretation.

5.12.2 Slope

The slope $\widehat{\beta}_1$ estimates the change in the conditional mean of Y for a one-unit increase in X:

$$\widehat{\beta}_1 \approx \mathbb{E}[Y \mid X = x + 1] - \mathbb{E}[Y \mid X = x]. \tag{5.73}$$

When the linear model is appropriate, this change is constant across x.

The units of the slope are

$$\frac{\text{units of } Y}{\text{units of } X}. \tag{5.74}$$

For example, if Y is test score and X is hours studied, then the slope is measured in test-score points per hour studied.

5.13 Association Is Not Causation

The slope is a statement about association unless the study design and assumptions justify a causal interpretation.

Suppose X is the height of mercury in a thermometer and Y is ambient temperature. A fitted regression may show a strong positive slope. But if we artificially raise the mercury level by one unit, we should not expect the ambient temperature to rise. The mercury height measures temperature; it does not cause temperature.

Before interpreting a slope causally, ask:

1. Was X randomly assigned or experimentally controlled?
2. Could another variable cause both X and Y?
3. Is there a plausible mechanism from X to Y?
4. Is the interpretation inside the range of observed X values?
5. Would changing X actually change Y, or only change how we measure Y?

In this course, unless a causal design has been established, interpret regression slopes as associational.

5.14 Variance and Covariance Notation

The notation difference between variance and covariance is worth understanding.

For a single variable X, variance is a second central moment:

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mu_X)^2]. \tag{5.75}$$

It has squared units. If X is measured in meters, then $\mathrm{Var}(X)$ is measured in square meters.

For two variables X and Y, covariance is a cross-moment:

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \tag{5.76}$$

It has product units. If X is measured in meters and Y is measured in seconds, then covariance is measured in meter-seconds.

This is why the sample variance is written with a square,

$$s_x^2, \tag{5.77}$$

while sample covariance is written with two subscripts,

$$s_{xy}. \tag{5.78}$$

The square in s_x^2 signals the squared unit structure of a variance. The subscripts in s_{xy} signal co-movement between two variables.

5.15 Biased and Unbiased Variance Estimators

Two common sample variance estimators are

$$\widehat{\sigma}_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2, \tag{5.79}$$

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2. \tag{5.80}$$

The first uses denominator n and is biased for the population variance in the usual independent-sample setting. The second uses denominator $n-1$ and is unbiased.

The unbiased estimator corrects the expected value, but it can have higher variance. This is a concrete example of the bias-variance tradeoff. Sometimes we intentionally accept bias to reduce variance or improve prediction performance.

Goal	Common choice	Reason
Classical inference	Unbiased estimators such as s_x^2	Supports standard inferential formulas
Maximum likelihood under normality	Denominator-n estimator	Matches the likelihood optimizer
Prediction-focused modeling	Sometimes biased estimators	May reduce MSE
Regularized regression	Intentionally biased estimates	Bias can reduce variance

Table 5.2: Biased estimators are not automatically bad; the choice depends on the objective.

For simple linear regression slope estimation, the denominator choice cancels if covariance and variance are computed consistently:

$$\widehat{\beta}_1 = \frac{s_{xy}}{s_x^2} = \frac{\widehat{\sigma}_{xy}}{\widehat{\sigma}_x^2}. \tag{5.81}$$

5.16 Worked Example: Computing the Least Squares Line by Hand

Consider three observations:

x_i	y_i
1	2
2	3
3	5

First compute the sample means:

$$\bar{x} = \frac{1+2+3}{3} = 2, \qquad \bar{y} = \frac{2+3+5}{3} = \frac{10}{3}. \tag{5.82}$$

Next compute the centered numerator:

$$\begin{aligned}
\sum_{i=1}^{3}(x_i-\bar{x})(y_i-\bar{y}) &= (1-2)\left(2-\frac{10}{3}\right) + (2-2)\left(3-\frac{10}{3}\right) + (3-2)\left(5-\frac{10}{3}\right) && (5.83)\\
&= (-1)\left(-\frac{4}{3}\right) + 0\left(-\frac{1}{3}\right) + (1)\left(\frac{5}{3}\right) && (5.84)\\
&= 3. && (5.85)
\end{aligned}$$

Then compute the centered denominator:

$$\sum_{i=1}^{3}(x_i-\bar{x})^2 = (-1)^2 + 0^2 + 1^2 = 2. \tag{5.86}$$

Thus,

$$\widehat{\beta}_1 = \frac{3}{2} = 1.5. \tag{5.87}$$

The intercept is

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1\bar{x} = \frac{10}{3} - 1.5(2) = \frac{1}{3}. \tag{5.88}$$

The least squares line is

$$\widehat{y} = \frac{1}{3} + 1.5x. \tag{5.89}$$

The fitted values and residuals are:

i	x_i	y_i	$\widehat{y}_i$	$e_i = y_i - \widehat{y}_i$
1	1	2	11/6	1/6
2	2	3	10/3	$-1/3$
3	3	5	29/6	1/6

Check the normal equations:

$$\begin{aligned}
\sum_{i=1}^{3} e_i &= \frac{1}{6} - \frac{1}{3} + \frac{1}{6} = 0, && (5.90)\\
\sum_{i=1}^{3} x_i e_i &= 1\left(\frac{1}{6}\right) + 2\left(-\frac{1}{3}\right) + 3\left(\frac{1}{6}\right) = 0. && (5.91)
\end{aligned}$$

The residuals are orthogonal to both the intercept column and the predictor column.

5.17 A Practical Interpretation Checklist

When interpreting a fitted SLR model,

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x, \tag{5.92}$$

work through the following checklist.

1. **Response:** What is Y, and what are its units?
2. **Predictor:** What is X, and what are its units?
3. **Intercept:** Is $X = 0$ meaningful and inside the data range?
4. **Slope:** What does "one unit of X" mean operationally?
5. **Association:** Does the slope describe association or causation?
6. **Range:** Are predictions being made inside the observed X range?
7. **Residuals:** Are residuals small enough for the model to be useful?
8. **Assumptions:** Which assumption set is needed for the claim being made?

Coding companion reference. Package-specific Python/R commands for fitting simple linear regression and producing quick residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the derivation, interpretation checklist, and common mistakes.

5.18 Common Mistakes

1. **Dropping the parentheses in the residual.** The residual is $y_i - (b_0 + b_1 x_i)$, not $y_i - b_0 + b_1 x_i$.
2. **Treating the intercept as always meaningful.** The intercept is only meaningful when $X = 0$ is meaningful in context.
3. **Calling the slope causal without design support.** Regression slopes are usually associational.
4. **Ignoring units.** The slope has units of response per predictor unit.
5. **Extrapolating outside the data range.** A fitted line may behave badly outside the observed X values.
6. **Confusing residuals with true errors.** Residuals estimate errors, but they are not the unobserved errors themselves.
7. **Forgetting the denominator condition.** If all x_i values are the same, the slope cannot be estimated.

5.19 Chapter Summary

Simple linear regression estimates the conditional mean of a quantitative response using one predictor. The model is

$$Y = \beta_0 + \beta_1 X + \epsilon. \tag{5.93}$$

The least squares estimators minimize the empirical mean squared error, equivalently the residual sum of squares. Calculus gives the normal equations, and solving them yields

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}, \tag{5.94}$$

$$\widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}. \tag{5.95}$$

The slope is the sample covariance of x and y scaled by the sample variance of x. The intercept anchors the line through the sample means. The residuals sum to zero and are orthogonal to the predictor column.

The main interpretive discipline is this: least squares gives the best line under a squared-error criterion, but the usefulness of that line depends on modeling assumptions, residual behavior, variable meaning, and whether the analyst is making an associational or causal claim.

5.20 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 5 use	**Math Foundations connection**	**MF textbook location**
Data columns as vectors	$\mathbb{R}^n$ vector notation, vector entries, and the ones vector $\mathbf{1}$	Ch02 Secs. 2.1, 2.2, 2.4, 2.6
Centered vectors and sums	Vector subtraction, scalar multiplication, and linear combinations	Ch03 Secs. 3.1, 3.2, 3.4, 3.5
Sample means as sums	Dot product examples, including $\mathbf{1}^T\mathbf{a} = \sum_i a_i$ and $(1/n)\mathbf{1}^T\mathbf{a}$ as a sample mean	Ch02 Sec. 2.4; Ch05 Secs. 5.1, 5.3
Least squares objective as a norm	$\mathbf{a}^T\mathbf{a} = \sum_i a_i^2$ and the Euclidean norm $\\|\mathbf{x}\\| = \sqrt{\mathbf{x}^T\mathbf{x}}$	Ch05 Secs. 5.1, 5.3, 5.5; Ch12 Sec. 12.9
Residual orthogonality	Dot products, angles, and the special case $\mathbf{a}^T\mathbf{b} = 0$ meaning orthogonality	Ch02 Sec. 2.6; Ch05 Secs. 5.1, 5.3, 5.8; Ch12 Sec. 12.12
Design matrix notation	Matrix size, transpose notation, and matrix-vector multiplication as row inner products / linear combinations of columns	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3; Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2

5.21 Practice and Workflow

Focused Study Checklist

Use this as a final check after working a simple linear regression problem.

1. Can you write the model $Y = \beta_0 + \beta_1 X + \epsilon$ and state which rung of the Big Three assumptions the task requires?
2. Can you compute $\widehat{\beta}_1$ and $\widehat{\beta}_0$ from the centered-sum formulas?
3. Can you write the fitted line and compute residuals $e_i = y_i - \widehat{y}_i$?
4. Can you interpret the slope with units and avoid over-interpreting the intercept?
5. Can you use $\sum e_i = 0$ and $\sum x_i e_i = 0$ as algebra checks rather than as evidence that the model is correct?
6. Can you separate association from causation in your conclusion?

Chapter 6

Simple Linear Regression: Fit Quality, R^2, and Hypothesis Testing

6.1 The Big Picture

The previous chapter built the simple linear regression model and derived the least squares line

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x. \tag{6.1}$$

The next question is not how to find the line. The next question is:

> Once we have a fitted line, how good is it, and is there statistical evidence that the predictor is useful?

This chapter introduces two related but different ideas.

1. **Fit quality:** How much variation in the response does the fitted line explain? This is measured by R^2.
2. **Statistical evidence:** Is the fitted relationship distinguishable from a zero-slope model after accounting for sample size and noise? This is tested with the F-test or, equivalently in simple linear regression, the slope t-test.

The difference matters. A model can have a high R^2 and still be based on too few observations to trust. A model can have a statistically significant slope but still explain little practical variation. Good analysis reports both fit quality and inferential evidence.

6.2 Review: The Mean-Only Baseline Model

Before using a predictor, we can always estimate a quantitative response Y by its sample mean:

$$\widehat{y}_i = \bar{y}, \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{6.2}$$

This is the *mean-only model*. It ignores all predictor values.

The prediction error for observation i under the mean-only model is

$$y_i - \bar{y}. \tag{6.3}$$

The mean squared error of the mean-only model is

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{6.4}$$

This is the denominator-n version of sample variance. Thus, for one-dimensional data, evaluating the mean-only estimate by MSE is essentially evaluating variance.

Math Foundations Connection. A response column $(y_1, \ldots, y_n)^T$ is a vector in $\mathbb{R}^n$, and the mean-only fitted vector can be written as $\bar{y}\mathbf{1}$. The sums of squared deviations below are squared Euclidean norms; see the Math Foundations textbook, Chapters 2, 3, and 5.

6.3 Adding a Predictor

When we observe paired data

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n), \tag{6.5}$$

we can compare two strategies:

1. predict all responses using $\bar{y}$;
2. predict responses using the fitted regression line

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i. \tag{6.6}$$

If X is useful, then the fitted line should have smaller squared prediction error than the mean-only model. If X is useless, then the fitted line will look like a horizontal line with slope near zero, and it will not improve much over the mean-only model.

This is the core intuition behind R^2 and the F-test:

> Compare the error from the mean-only model to the error from the fitted regression model.

6.4 Three Sums of Squares

Let

$$e_i = y_i - \widehat{y}_i \tag{6.7}$$

be the residual from the fitted regression line.

6.4.1 Total Sum of Squares

The *total sum of squares* measures how much the response varies around its sample mean:

$$\text{TSS} = \sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{6.8}$$

This is the squared error from the mean-only model.

6.4.2 Residual Sum of Squares

The *residual sum of squares* measures how much variation remains after fitting the regression line:

$$\text{RSS} = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 = \sum_{i=1}^{n} e_i^2. \tag{6.9}$$

This is the squared error from the fitted regression model.

6.4.3 Regression Sum of Squares

The *regression sum of squares*, sometimes called explained sum of squares, measures how much variation the model explains relative to the mean-only baseline:

$$\text{RegSS} = \text{TSS} - \text{RSS}\,. \tag{6.10}$$

When the regression model includes an intercept, least squares gives the decomposition

$$\boxed{\text{TSS} = \text{RegSS} + \text{RSS}\,.} \tag{6.11}$$

Math Foundations Connection. The decomposition in (6.11) is a geometry statement. It relies on the residual vector being orthogonal to the fitted-value direction after centering. Dot products, norms, and orthogonality are covered in the Math Foundations textbook, Chapters 2, 3, and 5.

6.5 R^2: Proportion of Variation Explained

The coefficient of determination is

$$\boxed{R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}.} \tag{6.12}$$

The numerator $\text{TSS} - \text{RSS}$ is the reduction in squared error achieved by using the predictor. The denominator TSS is the squared error we had before using the predictor.

Thus,

> R^2 is the proportion of the response variation explained by the regression model.

If $R^2 = 0$, then the fitted model does no better than the mean-only model. If $R^2 = 1$, then the residuals are all zero and the fitted line passes exactly through every observed point.

For simple linear regression with an intercept,

$$0 \leq R^2 \leq 1. \tag{6.13}$$

This interval property depends on comparing an intercept-containing regression model to the mean-only baseline. Some nonstandard models, such as no-intercept regressions, can produce different behavior.

6.6 Interpreting R^2

Suppose two models are used to predict pediatric weight.

- Model A uses wrist circumference and has $R^2 = 0.25$.
- Model B uses height and has $R^2 = 0.50$.

Then Model B explains twice as much of the observed weight variation as Model A, at least in the data used to fit the models.

The phrase *explains variation* should be read carefully. It does not automatically imply causation. It means that the fitted line reduces squared prediction error relative to the mean-only model.

6.7 R^2 and Correlation in Simple Linear Regression

For simple linear regression with an intercept, R^2 is equal to the squared sample correlation between x and y:

$$\boxed{R^2 = r_{xy}^2.} \tag{6.14}$$

To see why, define centered values

$$x_i^c = x_i - \bar{x}, \qquad y_i^c = y_i - \bar{y}. \tag{6.15}$$

Let

$$S_{xx} = \sum_{i=1}^{n} (x_i^c)^2, \qquad S_{yy} = \sum_{i=1}^{n} (y_i^c)^2, \qquad S_{xy} = \sum_{i=1}^{n} x_i^c y_i^c. \tag{6.16}$$

The least squares slope is

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}. \tag{6.17}$$

The centered fitted values satisfy

$$\widehat{y}_i - \bar{y} = \widehat{\beta}_1 (x_i - \bar{x}). \tag{6.18}$$

Therefore, the regression sum of squares is

$$\begin{aligned}
\text{RegSS} &= \sum_{i=1}^{n} (\widehat{y}_i - \bar{y})^2 && (6.19)\\
&= \sum_{i=1}^{n} \widehat{\beta}_1^2 (x_i - \bar{x})^2 && (6.20)\\
&= \widehat{\beta}_1^2 S_{xx} && (6.21)\\
&= \left(\frac{S_{xy}}{S_{xx}}\right)^2 S_{xx} && (6.22)\\
&= \frac{S_{xy}^2}{S_{xx}}. && (6.23)
\end{aligned}$$

Thus,

$$\begin{aligned}
R^2 &= \frac{\text{RegSS}}{\text{TSS}} && (6.24)\\
&= \frac{S_{xy}^2/S_{xx}}{S_{yy}} && (6.25)\\
&= \frac{S_{xy}^2}{S_{xx}S_{yy}} && (6.26)\\
&= \left(\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}\right)^2 && (6.27)\\
&= r_{xy}^2. && (6.28)
\end{aligned}$$

This equality is useful in SLR. It does not directly generalize to multiple linear regression as a single ordinary correlation between one predictor and the response.

6.8 Limitations of R^2

R^2 is useful, but it does not tell us everything.

6.8.1 A Perfect Fit Can Be Misleading

With two data points and a line with two parameters, we can fit the data perfectly as long as the two x values are different. Then

$$\text{RSS} = 0 \qquad \Rightarrow \qquad R^2 = 1. \tag{6.29}$$

This does not mean the model will predict future data well. It means only that a line can pass through two points.

6.8.2 R^2 Does Not Measure Statistical Evidence

R^2 measures in-sample fit. It does not directly answer:

- Is the slope distinguishable from zero?
- How much sampling uncertainty is in $\widehat{\beta}_1$?
- How noisy are the observations?
- How many observations were used?

These questions require hypothesis tests, confidence intervals, and residual diagnostics.

6.8.3 R^2 Does Not Check Model Assumptions

A high R^2 does not prove that the linear model assumptions are valid. Residual plots may still reveal curvature, nonconstant variance, outliers, or dependence among observations.

6.9 From Fit Quality to Hypothesis Testing

Fit quality asks:

> How much did the line reduce squared error compared to the mean-only model?

Hypothesis testing asks:

> Is that reduction large enough, relative to residual noise and sample size, to count as statistical evidence that the slope is not zero?

For SLR, the inferential question is usually

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{6.30}$$

The null hypothesis says that X does not improve the conditional mean of Y through a linear slope. The alternative says that the slope is nonzero.

Math Foundations Connection. Hypotheses are mathematical statements, and hypothesis testing depends on understanding logical alternatives and negation. Basic logical statements, negation, and truth-table reasoning are covered in the Math Foundations textbook, Chapter 1.

6.10 The F-Statistic

The F-statistic compares the mean-only model to the fitted regression model.

For a general nested-model comparison,

$$F = \frac{(\mathrm{SS}_{\text{null}} - \mathrm{SS}_{\text{full}})\,/(p_{\text{full}} - p_{\text{null}})}{\mathrm{SS}_{\text{full}}\,/(n - p_{\text{full}})}, \tag{6.31}$$

where:

- $\mathrm{SS}_{\text{null}}$ is the squared error from the simpler model;

- SS_{full} is the squared error from the larger model;
- p_{null} is the number of parameters in the simpler model;
- p_{full} is the number of parameters in the larger model.

For simple linear regression, the null model is the mean-only model, so

$$p_{\text{null}} = 1, \qquad SS_{\text{null}} = \text{TSS}\,. \tag{6.32}$$

The full model is the fitted line, so

$$p_{\text{full}} = 2, \qquad SS_{\text{full}} = \text{RSS}\,. \tag{6.33}$$

Therefore,

$$F = \frac{(\text{TSS} - \text{RSS})/(2-1)}{\text{RSS}\,/(n-2)} \tag{6.34}$$

$$= \boxed{\frac{\text{TSS} - \text{RSS}}{\text{RSS}\,/(n-2)}}. \tag{6.35}$$

The numerator is the improvement over the mean-only model. The denominator is the residual variation per leftover degree of freedom.

Math Foundations Connection. The phrase *leftover degrees of freedom* is a dimension idea. The residual variation lives in directions left over after fitting the intercept and slope. Linear independence, bases, and the fact that a linearly independent set of n-vectors can have at most n elements are introduced in the Math Foundations textbook, Chapters 2, 5, and 6.

6.11 The F-Statistic in Terms of R^2

Because

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}, \tag{6.36}$$

we can rewrite the SLR F-statistic as

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS}\,/(n-2)} \tag{6.37}$$

$$= \frac{R^2\,\text{TSS}}{(1-R^2)\,\text{TSS}\,/(n-2)} \tag{6.38}$$

$$= \boxed{\frac{R^2}{1-R^2}(n-2).} \tag{6.39}$$

This formula explains why sample size matters. If two simple linear regressions have the same R^2, the one with more observations will have a larger F-statistic. More data can make the same proportion of explained variation more statistically detectable.

6.12 The F-Distribution and the p-Value

Under Assumption 3 from Chapter 5—the Gaussian error model—and the null hypothesis $H_0 : \beta_1 = 0$,

$$F \sim F_{1,n-2}. \tag{6.40}$$

The first degree of freedom is the numerator degree of freedom:

$$p_{\text{full}} - p_{\text{null}} = 2 - 1 = 1. \tag{6.41}$$

The second degree of freedom is the residual degree of freedom:

$$n - p_{\text{full}} = n - 2. \tag{6.42}$$

After calculating the observed statistic F_{obs}, the p-value is

$$p\text{-value} = \Pr\left(F_{1,n-2} \geq F_{\text{obs}} \mid H_0\right). \tag{6.43}$$

A small p-value means that, if the true slope were zero and the model assumptions held, an improvement this large would be unlikely by random sampling variation alone.

6.13 Why the Assumptions Matter

The F-test relies on the same three-step regression assumption ladder introduced in Chapter 5.

1. Assumption 1 (mean-zero errors) supports the interpretation of the line as the conditional mean.
2. Assumption 2 (constant variance and uncorrelated errors) supports the standard error and variance calculations.
3. Assumption 3 (Gaussian errors) supports exact finite-sample t and F distributions.

If these assumptions are badly violated, the p-value may not have the advertised interpretation. This is why residual analysis is not optional.

6.14 The Slope t-Test

Most software output also reports a t-test for the slope. The slope test is

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{6.44}$$

The residual variance estimator is

$$\widehat{\sigma}^2 = \frac{\text{RSS}}{n-2}. \tag{6.45}$$

The standard error of the slope is

$$\text{SE}(\widehat{\beta}_1) = \sqrt{\frac{\widehat{\sigma}^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}} = \sqrt{\frac{\widehat{\sigma}^2}{S_{xx}}}. \tag{6.46}$$

The t-statistic is

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)}. \tag{6.47}$$

Under the null hypothesis and Assumption 3 (Gaussian errors),

$$t_{\text{obs}} \sim t_{n-2}. \tag{6.48}$$

6.15 Why the SLR F-Test and Slope t-Test Agree

In simple linear regression with an intercept,

$$\boxed{F = t^2.} \tag{6.49}$$

To see why, recall that

$$\text{RegSS} = \widehat{\beta}_1^2 S_{xx}. \tag{6.50}$$

Also,

$$\text{SE}(\widehat{\beta}_1)^2 = \frac{\widehat{\sigma}^2}{S_{xx}}, \qquad \widehat{\sigma}^2 = \frac{\text{RSS}}{n-2}. \tag{6.51}$$

Thus,

$$\begin{aligned}
t^2 &= \frac{\widehat{\beta}_1^2}{\text{SE}(\widehat{\beta}_1)^2} && (6.52)\\
&= \frac{\widehat{\beta}_1^2}{\widehat{\sigma}^2/S_{xx}} && (6.53)\\
&= \frac{\widehat{\beta}_1^2 S_{xx}}{\widehat{\sigma}^2} && (6.54)\\
&= \frac{\text{RegSS}}{\text{RSS}/(n-2)} && (6.55)\\
&= F. && (6.56)
\end{aligned}$$

Therefore, in SLR, the overall F-test for the model and the two-sided t-test for the slope produce the same conclusion. In multiple linear regression, the overall F-test and individual coefficient t-tests answer different questions.

6.16 Worked Example

Consider five observations:

x_i	y_i
1	2
2	3
3	5
4	4
5	6

The least squares line is

$$\widehat{y} = 1.3 + 0.9x. \tag{6.57}$$

The fitted values and residuals are:

i	x_i	y_i	$\widehat{y}_i$	e_i
1	1	2	2.2	-0.2
2	2	3	3.1	-0.1
3	3	5	4.0	1.0
4	4	4	4.9	-0.9
5	5	6	5.8	0.2

The sample mean is

$$\bar{y} = 4. \tag{6.58}$$

The total sum of squares is

$$\text{TSS} = (2-4)^2 + (3-4)^2 + (5-4)^2 + (4-4)^2 + (6-4)^2 = 10. \tag{6.59}$$

The residual sum of squares is

$$\text{RSS} = (-0.2)^2 + (-0.1)^2 + (1.0)^2 + (-0.9)^2 + (0.2)^2 \tag{6.60}$$
$$= 1.9. \tag{6.61}$$

Therefore,

$$R^2 = 1 - \frac{1.9}{10} = 0.81. \tag{6.62}$$

The model explains 81% of the observed variation in y.

The F-statistic is

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n-2)} \tag{6.63}$$
$$= \frac{10 - 1.9}{1.9/3} \tag{6.64}$$
$$\approx 12.79. \tag{6.65}$$

Under $H_0 : \beta_1 = 0$, this is compared to an $F_{1,3}$ distribution. Equivalently, the slope t-statistic is

$$t = \sqrt{F} \approx 3.58, \tag{6.66}$$

with the sign determined by the sign of $\widehat{\beta}_1$. Since the slope is positive, $t \approx 3.58$.

6.17 How to Read Software Output

Regression software usually reports both fit-quality and hypothesis-testing information.

6.17.1 Fit Quality Fields

Common fields include:

- **R-squared:** the R^2 value;
- **Residual standard error** or **scale**: an estimate of residual noise;
- **F-statistic:** the model-versus-null comparison;
- **Prob(F-statistic):** the p-value for the F-test.

6.17.2 Coefficient Table Fields

For the predictor row, common fields include:

- **coef:** $\widehat{\beta}_1$;
- **std err:** $\mathrm{SE}(\widehat{\beta}_1)$;
- **t:** the slope t-statistic;
- **P¿—t—:** the two-sided p-value for $H_0 : \beta_1 = 0$;
- **confidence interval:** an interval estimate for the slope.

In SLR, the p-value in the predictor row and the p-value for the model F-statistic correspond to the same slope question.

6.18 Decision Template for SLR Fit and Significance

When reporting a simple linear regression, use the following template.

1. **State the fitted model:**
$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x. \tag{6.67}$$
2. **Interpret the slope with units.**
3. **Report R^2:** say what proportion of response variation is explained by the predictor.
4. **State the hypothesis test:**
$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_A : \beta_1 \neq 0. \tag{6.68}$$
5. **Report the test statistic and p-value.**
6. **Translate the conclusion into context.**
7. **Check residuals before overclaiming.**
8. **Avoid causal language unless the design supports causation.**

Coding companion reference. Package-specific Python/R commands for computing R^2, F-statistics, p-values, and related output have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual guidance for reading software output.

6.19 Common Mistakes

1. **Treating high R^2 as proof of causation.** R^2 is about explained variation, not causal mechanism.
2. **Treating high R^2 as proof of validity.** A model can have high R^2 and still violate assumptions.
3. **Ignoring sample size.** The same R^2 can produce different statistical evidence depending on n.
4. **Confusing residual variance with response variance.** TSS measures response variation before the model; RSS measures variation left after the model.
5. **Forgetting degrees of freedom.** In SLR, residual variance uses denominator $n-2$ because two parameters are estimated.
6. **Thinking $R^2 = 1$ always means a useful model.** Two points can give a perfect line, but not a reliable model.
7. **Reading p-values without context.** A small p-value does not tell you effect size, practical importance, or model adequacy.

6.20 Chapter Summary

The mean-only model predicts every response using $\bar{y}$. Its squared error is the total sum of squares,

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2. \tag{6.69}$$

The fitted regression model predicts each response with

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i. \tag{6.70}$$

Its squared error is the residual sum of squares,

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2. \tag{6.71}$$

The coefficient of determination is

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}. \tag{6.72}$$

It measures the proportion of observed response variation explained by the fitted line.

The F-test compares the fitted line to the mean-only model:

$$F = \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{RSS}/(n-2)}. \tag{6.73}$$

Under $H_0 : \beta_1 = 0$ and Assumption 3 (Gaussian errors),

$$F \sim F_{1,n-2}. \tag{6.74}$$

In simple linear regression with an intercept, the slope t-test and model F-test are equivalent:

$$F = t^2. \tag{6.75}$$

The central distinction is this:

> R^2 measures how much the model fits the observed data; the F-test and t-test measure whether the observed improvement is statistically distinguishable from a zero-slope model under the same regression assumption ladder.

6.21 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 6 use	**Math Foundations connection**	**MF textbook location**
Mean-only fitted vector	$\mathbb{R}^n$ vector notation and the ones vector $\mathbf{1}$	Ch02 Secs. 2.1, 2.4
TSS and RSS as squared lengths	Dot products, $\mathbf{a}^T\mathbf{a} = \sum_i a_i^2$, and Euclidean norm $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T\mathbf{x}}$	Ch05 Secs. 5.1, 5.3, 5.5
Sum-of-squares decomposition	Dot products, norms, and orthogonality of fitted and residual components	Ch02 Sec. 2.6; Ch05 Secs. 5.1, 5.3, 5.5, 5.8; Ch12 Sec. 12.12
Residual degrees of freedom	Linear independence, basis size, and leftover dimension intuition	Ch02 Sec. 2.6; Ch06 Secs. 6.3, 6.4, 6.5
Hypothesis statements	Logic statements, negation, and truth-table reasoning as the foundation for null-versus-alternative statements	Ch01 Sec. 1.1
Software/design-matrix bridge	Matrix notation, transpose notation, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2

6.22 Practice and Workflow

Focused Study Checklist

Use this as a final check after analyzing fit quality and statistical evidence.

1. Can you identify the mean-only baseline and compute TSS?
2. Can you compute residuals, RSS, and $R^2 = 1 - \text{RSS} / \text{TSS}$?
3. Can you explain R^2 as fit quality, not as proof that the model is useful or causal?
4. Can you state the slope test hypotheses $H_0 : \beta_1 = 0$ and $H_A : \beta_1 \neq 0$?
5. Can you connect the model F-test to the slope t-test in SLR through $F = t^2$?
6. Can you conclude in context while separating fit, statistical evidence, and residual diagnostics?

Chapter 7

Linear Regression as Projection: Design Matrix and Hat Matrix

7.1 The Big Picture

The last two chapters developed simple linear regression using calculus, sums of squares, and hypothesis tests. This chapter rebuilds the same least squares estimator using linear algebra. The payoff is substantial:

> Least squares regression is an orthogonal projection of the response vector onto the subspace spanned by the columns of the design matrix.

This viewpoint explains why the normal equations appear, why residuals are orthogonal to fitted values, why the fitted values can be written with the hat matrix, and why the same formula works for multiple linear regression.

The main objects in this chapter are:

$$\mathbf{y} \in \mathbb{R}^n \qquad \text{the response vector,} \tag{7.1}$$

$$\mathbf{X} \in \mathbb{R}^{n\times(p+1)} \qquad \text{the design matrix,} \tag{7.2}$$

$$\widehat{\boldsymbol{\beta}} \in \mathbb{R}^{p+1} \qquad \text{the least squares coefficient vector,} \tag{7.3}$$

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} \qquad \text{the fitted-value vector,} \tag{7.4}$$

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} \qquad \text{the residual vector,} \tag{7.5}$$

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \qquad \text{the hat matrix.} \tag{7.6}$$

The key formula is

$$\boxed{\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}.} \tag{7.7}$$

The matrix $\mathbf{H}$ is called the hat matrix because it puts the hat on $\mathbf{y}$: it turns observed responses into fitted responses.

Notation bridge. From this chapter onward, bold lowercase vectors such as $\mathbf{y}$ denote observed sample vectors from the data in hand, while bold uppercase symbols such as $\mathbf{X}$ denote matrices.

Chapter 10 temporarily writes $\mathbf{Y}$ when the full response vector is being treated as random conditional on a fixed design matrix. The front-matter notation guide gives the book-wide convention.

Math Foundations Connection. This chapter uses vector notation in $\mathbb{R}^n$, the ones vector $\mathbf{1}$, linear combinations, dot products, norms, orthogonality, matrix notation, transpose notation, and matrix-vector multiplication. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, and 7.

7.2 Vectors Behind a Regression Scatterplot

A simple linear regression scatterplot displays points

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \tag{7.8}$$

In matrix language, we separate the observed predictor values and response values into vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \qquad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}. \tag{7.9}$$

The scatterplot is drawn in the usual (x, y)-plane, but the regression calculation takes place in $\mathbb{R}^n$. The response vector $\mathbf{y}$ is a point or vector in n-dimensional observation space. The fitted vector $\widehat{\mathbf{y}}$ is another vector in that same space.

For simple linear regression, every fitted vector has the form

$$\widehat{\mathbf{y}} = \widehat{\beta}_0 \mathbf{1} + \widehat{\beta}_1 \mathbf{x}. \tag{7.10}$$

Thus, the fitted values live in the set of all linear combinations of $\mathbf{1}$ and $\mathbf{x}$. This set is the plane

$$\mathrm{Col}(\mathbf{X}) = \mathrm{span}\{\mathbf{1}, \mathbf{x}\} \tag{7.11}$$

inside $\mathbb{R}^n$, provided $\mathbf{1}$ and $\mathbf{x}$ are linearly independent.

Important interpretation. A regression line in the scatterplot corresponds to a vector $\widehat{\mathbf{y}}$ on a plane in $\mathbb{R}^n$. We are not projecting points in the two-dimensional scatterplot. We are projecting the full response vector $\mathbf{y}$ onto the subspace of fitted-value vectors.

7.3 The Linear Regression Problem as an Overdetermined System

Suppose we have three data points. A perfect simple linear model would satisfy

$$y_1 = \beta_0 + \beta_1 x_1, \tag{7.12}$$
$$y_2 = \beta_0 + \beta_1 x_2, \tag{7.13}$$
$$y_3 = \beta_0 + \beta_1 x_3. \tag{7.14}$$

There are three equations and only two unknowns. With more than two points, the system is usually overdetermined. In vector form, the exact system would be

$$\mathbf{y} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}. \tag{7.15}$$

Usually $\mathbf{y} \notin \operatorname{span}\{\mathbf{1}, \mathbf{x}\}$. Therefore, no exact solution exists.

Least squares solves the nearest possible problem:

> Find the vector in $\operatorname{span}\{\mathbf{1}, \mathbf{x}\}$ that is closest to $\mathbf{y}$ in squared Euclidean distance.

That nearest vector is the orthogonal projection of $\mathbf{y}$ onto the fitted-value plane.

7.4 Projection Onto One Vector

Before projecting onto the two-dimensional regression plane, recall projection onto a single nonzero vector $\mathbf{a}$. The projection of $\mathbf{y}$ onto the line spanned by $\mathbf{a}$ is

$$\operatorname{proj}_{\mathbf{a}}(\mathbf{y}) = \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}. \tag{7.16}$$

The residual from this projection is

$$\mathbf{r} = \mathbf{y} - \operatorname{proj}_{\mathbf{a}}(\mathbf{y}). \tag{7.17}$$

It is orthogonal to $\mathbf{a}$:

$$\mathbf{a}^T \mathbf{r} = \mathbf{a}^T \left(\mathbf{y} - \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a} \right) \tag{7.18}$$

$$= \mathbf{a}^T \mathbf{y} - \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}^T \mathbf{a} \tag{7.19}$$

$$= 0. \tag{7.20}$$

Least squares regression is the same idea, but the target is not a line. The target is the column space of a design matrix.

Math Foundations Connection. The projection formula uses dot products, squared lengths, and orthogonality. Dot products and Euclidean norms are covered in the Math Foundations textbook, Chapters 5 and 12.

7.5 The Design Matrix for Simple Linear Regression

For simple linear regression, define the design matrix

$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x} \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n \times 2}. \tag{7.21}$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \tag{7.22}$$

Then the simple linear regression model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{7.23}$$

The fitted values are

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}. \tag{7.24}$$

The columns of $\mathbf{X}$ are directions in $\mathbb{R}^n$. In SLR, those directions are $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\widehat{\mathbf{y}}$ is a linear combination of those columns.

7.6 The Normal Equations from Orthogonality

The least squares residual vector is

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}. \tag{7.25}$$

The orthogonal projection condition says that the residual vector must be orthogonal to every column of $\mathbf{X}$. For SLR, this means

$$\mathbf{1}^T\mathbf{e} = 0, \tag{7.26}$$

$$\mathbf{x}^T\mathbf{e} = 0. \tag{7.27}$$

Stacking these two equations gives

$$\mathbf{X}^T\mathbf{e} = \mathbf{0}. \tag{7.28}$$

Substitute $\mathbf{e} = \mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}$:

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{0}, \tag{7.29}$$

$$\mathbf{X}^T\mathbf{y} - \mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{0}, \tag{7.30}$$

$$\boxed{\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}.} \tag{7.31}$$

These are the normal equations.

If $\mathbf{X}^T\mathbf{X}$ is invertible, then

$$\boxed{\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.} \tag{7.32}$$

The invertibility condition is equivalent to the columns of $\mathbf{X}$ being linearly independent. For SLR with an intercept, this requires the observed x_i's not all being equal.

Math Foundations Connection. The step from $\mathbf{X}^T\mathbf{e} = \mathbf{0}$ to the normal equations uses matrix transpose notation and matrix-vector multiplication. Matrix-vector multiplication can be read both as row inner products and as column linear combinations; see the Math Foundations textbook, Chapters 3, 5, 7, and 12.

7.7 Recovering the Familiar SLR Formulas

The matrix formula in (7.32) is not a different estimator. It is the same least squares estimator from the calculus derivation.

For SLR,

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} \mathbf{1}^T\mathbf{1} & \mathbf{1}^T\mathbf{x} \\ \mathbf{x}^T\mathbf{1} & \mathbf{x}^T\mathbf{x} \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{bmatrix}, \tag{7.33}$$

and

$$\mathbf{X}^T\mathbf{y} = \begin{bmatrix} \mathbf{1}^T\mathbf{y} \\ \mathbf{x}^T\mathbf{y} \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{bmatrix}. \tag{7.34}$$

Solving the two normal equations gives

$$\widehat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \qquad \widehat{\beta}_0 = \bar{y} - \widehat{\beta}_1 \bar{x}. \tag{7.35}$$

Thus, the matrix formula and the scalar SLR formula are two views of the same least squares solution.

7.8 The Hat Matrix

Start with the fitted values:

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}. \tag{7.36}$$

Now substitute the least squares estimator from (7.32):

$$\widehat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}. \tag{7.37}$$

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.} \tag{7.38}$$

Then

$$\boxed{\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}.} \tag{7.39}$$

The hat matrix is an $n \times n$ matrix. It maps observed response vectors in $\mathbb{R}^n$ to fitted-value vectors in the column space of $\mathbf{X}$:

$$\mathbf{H} : \mathbb{R}^n \to \mathrm{Col}(\mathbf{X}). \tag{7.40}$$

7.9 Properties of the Hat Matrix

Assume $\mathbf{X}$ has full column rank, so $\mathbf{X}^T\mathbf{X}$ is invertible.

7.9.1 Symmetry

The hat matrix is symmetric:

$$\boxed{\mathbf{H}^T = \mathbf{H}.} \tag{7.41}$$

Proof:

$$\mathbf{H}^T = \left[\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\right]^T \tag{7.42}$$
$$= \left(\mathbf{X}^T\right)^T \left[(\mathbf{X}^T\mathbf{X})^{-1}\right]^T \mathbf{X}^T \tag{7.43}$$
$$= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{7.44}$$
$$= \mathbf{H}. \tag{7.45}$$

The third line uses the fact that $\mathbf{X}^T\mathbf{X}$ is symmetric, so its inverse is also symmetric.

7.9.2 Idempotence

The hat matrix is idempotent:

$$\boxed{\mathbf{H}^2 = \mathbf{H}.} \tag{7.46}$$

Proof:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{7.47}$$
$$= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}(\mathbf{X}^T\mathbf{X})(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{7.48}$$
$$= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{7.49}$$
$$= \mathbf{H}. \tag{7.50}$$

Idempotence means that once a vector has already been projected into the fitted-value subspace, projecting it again does not change it:

$$\mathbf{H}\widehat{\mathbf{y}} = \mathbf{H}(\mathbf{H}\mathbf{y}) = \mathbf{H}^2\mathbf{y} = \mathbf{H}\mathbf{y} = \widehat{\mathbf{y}}. \tag{7.51}$$

7.9.3 Orthogonal Projection

A matrix that is both symmetric and idempotent is an orthogonal projection matrix. Therefore, $\mathbf{H}$ is the orthogonal projection onto $\mathrm{Col}(\mathbf{X})$.

This is the geometric core of least squares:

$$\boxed{\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y} \text{ is the closest vector to } \mathbf{y} \text{ in } \mathrm{Col}(\mathbf{X}).} \tag{7.52}$$

7.9.4 Influence

Because

$$\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \tag{7.53}$$

we have componentwise

$$\widehat{y}_i = \sum_{j=1}^{n} H_{ij} y_j. \tag{7.54}$$

Therefore,

$$\boxed{\frac{\partial \widehat{y}_i}{\partial y_j} = H_{ij}.} \tag{7.55}$$

The entry H_{ij} measures how much the observed response y_j influences the fitted response $\widehat{y}_i$. The diagonal entries H_{ii} are called leverages. A large leverage value means observation i's predictor row has unusual position relative to the design matrix.

7.10 The Residual Maker Matrix

The hat matrix makes fitted values. The complementary matrix makes residuals.

Define

$$\mathbf{M} = \mathbf{I} - \mathbf{H}. \tag{7.56}$$

Then

$$\begin{aligned} \mathbf{e} &= \mathbf{y} - \widehat{\mathbf{y}} && (7.57) \\ &= \mathbf{y} - \mathbf{H}\mathbf{y} && (7.58) \\ &= (\mathbf{I} - \mathbf{H})\mathbf{y} && (7.59) \\ &= \mathbf{M}\mathbf{y}. && (7.60) \end{aligned}$$

The matrix $\mathbf{M}$ is also symmetric and idempotent:

$$\mathbf{M}^T = \mathbf{M}, \qquad \mathbf{M}^2 = \mathbf{M}. \tag{7.61}$$

It projects $\mathbf{y}$ onto the residual subspace, which is the orthogonal complement of $\mathrm{Col}(\mathbf{X})$.

Because $\mathbf{X}^T\mathbf{e} = \mathbf{0}$, the residual vector is orthogonal to every column of the design matrix. Since $\widehat{\mathbf{y}} \in \mathrm{Col}(\mathbf{X})$, the residual vector is also orthogonal to the fitted-value vector:

$$\widehat{\mathbf{y}}^T\mathbf{e} = 0. \tag{7.62}$$

This gives the geometric decomposition

$$\mathbf{y} = \widehat{\mathbf{y}} + \mathbf{e}, \qquad \widehat{\mathbf{y}} \perp \mathbf{e}. \tag{7.63}$$

7.11 Dimensions and Degrees of Freedom

If $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$ has full column rank, then

$$\dim \mathrm{Col}(\mathbf{X}) = p + 1. \tag{7.64}$$

The fitted values live in this $(p+1)$-dimensional subspace. The residuals live in the orthogonal complement, whose dimension is

$$n - (p+1) = n - p - 1. \tag{7.65}$$

This is the linear algebra reason that the residual variance estimator in multiple linear regression divides by $n - p - 1$.

For SLR, $p = 1$, so the fitted-value subspace has dimension 2, and the residual subspace has dimension

$$n - 2. \tag{7.66}$$

Math Foundations Connection. The dimension claim depends on linear independence and bases. A set of linearly independent n-vectors can have at most n elements, and a basis provides unique linear-combination coordinates. See the Math Foundations textbook, Chapters 2 and 6.

7.12 Multiple Linear Regression as the Same Projection

The design matrix generalizes immediately from simple linear regression to multiple linear regression. With predictors $\mathbf{x}_1, \ldots, \mathbf{x}_p$, define

$$\mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x}_1 & \cdots & \mathbf{x}_p \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}. \tag{7.67}$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}. \tag{7.68}$$

The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{7.69}$$

The least squares solution is still

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}, \tag{7.70}$$

and the fitted values are still

$$\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T. \tag{7.71}$$

The only change is the subspace. In SLR, $\mathrm{Col}(\mathbf{X})$ is usually a two-dimensional plane in $\mathbb{R}^n$. In MLR, $\mathrm{Col}(\mathbf{X})$ is usually a $(p+1)$-dimensional subspace in $\mathbb{R}^n$.

7.13 Design Matrix as a Modeling Choice

The design matrix tells the model what directions it is allowed to use to approximate $\mathbf{y}$. Choosing a model means choosing columns for $\mathbf{X}$.

Examples:

$$\text{SLR:} \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x} \end{bmatrix}, \tag{7.72}$$

$$\text{Polynomial model:} \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x} & \mathbf{x}^2 & \mathbf{x}^3 \end{bmatrix}, \tag{7.73}$$

$$\text{MLR:} \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x}_1 & \cdots & \mathbf{x}_p \end{bmatrix}, \tag{7.74}$$

$$\text{Interaction model:} \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x}_1 & \mathbf{x}_2 & \mathbf{x}_1 \odot \mathbf{x}_2 \end{bmatrix}. \tag{7.75}$$

Here $\odot$ means elementwise multiplication, so the interaction column has entries $x_{i1}x_{i2}$.

The same least squares machinery works for all of these models as long as the model is linear in the coefficients. This is why regression can fit curved relationships using transformed columns while still being a linear model in parameters.

7.14 Worked Example: Projection in SLR

Consider three observations:

x_i	y_i
-1	1
0	0
1	2

Then

$$\mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & -1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}. \tag{7.76}$$

Compute

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \mathbf{X}^T\mathbf{y} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}. \tag{7.77}$$

Therefore,

$$\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}. \tag{7.78}$$

The fitted line is

$$\widehat{y} = 1 + \frac{1}{2}x. \tag{7.79}$$

The fitted-value vector is

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 1/2 \\ 1 \\ 3/2 \end{bmatrix}. \tag{7.80}$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = \begin{bmatrix} 1/2 \\ -1 \\ 1/2 \end{bmatrix}. \tag{7.81}$$

Check the orthogonality conditions:

$$\mathbf{1}^T\mathbf{e} = \frac{1}{2} - 1 + \frac{1}{2} = 0, \tag{7.82}$$

$$\mathbf{x}^T\mathbf{e} = (-1)\left(\frac{1}{2}\right) + 0(-1) + 1\left(\frac{1}{2}\right) = 0. \tag{7.83}$$

Thus, the residual vector is orthogonal to both columns of $\mathbf{X}$. This confirms that $\widehat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto $\operatorname{span}\{\mathbf{1}, \mathbf{x}\}$.

For this example, the hat matrix is

$$\mathbf{H} = \begin{bmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & 5/6 \end{bmatrix}. \tag{7.84}$$

Then

$$\mathbf{H}\mathbf{y} = \begin{bmatrix} 1/2 \\ 1 \\ 3/2 \end{bmatrix} = \widehat{\mathbf{y}}. \tag{7.85}$$

Coding companion reference. Package-specific Python/R commands for building design matrices, hat matrices, fitted values, residuals, and leverage values have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the projection geometry and matrix interpretation.

7.15 Common Mistakes

1. **Projecting the scatterplot instead of the response vector.** The projection takes place in $\mathbb{R}^n$, not in the two-dimensional scatterplot.
2. **Forgetting the intercept column.** With an intercept, the first column of $\mathbf{X}$ is $\mathbf{1}$. Removing the intercept changes the projection subspace.
3. **Assuming $(\mathbf{X}^T\mathbf{X})^{-1}$ always exists.** It exists only when the columns of $\mathbf{X}$ are linearly independent.
4. **Confusing random errors and residuals.** The model errors $\boldsymbol{\epsilon}$ are unobserved. The residuals $\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}$ are computed after fitting the model.
5. **Thinking the hat matrix estimates coefficients.** The hat matrix maps $\mathbf{y}$ to $\widehat{\mathbf{y}}$. The coefficient estimator is $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$.
6. **Ignoring design-matrix columns.** Transformed predictors, interaction terms, and categorical encodings are all columns of $\mathbf{X}$.
7. **Treating high leverage as a residual problem only.** Leverage comes from $\mathbf{X}$, not from y. It measures unusual predictor geometry.

7.16 Chapter Summary

Simple linear regression can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \mathbf{X} = \begin{bmatrix} \mathbf{1} & \mathbf{x} \end{bmatrix}. \tag{7.86}$$

The fitted values are linear combinations of the columns of $\mathbf{X}$:

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}. \tag{7.87}$$

Least squares chooses the fitted vector in $\mathrm{Col}(\mathbf{X})$ closest to $\mathbf{y}$. The residual vector is orthogonal to every column of $\mathbf{X}$, so

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}) = \mathbf{0}. \tag{7.88}$$

This gives the normal equations

$$\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}. \tag{7.89}$$

If $\mathbf{X}^T\mathbf{X}$ is invertible, then

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}. \tag{7.90}$$

The fitted values can be written as

$$\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T. \tag{7.91}$$

The hat matrix is symmetric and idempotent, so it is an orthogonal projection matrix. The diagonal entries of $\mathbf{H}$ are leverage values, and the full matrix entries describe how observed responses influence fitted responses.

The same projection formula generalizes immediately to multiple linear regression by adding more columns to the design matrix. In MLR, when $\mathbf{X}$ has full column rank, the fitted values are the projection of $\mathbf{y}$ onto the $(p+1)$-dimensional column space of $\mathbf{X}$, and the residuals live in the $(n-p-1)$-dimensional orthogonal complement. If $\operatorname{rank}(\mathbf{X}) = r$, replace $p+1$ by r and $n-p-1$ by $n-r$. Chapter 8 uses these dimensions to explain degrees of freedom and identifiability.

7.17 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 7 use	**Math Foundations connection**	**MF textbook location**
Response vectors and predictor vectors	$\mathbb{R}^n$ vector notation, coordinates, and the ones vector $\mathbf{1}$	Ch02 Secs. 2.1, 2.2, 2.4, 2.6; Ch06 Sec. 6.5
Fitted values as linear combinations	Scalar-vector multiplication and linear combinations of vectors	Ch03 Secs. 3.4, 3.5; Ch07 Sec. 7.6.2
Least squares objective	Euclidean norms, squared lengths, and distances	Ch05 Secs. 5.5, 5.7; Ch12 Sec. 12.9
Residual orthogonality	Dot products and the zero-dot-product condition for orthogonality	Ch02 Sec. 2.6; Ch05 Secs. 5.1, 5.3, 5.8; Ch12 Sec. 12.12
Normal equations	Matrix-vector multiplication as row inner products and column linear combinations	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3, 5.5; Ch07 Secs. 7.1, 7.6.1, 7.6.2; Ch12 Sec. 12.11
Full rank and invertibility intuition	Linear independence, bases, and orthonormal-basis ideas	Ch05 Sec. 5.5; Ch06 Secs. 6.3, 6.5; Ch09 Sec. 9.1; Ch10 Secs. 10.3, 10.5
Computational bridge	Python vector and matrix operations with NumPy arrays	Ch07 Sec. 7.7
Normal-equation bridge	Least squares as minimizing $\|\mathbf{Ax}-\mathbf{b}\|^2$, the normal equations, and gradient-based reasoning	Ch05 Sec. 5.5; Ch12 Secs. 12.4, 12.9, 12.11

7.18 Practice and Workflow

Focused Study Checklist

Use this as a final check after translating a regression problem into projection language.

1. Can you build the design matrix $\mathbf{X}$ with the intercept column and state its dimensions?
2. Can you explain why least squares projects $\mathbf{y}$ onto $\operatorname{Col}(\mathbf{X})$?
3. Can you derive or recognize the normal equations $\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}$?

4. Can you write $\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}$?
5. Can you explain what symmetry and idempotence of $\mathbf{H}$ mean for projection?
6. Can you state the geometric meaning of $\mathbf{e} \perp \mathrm{Col}(\mathbf{X})$?

Chapter 8

Degrees of Freedom, Subspaces, and Identifiability

8.1 The Big Picture

The previous chapter showed that linear regression is an orthogonal projection. The response vector $\mathbf{y} \in \mathbb{R}^n$ is projected onto the column space of the design matrix $\mathbf{X}$. The fitted values live in one subspace, and the residuals live in the orthogonal complement.

This chapter uses that geometry to explain three ideas that appear repeatedly in regression:

1. **Degrees of freedom**: how many independent dimensions remain after we estimate model parameters.
2. **Subspaces**: where fitted values, residuals, coefficient information, and non-identifiable directions live.
3. **Identifiability**: when regression coefficients are uniquely determined by the data, and when many coefficient vectors produce the same fitted values.

The central message is:

> Degrees of freedom are leftover dimensions. Identifiability fails when the design matrix has a nonzero null space.

For a full-rank multiple linear regression model with p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n\times(p+1)}, \qquad \operatorname{rank}(\mathbf{X}) = p+1, \tag{8.1}$$

and the residual degrees of freedom are

$$\boxed{n-p-1.} \tag{8.2}$$

This is why the denominator of the residual variance estimator and the denominator degrees of freedom in the overall regression F-test use $n-p-1$.

Math Foundations Connection. This chapter uses vector notation, the ones vector, dot products, orthogonality, matrix dimensions, matrix transposes, linear independence, bases, matrix-vector multiplication, left-inverse intuition, and null-space/subspace ideas. These are covered in the Math Foundations textbook, Chapters 2, 5, 6, 7, 10, and 11.

8.2 A First Definition: Degrees of Freedom as Independent Movement

In data analysis, degrees of freedom are often introduced as a formula to memorize. For example:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{8.3}$$

The natural question is: why $n-1$?

The geometric answer is that n observations define a vector in $\mathbb{R}^n$. Before estimating anything, the vector $\mathbf{y}$ has n independent coordinates:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n. \tag{8.4}$$

Once we estimate the mean $\bar{y}$, the residual deviations must satisfy one linear constraint:

$$\sum_{i=1}^{n}(y_i - \bar{y}) = 0. \tag{8.5}$$

One independent constraint removes one dimension. Therefore, the residual variation around the sample mean has $n-1$ degrees of freedom.

Reusable intuition. Each independently estimated parameter usually consumes one dimension. The leftover dimensions are available for residual variation.

8.3 Decomposing a Data Vector Around the Sample Mean

The simplest projection problem is the sample mean. Let

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n. \tag{8.6}$$

The sample mean vector is

$$\bar{y}\mathbf{1}. \tag{8.7}$$

The residual vector around the sample mean is

$$\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}. \tag{8.8}$$

Therefore,

$$\boxed{\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{y} - \bar{y}\mathbf{1})\,.} \tag{8.9}$$

This is a decomposition of the data vector into two orthogonal pieces:

1. $\bar{y}\mathbf{1}$, the projection onto the one-dimensional subspace span$\{\mathbf{1}\}$, and
2. $\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}$, the residual vector in the orthogonal complement.

To verify orthogonality, compute

$$\begin{aligned} \mathbf{1}^T\mathbf{r} &= \mathbf{1}^T(\mathbf{y} - \bar{y}\mathbf{1}) && (8.10)\\ &= \sum_{i=1}^{n} y_i - \bar{y}\sum_{i=1}^{n} 1 && (8.11)\\ &= n\bar{y} - n\bar{y} && (8.12)\\ &= 0. && (8.13) \end{aligned}$$

Thus,

$$\mathbf{r} \in \text{span}\{\mathbf{1}\}^{\perp}. \tag{8.14}$$

Since span$\{\mathbf{1}\}$ has dimension 1, its orthogonal complement in $\mathbb{R}^n$ has dimension

$$n - 1. \tag{8.15}$$

This is the geometric reason for the $n-1$ denominator in sample variance.

8.4 Worked Example: Mean Decomposition with Three Observations

Suppose

$$\mathbf{y} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}. \tag{8.16}$$

The sample mean is

$$\bar{y} = \frac{2+5+8}{3} = 5. \tag{8.17}$$

Then

$$\bar{y}\mathbf{1} = 5\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \tag{8.18}$$

and

$$\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}. \tag{8.19}$$

Check the residual constraint:

$$\mathbf{1}^T\mathbf{r} = (-3) + 0 + 3 = 0. \tag{8.20}$$

The residual vector lives on the plane

$$\left\{\mathbf{v} \in \mathbb{R}^3 : \mathbf{1}^T\mathbf{v} = 0\right\}. \tag{8.21}$$

That plane has dimension 2, which equals $n - 1 = 3 - 1$.

8.5 The Mean Projection Matrix

The sample mean is itself a projection. Define

$$\mathbf{H}_0 = \frac{1}{n}\mathbf{1}\mathbf{1}^T. \tag{8.22}$$

Then

$$\mathbf{H}_0\mathbf{y} = \frac{1}{n}\mathbf{1}\mathbf{1}^T\mathbf{y} = \frac{1}{n}\mathbf{1}\sum_{i=1}^{n} y_i = \bar{y}\mathbf{1}. \tag{8.23}$$

The residual-maker matrix for the mean-only model is

$$\mathbf{M}_0 = \mathbf{I} - \mathbf{H}_0. \tag{8.24}$$

Then

$$\mathbf{M}_0\mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}. \tag{8.25}$$

The matrix $\mathbf{H}_0$ is symmetric and idempotent, so it is an orthogonal projection matrix. Its rank and trace are both 1:

$$\operatorname{rank}(\mathbf{H}_0) = \operatorname{tr}(\mathbf{H}_0) = 1. \tag{8.26}$$

Likewise,

$$\operatorname{rank}(\mathbf{M}_0) = \operatorname{tr}(\mathbf{M}_0) = n - 1. \tag{8.27}$$

This trace perspective becomes useful in regression because the hat matrix for linear regression is also a symmetric idempotent projection matrix.

8.6 Regression Decomposes $\mathbb{R}^n$ Into Two Orthogonal Pieces

For multiple linear regression with an intercept and p predictors, write

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \mathbf{X} \in \mathbb{R}^{n\times(p+1)}. \tag{8.28}$$

The fitted values are

$$\widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T, \tag{8.29}$$

when $\mathbf{X}$ has full column rank.

The residuals are

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{M}\mathbf{y}. \tag{8.30}$$

Regression decomposes the observation space as

$$\boxed{\mathbb{R}^n = \mathrm{Col}(\mathbf{X}) \oplus \mathrm{Null}(\mathbf{X}^T).} \tag{8.31}$$

Here $\oplus$ means a direct sum: every vector in $\mathbb{R}^n$ can be written uniquely as a piece in $\mathrm{Col}(\mathbf{X})$ plus a piece in $\mathrm{Null}(\mathbf{X}^T)$. These two pieces are orthogonal.

In regression terms:

$$\widehat{\mathbf{y}} \in \mathrm{Col}(\mathbf{X}), \tag{8.32}$$

$$\mathbf{e} \in \mathrm{Null}(\mathbf{X}^T), \tag{8.33}$$

$$\widehat{\mathbf{y}}^T\mathbf{e} = 0. \tag{8.34}$$

The fitted values live in the column space. The residuals live in the left null space.

8.7 Residual Degrees of Freedom

Assume $\mathbf{X}$ has full column rank. Then

$$\mathrm{rank}(\mathbf{X}) = p + 1. \tag{8.35}$$

Therefore,

$$\dim \mathrm{Col}(\mathbf{X}) = p + 1. \tag{8.36}$$

The column space consumes $p + 1$ dimensions of the original n-dimensional observation space. The leftover dimensions are the residual degrees of freedom:

$$\boxed{\mathrm{df}_{\mathrm{resid}} = n - (p + 1) = n - p - 1.} \tag{8.37}$$

This explains the residual variance estimator:

$$\widehat{\sigma}^2 = \frac{\mathrm{RSS}}{n - p - 1} = \frac{\mathbf{e}^T\mathbf{e}}{n - p - 1}. \tag{8.38}$$

It also explains the denominator degrees of freedom in the overall F-test for MLR.

Special cases.

Model	**Fitted subspace dimension**	**Residual df**
Mean-only model	1	$n - 1$
SLR with intercept	2	$n - 2$
MLR with intercept and p predictors	$p + 1$	$n - p - 1$
Full rank model with design rank r	r	$n - r$

8.8 A Subtle Point: Model Degrees of Freedom Versus Fitted-Subspace Dimension

There are two closely related uses of degrees of freedom in regression:

1. **Fitted-subspace dimension**: with an intercept and p predictors, $\mathrm{Col}(\mathbf{X})$ has dimension $p+1$.
2. **Model improvement degrees of freedom**: when comparing the full model to the intercept-only model, the predictors add p dimensions beyond the mean direction.

That is why an ANOVA-style regression decomposition often uses

$$\text{Total df} = n-1, \qquad \text{Regression df} = p, \qquad \text{Residual df} = n-p-1. \tag{8.39}$$

The $n-1$ total degrees of freedom already subtract the mean direction. The regression degrees of freedom p count the additional fitted directions beyond the intercept-only model.

8.9 Trace of the Hat Matrix

For a full-rank linear regression model, the hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T. \tag{8.40}$$

It is symmetric and idempotent. For symmetric idempotent matrices, the trace equals the rank. Therefore,

$$\boxed{\mathrm{tr}(\mathbf{H}) = \mathrm{rank}(\mathbf{H}) = \mathrm{rank}(\mathbf{X}).} \tag{8.41}$$

If $\mathbf{X}$ has full column rank, this becomes

$$\mathrm{tr}(\mathbf{H}) = p+1. \tag{8.42}$$

The residual-maker matrix is $\mathbf{M} = \mathbf{I} - \mathbf{H}$, so

$$\begin{aligned} \mathrm{tr}(\mathbf{M}) &= \mathrm{tr}(\mathbf{I}) - \mathrm{tr}(\mathbf{H}) && (8.43)\\ &= n-(p+1) && (8.44)\\ &= n-p-1. && (8.45) \end{aligned}$$

This is another way to see residual degrees of freedom.

Connection to leverage. The diagonal entries of $\mathbf{H}$, denoted h_{ii}, are leverages. Since the trace is the sum of diagonal entries,

$$\sum_{i=1}^{n} h_{ii} = p+1. \tag{8.46}$$

The average leverage is therefore

$$\bar{h} = \frac{p+1}{n}. \tag{8.47}$$

Large leverage means an observation is using more than its average share of the fitted-subspace degrees of freedom.

8.10 The Big Four Subspaces of the Design Matrix

Let

$$\mathbf{X} \in \mathbb{R}^{n\times(p+1)} \tag{8.48}$$

with rank r. Linear algebra associates four fundamental subspaces with $\mathbf{X}$. In regression, these subspaces have direct meanings.

Subspace	**Where it lives**	**Regression meaning**	**Dimension**
$\mathrm{Col}(\mathbf{X})$	$\mathbb{R}^n$	Fitted values $\widehat{\mathbf{y}}$	r
$\mathrm{Null}(\mathbf{X}^T)$	$\mathbb{R}^n$	Residual vectors $\mathbf{e}$	$n-r$
$\mathrm{Row}(\mathbf{X}) = \mathrm{Col}(\mathbf{X}^T)$	$\mathbb{R}^{p+1}$	Identifiable coefficient directions	r
$\mathrm{Null}(\mathbf{X})$	$\mathbb{R}^{p+1}$	Non-identifiable coefficient directions	$(p+1)-r$

The first two subspaces live in observation space $\mathbb{R}^n$. The last two live in coefficient space $\mathbb{R}^{p+1}$. This distinction prevents a common mistake: residuals and fitted values are vectors with n entries, while coefficient directions have $p+1$ entries.

8.11 Column Space and Left Null Space

The fitted values satisfy

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}}. \tag{8.49}$$

Therefore, $\widehat{\mathbf{y}}$ is a linear combination of the columns of $\mathbf{X}$:

$$\widehat{\mathbf{y}} \in \mathrm{Col}(\mathbf{X}). \tag{8.50}$$

The residual vector satisfies the normal equations:

$$\mathbf{X}^T\mathbf{e} = \mathbf{0}. \tag{8.51}$$

Therefore,

$$\mathbf{e} \in \mathrm{Null}(\mathbf{X}^T). \tag{8.52}$$

Since $\mathrm{Null}(\mathbf{X}^T)$ is the orthogonal complement of $\mathrm{Col}(\mathbf{X})$, every residual vector is orthogonal to every fitted-value direction.

This is the geometric core of regression diagnostics:

$$\mathbf{y} = \widehat{\mathbf{y}} + \mathbf{e}, \qquad \widehat{\mathbf{y}} \perp \mathbf{e}. \tag{8.53}$$

8.12 Row Space and Null Space

The row space and null space of $\mathbf{X}$ live in coefficient space. They answer a different question:

> Which coefficient directions change fitted values, and which coefficient directions are invisible to the data?

A vector $\mathbf{v} \in \mathbb{R}^{p+1}$ is in the null space of $\mathbf{X}$ if

$$\mathbf{X}\mathbf{v} = \mathbf{0}. \tag{8.54}$$

If $\mathbf{v} \neq \mathbf{0}$, then changing the coefficient vector in the direction $\mathbf{v}$ does not change fitted values:

$$\mathbf{X}(\boldsymbol{\beta} + c\mathbf{v}) = \mathbf{X}\boldsymbol{\beta} + c\mathbf{X}\mathbf{v} = \mathbf{X}\boldsymbol{\beta} \tag{8.55}$$

for any scalar c. Therefore, $\mathbf{v}$ is a non-identifiable direction.

The row space contains coefficient directions that the data can see. The null space contains coefficient directions that the data cannot see.

8.13 Identifiability

A parameter is identifiable when different parameter values imply different observable behavior. In linear regression, the coefficient vector $\boldsymbol{\beta}$ is identifiable when the mapping

$$\boldsymbol{\beta} \mapsto \mathbf{X}\boldsymbol{\beta} \tag{8.56}$$

is one-to-one.

This happens exactly when

$$\mathrm{Null}(\mathbf{X}) = \{\mathbf{0}\}. \tag{8.57}$$

Equivalently, the columns of $\mathbf{X}$ must be linearly independent.

For $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$, the following statements are equivalent:

1. $\boldsymbol{\beta}$ is identifiable.
2. $\mathrm{Null}(\mathbf{X}) = \{\mathbf{0}\}$.
3. The columns of $\mathbf{X}$ are linearly independent.
4. $\mathrm{rank}(\mathbf{X}) = p + 1$.
5. $\mathbf{X}^T\mathbf{X}$ is invertible.
6. The least squares coefficient vector

   $$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} \tag{8.58}$$

   is unique.

Why $\mathbf{X}^T\mathbf{X}$ matters. For any $\mathbf{v}$,

$$\mathbf{v}^T\mathbf{X}^T\mathbf{X}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2. \tag{8.59}$$

If $\mathbf{X}\mathbf{v} = \mathbf{0}$ for some nonzero $\mathbf{v}$, then $\mathbf{X}^T\mathbf{X}$ cannot be invertible. Conversely, if $\mathbf{X}^T\mathbf{X}$ is singular, then there is a nonzero $\mathbf{v}$ such that $\mathbf{X}\mathbf{v} = \mathbf{0}$. Thus, $\mathbf{X}^T\mathbf{X}$ is invertible exactly when $\mathbf{X}$ has no nonzero null directions.

8.14 What Fails When the Model Is Not Identified?

When $\mathrm{Null}(\mathbf{X}) \neq \{\mathbf{0}\}$, there are infinitely many coefficient vectors that produce the same fitted values. If $\mathbf{v} \in \mathrm{Null}(\mathbf{X})$, then

$$\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}(\widehat{\boldsymbol{\beta}} + c\mathbf{v}) \tag{8.60}$$

for every scalar c. The data cannot distinguish $\widehat{\boldsymbol{\beta}}$ from $\widehat{\boldsymbol{\beta}} + c\mathbf{v}$.

Two important consequences follow:

1. The coefficient vector is not unique.
2. The fitted values may still be unique, because the projection of $\mathbf{y}$ onto $\mathrm{Col}(\mathbf{X})$ is unique.

Thus, non-identifiability is primarily a coefficient problem. The model may still make fitted predictions, but the individual coefficients cannot be interpreted uniquely.

8.15 Non-Identifiability Example: Celsius and Fahrenheit

Suppose a model includes an intercept, temperature measured in Celsius, and temperature measured in Fahrenheit:

$$y_i = \beta_0 + \beta_C C_i + \beta_F F_i + \epsilon_i. \tag{8.61}$$

The two temperature predictors are exactly related:

$$F_i = \frac{9}{5}C_i + 32. \tag{8.62}$$

The design matrix is

$$\mathbf{X} = \begin{bmatrix} 1 & C_1 & F_1 \\ 1 & C_2 & F_2 \\ \vdots & \vdots & \vdots \\ 1 & C_n & F_n \end{bmatrix}. \tag{8.63}$$

The columns are linearly dependent because

$$\mathbf{x}_F - \frac{9}{5}\mathbf{x}_C - 32\mathbf{1} = \mathbf{0}. \tag{8.64}$$

Equivalently,

$$\mathbf{X}\begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} = \mathbf{0}. \tag{8.65}$$

Therefore,

$$\begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} \in \text{Null}(\mathbf{X}). \tag{8.66}$$

This means that coefficient vectors can move in this direction without changing predictions.

For example, the coefficient vector

$$\boldsymbol{\beta} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} \tag{8.67}$$

gives

$$\widehat{y} = 10 + 2C. \tag{8.68}$$

But adding the null-space vector gives

$$\boldsymbol{\beta}^{\star} = \begin{bmatrix} 10 \\ 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -32 \\ -9/5 \\ 1 \end{bmatrix} = \begin{bmatrix} -22 \\ 1/5 \\ 1 \end{bmatrix}. \tag{8.69}$$

This gives

$$\begin{aligned} \widehat{y}^{\star} &= -22 + \frac{1}{5}C + F & (8.70) \\ &= -22 + \frac{1}{5}C + \left(\frac{9}{5}C + 32\right) & (8.71) \\ &= 10 + 2C. & (8.72) \end{aligned}$$

The predictions are identical, but the coefficient interpretations are different. Therefore, the individual Celsius and Fahrenheit coefficients are not identified.

8.16 Common Sources of Non-Identifiability

Exact non-identifiability usually comes from exact linear dependence among columns of the design matrix. Common examples include:

1. **Duplicate variables**: including the same predictor twice.
2. **Unit conversions**: including meters and feet, or Celsius and Fahrenheit, in a model with an intercept.
3. **The dummy variable trap**: including an intercept and all category indicator columns.
4. **Derived totals**: including all component variables and their exact sum.
5. **Too many predictors**: if $p + 1 > n$, then $\mathbf{X}$ cannot have full column rank.

Dummy variable trap. Suppose a categorical variable has three levels: Army, Navy, and Marine Corps. Let $\mathbf{d}_A, \mathbf{d}_N, \mathbf{d}_M$ be the three indicator columns. For every row,

$$d_{iA} + d_{iN} + d_{iM} = 1. \tag{8.73}$$

Thus,

$$\mathbf{d}_A + \mathbf{d}_N + \mathbf{d}_M = \mathbf{1}. \tag{8.74}$$

If the design matrix includes $\mathbf{1}$, $\mathbf{d}_A$, $\mathbf{d}_N$, and $\mathbf{d}_M$, then its columns are linearly dependent. The usual repair is to include the intercept and drop one category indicator as the reference group.

8.17 Near Non-Identifiability: Multicollinearity

Exact linear dependence causes exact non-identifiability. Near linear dependence causes instability. This is multicollinearity.

If one column of $\mathbf{X}$ is almost a linear combination of other columns, then $\mathbf{X}^T\mathbf{X}$ may still be invertible, but it is close to singular. The least squares coefficients exist, but small changes in the data can produce large changes in the coefficient estimates.

This matters because:

1. coefficient standard errors become large,
2. signs and magnitudes can be unstable,
3. individual coefficient interpretations become fragile, and
4. predictions may remain reasonable even while coefficients are hard to interpret.

Operational distinction.

Condition	**Design-matrix geometry**	**Consequence**
Exact dependence	Nonzero $\mathrm{Null}(\mathbf{X})$	Coefficients not unique
Near dependence	Columns nearly dependent	Coefficients unstable
Full rank and well-conditioned	Columns independent and not nearly redundant	Coefficients more stable

8.18 Worked Example: Counting Regression Degrees of Freedom

Suppose we have $n = 40$ observations and fit an MLR model with an intercept and $p = 5$ predictors. If the design matrix has full column rank, then

$$\mathbf{X} \in \mathbb{R}^{40\times 6}, \qquad \mathrm{rank}(\mathbf{X}) = 6. \tag{8.75}$$

The fitted values live in a 6-dimensional subspace:

$$\dim \mathrm{Col}(\mathbf{X}) = 6. \tag{8.76}$$

The residuals live in the orthogonal complement:

$$\dim \mathrm{Null}(\mathbf{X}^T) = 40 - 6 = 34. \tag{8.77}$$

Therefore,

$$\boxed{\mathrm{df}_{\mathrm{resid}} = 34.} \tag{8.78}$$

For the overall F-test comparing the full model to the intercept-only model,

$$\text{numerator df} = 5, \qquad \text{denominator df} = 34. \tag{8.79}$$

The numerator has 5 degrees of freedom because the full model adds five predictor directions beyond the mean-only model.

Coding companion reference. Package-specific Python/R commands for checking rank, residual degrees of freedom, and non-identifiability have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the degrees-of-freedom and subspace interpretation.

8.19 Common Mistakes

1. **Forgetting the intercept.** With an intercept and p predictors, the design matrix has $p+1$ columns, not p.
2. **Mixing up fitted-space dimension and regression F-test numerator df.** The fitted subspace has dimension $p+1$, but the overall F-test numerator df is p when comparing to the intercept-only model.
3. **Thinking residuals live in predictor space.** Residuals are n-vectors. They live in $\mathbb{R}^n$, not in $\mathbb{R}^p$.
4. **Thinking coefficient null-space directions are residual directions.** $\text{Null}(\mathbf{X})$ lives in coefficient space, while $\text{Null}(\mathbf{X}^T)$ lives in observation space.
5. **Using $(\mathbf{X}^T\mathbf{X})^{-1}$ when $\mathbf{X}$ is rank deficient.** If columns are dependent, the inverse does not exist.
6. **Assuming fitted values are not unique when coefficients are not unique.** Coefficients may be non-unique while the projection $\widehat{\mathbf{y}}$ is still unique.
7. **Checking only pairwise correlations for identifiability.** A column can be a combination of several others even if no single pair looks perfectly correlated.
8. **Including every dummy variable plus an intercept.** This creates exact linear dependence.
9. **Treating multicollinearity as a prediction failure only.** Multicollinearity often harms coefficient interpretation before it visibly harms prediction.

8.20 Chapter Summary

Degrees of freedom are best understood as dimensions. An n-vector of observations begins with n dimensions. Estimating the sample mean projects $\mathbf{y}$ onto $\text{span}\{\mathbf{1}\}$, leaving residual variation in an $(n-1)$-dimensional orthogonal complement.

Regression generalizes the same idea. The design matrix $\mathbf{X}$ defines the fitted-value subspace. If $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$ has full column rank, then fitted values live in a $(p+1)$-dimensional column space, and residuals live in an $(n-p-1)$-dimensional orthogonal complement. Thus,

$$\text{df}_{\text{resid}} = n - p - 1. \tag{8.80}$$

The four fundamental subspaces of $\mathbf{X}$ organize the regression geometry:

$$\mathrm{Col}(\mathbf{X}) \quad \text{contains fitted values,} \tag{8.81}$$
$$\mathrm{Null}(\mathbf{X}^T) \quad \text{contains residuals,} \tag{8.82}$$
$$\mathrm{Row}(\mathbf{X}) \quad \text{contains identifiable coefficient directions,} \tag{8.83}$$
$$\mathrm{Null}(\mathbf{X}) \quad \text{contains non-identifiable coefficient directions.} \tag{8.84}$$

Identifiability requires $\mathrm{Null}(\mathbf{X}) = \{\mathbf{0}\}$. If the design matrix has linearly dependent columns, then infinitely many coefficient vectors can produce the same fitted values. The Celsius/Fahrenheit example shows this directly: including an intercept, Celsius, and Fahrenheit creates exact linear dependence because Fahrenheit is an affine transformation of Celsius.

8.21 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 8 use	**Math Foundations connection**	**MF textbook location**
Data vectors and mean vectors	Vector notation in $\mathbb{R}^n$, coordinates, and the ones vector $\mathbf{1}$	Ch02 Secs. 2.1, 2.2, 2.4, 2.6; Ch06 Sec. 6.5
Mean and residual decomposition	Dot products, $\mathbf{1}^T\mathbf{a} = \sum a_i$, and sample-mean notation	Ch02 Secs. 2.4, 2.6; Ch05 Secs. 5.1, 5.3
Orthogonal complements	Euclidean norms, distances, angles, and the condition $\mathbf{a}^T\mathbf{b} = 0$ for orthogonality	Ch04 Secs. 4.1, 4.2, 4.4; Ch05 Secs. 5.5, 5.7, 5.8
Column space and fitted values	Matrix-vector multiplication as row inner products and as a linear combination of columns	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3; Ch07 Secs. 7.1, 7.6.1, 7.6.2; Ch11 Sec. 11.2
Rank, basis, and full column rank intuition	Linear independence, bases, and unique linear-combination representations	Ch06 Secs. 6.3, 6.5; Ch10 Sec. 10.5
Matrix dimensions and transpose notation	Matrix sizes, tall/wide matrices, identity matrices, transposes, and symmetry	Ch07 Secs. 7.1, 7.3, 7.4
Left-inverse and identifiability bridge	Left inverse, linearly independent columns, the trivial-solution criterion, and the equivalence between full column rank and invertibility of $\mathbf{A}^T\mathbf{A}$	Ch06 Sec. 6.3; Ch10 Secs. 10.1, 10.3, 10.5, 10.6; Ch11 Sec. 11.4
Computational check	Python/NumPy matrix operations, matrix shapes, and transposes	Ch07 Secs. 7.1, 7.4, 7.7

8.22 Practice and Workflow

Focused Study Checklist

Use this as a final check after a degrees-of-freedom or identifiability problem.

1. Can you identify n, p, and the dimensions of $\mathbf{X}$?
2. Can you determine $r = \text{rank}(\mathbf{X})$?
3. Can you state fitted-space degrees of freedom r and residual degrees of freedom $n - r$?
4. In the full-rank intercept model, can you simplify residual degrees of freedom to $n - p - 1$?
5. Can you explain why identifiability requires $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$?
6. Can you recognize exact dependence, such as redundant columns or the dummy-variable trap?
7. Can you distinguish exact non-identifiability from near dependence or multicollinearity?

Chapter 9

Multiple Linear Regression in Practice: Predictors, Hyperplanes, and the Advertising Example

9.1 The Big Picture

The previous chapters built the geometry of least squares. We now use that geometry in a practical multiple linear regression setting. The goal is to move from the algebra

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{9.1}$$

to the workflow an analyst actually uses when fitting, interpreting, and defending a regression model.

The central message of this chapter is:

> Multiple linear regression is the same projection problem as simple linear regression, but the fitted values now live in a higher-dimensional subspace determined by multiple predictors.

The practical challenge is that the model is still mathematically simple, but it becomes harder to visualize and harder to interpret casually. With one predictor, the least squares fit is a line. With two predictors, the fit is a plane. With three or more predictors, the fit is a hyperplane. The word changes, but the algebra stays the same.

The main questions in this chapter are:

1. Is at least one predictor useful in predicting the response?
2. Do all predictors help, or is only a subset useful?
3. How well does the model fit the data?
4. Given new predictor values, what response should we predict, and how accurate is that prediction?

We will use the `Advertising.csv` data as the running example. The response is `Sales`, and the predictors are advertising budgets for `TV`, `Radio`, and `Newspaper` across 200 markets.

We keep the Chapter 7 sample-level notation: $\mathbf{y}$ is the observed response vector for the current data set, $\mathbf{X}$ is the design matrix, and $\widehat{\mathbf{y}}$ is the fitted vector.

Math Foundations Connection. This chapter uses vector notation, matrix dimensions, matrix-vector multiplication, column-space ideas, linear combinations, dot products, transposes, orthogonality, and rank/full-column-rank intuition. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, 7, 10, 11, and 12.

9.2 Simple Linear Regression and Multiple Linear Regression Use the Same Model Form

In simple linear regression, there is one predictor. For observation i, the model is

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \tag{9.2}$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{9.3}$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \tag{9.4}$$

In multiple linear regression with p predictors, observation i has predictor values

$$x_{i1}, x_{i2}, \ldots, x_{ip}. \tag{9.5}$$

The model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i. \tag{9.6}$$

In vector form, it is still

$$\boxed{\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.} \tag{9.7}$$

The only change is the number of columns in $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n\times(p+1)}. \tag{9.8}$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \in \mathbb{R}^{p+1}. \tag{9.9}$$

Key point. The notation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ is not unique to multiple regression. It already describes simple linear regression. Multiple regression simply adds more predictor columns to the design matrix.

9.3 The Design Matrix View

The design matrix stores the predictors in a way that lets one matrix-vector multiplication generate all fitted values at once. For a coefficient vector $\boldsymbol{\beta}$,

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{1p} \\ \beta_0 + \beta_1 x_{21} + \cdots + \beta_p x_{2p} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \cdots + \beta_p x_{np} \end{bmatrix}. \tag{9.10}$$

Thus, the ith fitted value is the inner product of row i of $\mathbf{X}$ with $\boldsymbol{\beta}$:

$$\widehat{y}_i = \mathbf{x}_{i,:}^T \widehat{\boldsymbol{\beta}}. \tag{9.11}$$

There is an equally important column-space interpretation:

$$\widehat{\mathbf{y}} = \widehat{\beta}_0 \mathbf{1} + \widehat{\beta}_1 \mathbf{x}_1 + \widehat{\beta}_2 \mathbf{x}_2 + \cdots + \widehat{\beta}_p \mathbf{x}_p. \tag{9.12}$$

Here $\mathbf{x}_j$ is the jth predictor column. Therefore,

$$\widehat{\mathbf{y}} \in \mathrm{Col}(\mathbf{X}). \tag{9.13}$$

Regression chooses the linear combination of design-matrix columns that gets closest to $\mathbf{y}$ in squared-error distance.

Looking ahead. The columns of a design matrix do not have to be only raw numeric predictors. Later chapters use the same matrix machinery after adding encoded categorical predictors, transformed predictors, and interaction columns. The least-squares problem stays the same; what changes is the analyst's responsibility to choose columns that are interpretable, identifiable, and defensible.

9.4 What Changes Geometrically? Lines, Planes, and Hyperplanes

A simple linear regression fit looks like a line in a two-dimensional scatterplot. With two predictors, the fitted relationship is a plane:

$$\widehat{Y} = \widehat{\beta}_0 + \widehat{\beta}_1 X_1 + \widehat{\beta}_2 X_2. \tag{9.14}$$

This plane lives in the ordinary three-dimensional plotting space with axes X_1, X_2, and Y.

With three predictors, the fitted relationship is

$$\widehat{Y} = \widehat{\beta}_0 + \widehat{\beta}_1 X_1 + \widehat{\beta}_2 X_2 + \widehat{\beta}_3 X_3. \tag{9.15}$$

This is a hyperplane. It is not easy to draw because the graph would need one response axis plus three predictor axes.

With p predictors, the fitted relationship is

$$\boxed{\widehat{Y} = \widehat{\beta}_0 + \sum_{j=1}^{p} \widehat{\beta}_j X_j.} \tag{9.16}$$

This is a p-dimensional hyperplane in $(p+1)$-dimensional predictor-response plotting space.

Do not confuse two spaces. There are two geometric pictures in play:

1. The **predictor-response plot**: a line, plane, or hyperplane in the space with predictor axes and a response axis.
2. The **least-squares projection**: an orthogonal projection of $\mathbf{y} \in \mathbb{R}^n$ onto $\mathrm{Col}(\mathbf{X}) \subseteq \mathbb{R}^n$.

Both are useful. The first helps us visualize the model. The second explains the algebra.

9.5 Least Squares Estimation in Multiple Regression

The fitted coefficient vector minimizes the residual sum of squares:

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\mathbf{b} \in \mathbb{R}^{p+1}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2. \tag{9.17}$$

The normal equations are

$$\mathbf{X}^T \left(\mathbf{y} - \mathbf{X}\widehat{\boldsymbol{\beta}}\right) = \mathbf{0}. \tag{9.18}$$

Equivalently,

$$\mathbf{X}^T\mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{y}. \tag{9.19}$$

If $\mathbf{X}$ has full column rank, then $\mathbf{X}^T\mathbf{X}$ is invertible, and

$$\boxed{\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.} \tag{9.20}$$

The fitted values are

$$\widehat{\mathbf{y}} = \mathbf{X}\widehat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} = \mathbf{H}\mathbf{y}, \tag{9.21}$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{9.22}$$

is the hat matrix.

The residuals are

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \tag{9.23}$$

The residuals are orthogonal to every column of $\mathbf{X}$:

$$\boxed{\mathbf{X}^T\mathbf{e} = \mathbf{0}.} \tag{9.24}$$

This remains true whether there is one predictor or many.

9.6 The Advertising Data

The `Advertising.csv` data set contains observations from 200 markets. The selected response variable is

$$\texttt{Sales}, \tag{9.25}$$

measured in thousands of units sold. The predictors are advertising budgets, measured in thousands of dollars:

$$\texttt{TV}, \qquad \texttt{Radio}, \qquad \texttt{Newspaper}. \tag{9.26}$$

A natural multiple linear regression model is

$$\texttt{Sales}_i = \beta_0 + \beta_1 \texttt{TV}_i + \beta_2 \texttt{Radio}_i + \beta_3 \texttt{Newspaper}_i + \epsilon_i. \tag{9.27}$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{9.28}$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \texttt{TV}_1 & \texttt{Radio}_1 & \texttt{Newspaper}_1 \\ 1 & \texttt{TV}_2 & \texttt{Radio}_2 & \texttt{Newspaper}_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \texttt{TV}_{200} & \texttt{Radio}_{200} & \texttt{Newspaper}_{200} \end{bmatrix}. \tag{9.29}$$

Here

$$n = 200, \qquad p = 3, \qquad \mathbf{X} \in \mathbb{R}^{200\times 4}. \tag{9.30}$$

If the columns are linearly independent, then the residual degrees of freedom are

$$n - p - 1 = 200 - 3 - 1 = 196. \tag{9.31}$$

9.7 A Scatterplot Matrix Is the First Reconnaissance Tool

With one predictor, a scatterplot can show the relationship between the predictor and response. With three predictors, we need to inspect several pairwise relationships. A scatterplot matrix displays the pairwise relationships among

$$\texttt{Sales}, \quad \texttt{TV}, \quad \texttt{Radio}, \quad \texttt{Newspaper}. \tag{9.32}$$

A scatterplot matrix helps answer two first-pass questions:

1. Which predictors appear most strongly associated with `Sales`?
2. Which predictors appear most strongly associated with each other?

The first question is about predictive signal. The second question is about collinearity. A predictor can look useful by itself but become less useful once another correlated predictor is included in the model.

Important caution. Pairwise plots are not the final analysis. They are reconnaissance. Multiple regression coefficients are interpreted after holding the other included predictors fixed. Pairwise plots do not automatically show that adjusted relationship.

9.8 Question 1: Is At Least One Predictor Useful?

The first model-level question is whether at least one predictor is useful in predicting the response. For the Advertising model,

$$\texttt{Sales}_i = \beta_0 + \beta_1 \texttt{TV}_i + \beta_2 \texttt{Radio}_i + \beta_3 \texttt{Newspaper}_i + \epsilon_i, \tag{9.33}$$

the null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0. \tag{9.34}$$

The alternative is

$$H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \text{ is nonzero.} \tag{9.35}$$

The test statistic is the overall regression F-statistic:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}\,/(n-p-1)}. \tag{9.36}$$

Here

$$\text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2, \tag{9.37}$$

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2. \tag{9.38}$$

For the Advertising full model, the numerator degrees of freedom are $p = 3$, and the denominator degrees of freedom are $n - p - 1 = 196$. A large F-statistic provides evidence that at least one advertising budget is associated with `Sales`.

9.9 Question 2: Which Predictors Help?

Once the global F-test suggests that at least one predictor is useful, the next question is which predictors matter after adjusting for the others.

For each coefficient β_j, the usual individual test is

$$H_0 : \beta_j = 0 \qquad \text{versus} \qquad H_A : \beta_j \neq 0. \tag{9.39}$$

The corresponding t-statistic has the form

$$t_j = \frac{\widehat{\beta}_j}{\text{SE}(\widehat{\beta}_j)}. \tag{9.40}$$

In the full Advertising model, the fitted coefficient estimates are commonly reported approximately as

Term	**Estimate**	**Std. error**	**t-statistic**	**p-value**
Intercept	2.939	0.312	9.42	very small
TV	0.046	0.0014	32.81	very small
Radio	0.189	0.0086	21.89	very small
Newspaper	-0.001	0.0059	-0.18	0.86

These results suggest that `TV` and `Radio` are useful after adjusting for the other included predictors. `Newspaper` does not appear useful after adjusting for `TV` and `Radio`.

Interpretation of the coefficient for `TV`. Holding `Radio` and `Newspaper` fixed, an additional 1 thousand dollars in `TV` advertising is associated with about 0.046 thousand additional units sold. That is about 46 additional units sold.

Interpretation of the coefficient for `Radio`. Holding `TV` and `Newspaper` fixed, an additional 1 thousand dollars in `Radio` advertising is associated with about 0.189 thousand additional units sold. That is about 189 additional units sold.

Interpretation of the coefficient for `Newspaper`. Holding `TV` and `Radio` fixed, an additional 1 thousand dollars in `Newspaper` advertising is associated with an estimated change near zero. In this model, `Newspaper` does not add meaningful predictive information after accounting for the other two media.

9.10 Why a Predictor Can Look Useful Alone but Not in the Full Model

A simple linear regression of `Sales` on `Newspaper` can show a positive association. But the full multiple regression model can assign `Newspaper` a coefficient near zero.

This is not a contradiction. It is a lesson about adjustment.

A simple regression asks:

> As `Newspaper` changes, how does `Sales` tend to change, ignoring other predictors?

A multiple regression coefficient asks:

> As `Newspaper` changes, how does `Sales` tend to change while `TV` and `Radio` are held fixed?

If `Newspaper` budgets are correlated with `Radio` budgets, then `Newspaper` may appear useful by itself because it is partly acting as a surrogate for `Radio`. Once `Radio` is included directly, `Newspaper` may no longer have its own contribution.

Reusable diagnostic phrase. Simple regression measures marginal association. Multiple regression measures adjusted association.

9.11 Question 3: How Well Does the Model Fit?

Two common fit summaries are the residual standard error and R^2.

The residual standard error is

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n - p - 1}}. \tag{9.41}$$

It measures a typical residual size in the response units. For the Advertising full model, RSE is about 1.69. Since `Sales` is measured in thousands of units, this corresponds to a typical residual size of about 1,690 units.

The coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}. \tag{9.42}$$

It measures the proportion of sample variation in $\mathbf{y}$ explained by the fitted model. For the Advertising full model, R^2 is about 0.897, meaning the model explains about 89.7 percent of the sample variation in `Sales`.

Important limitation. Training-set R^2 does not prove that the model will predict well on new data. It also does not prove that the model assumptions are correct. It is a fit summary, not a full validation report.

9.12 Adjusted R^2 as a Quick Complexity Penalty

Ordinary R^2 cannot decrease when a predictor is added to the model, because adding a column to $\mathbf{X}$ can only reduce or maintain the residual sum of squares. This can make ordinary R^2 too generous when comparing models with different numbers of predictors.

Adjusted R^2 introduces a degrees-of-freedom penalty:

$$R^2_{\text{adj}} = 1 - \frac{\text{RSS}\,/(n - p - 1)}{\text{TSS}\,/(n - 1)}. \tag{9.43}$$

Unlike ordinary R^2, adjusted R^2 can decrease when a predictor is added. It asks whether the reduction in RSS was large enough to justify the extra parameter.

Practical use. Adjusted R^2 is a quick model-comparison tool, but it should not replace out-of-sample validation, residual diagnostics, or domain reasoning.

9.13 Question 4: What Should We Predict and How Accurate Is the Prediction?

For new predictor values

$$\mathbf{x}_0 = \begin{bmatrix} 1 & x_{01} & x_{02} & \cdots & x_{0p} \end{bmatrix}^T, \tag{9.44}$$

the point prediction is

$$\boxed{\widehat{y}_0 = \mathbf{x}_0^T \widehat{\boldsymbol{\beta}}.} \tag{9.45}$$

For the Advertising model, suppose a market has

$$\texttt{TV} = 100, \qquad \texttt{Radio} = 20, \qquad \texttt{Newspaper} = 30, \tag{9.46}$$

where each value is in thousands of dollars. Using the approximate full-model estimates,

$$\widehat{\texttt{Sales}} \approx 2.939 + 0.046(100) + 0.189(20) - 0.001(30) \tag{9.47}$$
$$= 2.939 + 4.600 + 3.780 - 0.030 \tag{9.48}$$
$$= 11.289. \tag{9.49}$$

The predicted sales are about 11.289 thousand units, or 11,289 units.

There are two common uncertainty intervals:

1. A **confidence interval for the mean response**: uncertainty about the average sales among many markets with these same predictor values.
2. A **prediction interval for a new response**: uncertainty about the sales in one individual new market.

The prediction interval is wider because it includes both uncertainty in the estimated mean and the irreducible noise for a single future market.

9.14 Worked Example: Interpreting a Full Advertising Model

Assume the fitted model is approximately

$$\widehat{\texttt{Sales}} = 2.939 + 0.046\texttt{TV} + 0.189\texttt{Radio} - 0.001\texttt{Newspaper}. \tag{9.50}$$

All variables are in their original Advertising units: `Sales` in thousands of units and budgets in thousands of dollars.

Example 1: Increase TV while holding other media fixed

Suppose `TV` increases by 10, meaning 10 thousand dollars, while `Radio` and `Newspaper` are held fixed. Then

$$\Delta\widehat{\texttt{Sales}} = 0.046(10) = 0.460. \tag{9.51}$$

The predicted increase is about 0.460 thousand units, or 460 units.

Example 2: Increase Radio while holding other media fixed

Suppose `Radio` increases by 10, meaning 10 thousand dollars, while `TV` and `Newspaper` are held fixed. Then

$$\Delta\widehat{\texttt{Sales}} = 0.189(10) = 1.890. \tag{9.52}$$

The predicted increase is about 1.890 thousand units, or 1,890 units.

Example 3: Increase all three budgets

Suppose the budgets change by

$$\Delta\texttt{TV} = 10, \qquad \Delta\texttt{Radio} = 10, \qquad \Delta\texttt{Newspaper} = 10. \tag{9.53}$$

Then

$$\begin{aligned}\Delta\widehat{\texttt{Sales}} &= 0.046(10) + 0.189(10) - 0.001(10) &(9.54)\\ &= 0.460 + 1.890 - 0.010 &(9.55)\\ &= 2.340. &(9.56)\end{aligned}$$

The predicted increase is about 2.340 thousand units, or 2,340 units.

Operational interpretation. The fitted model suggests that, after adjusting for the other media, `Radio` has the largest estimated marginal effect per thousand dollars among the three predictors in this model. That does not automatically mean all future spending should go to `Radio`; the model is local to the observed data, linear in the included predictors, and does not yet account for nonlinear saturation, interactions, campaign constraints, or causal design.

9.15 Assumptions Behind the Practical Regression Workflow

For estimating a useful least squares line or hyperplane, the weakest core condition is often written as

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}. \tag{9.57}$$

This says that the included predictors capture the conditional mean structure, leaving errors with mean zero after conditioning on the design matrix.

For the usual standard errors, t-tests, and F-tests, a common working assumption is

$$\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2\mathbf{I}. \tag{9.58}$$

This says the errors have constant variance and are uncorrelated across observations.

For exact finite-sample normal-theory inference, we often further assume

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I}). \tag{9.59}$$

This assumption is stronger than what is needed for least squares estimation itself, but it supports exact t- and F-distribution calculations in the classical model.

9.16 Why Graphics Become Difficult in Higher Dimensions

In simple linear regression, one scatterplot can show the predictor, response, fitted line, and residual pattern. In multiple regression, one plot cannot show everything.

With two predictors, a 3D plot can show a regression plane, but even then it can be hard to see residual patterns, leverage, and interactions. With three or more predictors, no ordinary static plot can show the full fitted hyperplane.

Therefore, the practical workflow shifts toward:

1. scatterplot matrices for pairwise reconnaissance,
2. correlation matrices for predictor relationships,
3. coefficient tables for adjusted effects,
4. residual plots for assumption checks,
5. partial residual or component-plus-residual plots for predictor-specific patterns,
6. test-set or cross-validation performance for generalization.

9.17 A Practical Modeling Workflow for the Advertising Example

A defensible multiple linear regression workflow for the Advertising data could look like this:

1. State the response and predictor units.
2. Create scatterplots of `Sales` against each predictor.
3. Create a scatterplot matrix for `Sales`, `TV`, `Radio`, and `Newspaper`.
4. Check predictor-predictor relationships, especially possible collinearity.
5. Fit the full model with all three predictors.
6. Use the overall F-test to ask whether at least one predictor is useful.
7. Use coefficient tests and domain reasoning to ask which predictors contribute after adjustment.
8. Compare simpler models such as `Sales` on `TV` and `Radio` only.
9. Report RSE and R^2, but do not stop there.
10. Check residual plots for nonlinearity, heteroskedasticity, and unusual observations.
11. Use prediction or confidence intervals depending on whether the question is about one future market or the mean response at a predictor setting.
12. Document all modeling decisions so the analysis is reproducible and defensible.

Coding companion reference. Package-specific Python/R commands for scatterplot matrices, full and reduced multiple-regression models, prediction intervals, and residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the applied modeling workflow and interpretation guidance.

9.18 Common Mistakes

1. **Forgetting that coefficients are adjusted effects.** In MLR, $\widehat{\beta}_j$ is interpreted while holding the other included predictors fixed.
2. **Treating pairwise plots as final proof.** Scatterplot matrices are useful reconnaissance, but they do not replace adjusted model interpretation.

3. **Confusing units.** In the Advertising data, `Sales` is in thousands of units and budgets are in thousands of dollars.
4. **Saying a nonsignificant coefficient proves no relationship.** It means the fitted model did not provide strong evidence for an adjusted linear effect under the model assumptions.
5. **Ignoring correlation among predictors.** Predictor-predictor relationships can change coefficient interpretation and standard errors.
6. **Using ordinary R^2 alone to choose variables.** Ordinary R^2 increases or stays the same when predictors are added.
7. **Mixing confidence intervals and prediction intervals.** Confidence intervals are for mean response; prediction intervals are for individual future responses.
8. **Drawing the hyperplane but skipping residual checks.** A fitted plane can still be the wrong shape if the true relationship is nonlinear or contains interactions.
9. **Assuming predictive association is causal.** Regression on observational data can estimate associations, but causal claims require design, assumptions, or additional evidence.

9.19 Chapter Summary

Multiple linear regression extends simple linear regression by adding predictor columns to the design matrix. The model remains

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{9.60}$$

With p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n\times(p+1)}. \tag{9.61}$$

Least squares chooses the coefficient vector that minimizes squared residual length:

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\mathbf{b}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2. \tag{9.62}$$

If $\mathbf{X}$ has full column rank,

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}. \tag{9.63}$$

The fitted values live in $\mathrm{Col}(\mathbf{X})$, and residuals are orthogonal to every column of $\mathbf{X}$. With one predictor, the fitted relationship is a line. With two predictors, it is a plane. With three or more predictors, it is a hyperplane.

The Advertising example illustrates the main practical questions in MLR. The overall F-test asks whether at least one predictor is useful. Individual coefficient tests ask whether a predictor contributes after adjusting for the others. RSE and R^2 summarize model fit. Prediction and confidence intervals quantify uncertainty for new predictor settings.

Most importantly, MLR coefficients are adjusted effects. A predictor can look useful in a simple regression but lose usefulness in a multiple regression after another correlated predictor is included.

9.20 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 9 use	Math Foundations connection	MF textbook location
Response and predictor vectors	Vector notation in $\mathbb{R}^n$, coordinates, and vector entries	Ch02 Secs. 2.1, 2.2, 2.6; Ch06 Sec. 6.5
Column-space interpretation	Linear combinations of vectors, span/column-space ideas, and standard-basis intuition	Ch03 Sec. 3.5; Ch06 Secs. 6.1, 6.5; Ch11 Sec. 11.2; Ch12 Sec. 12.12
Normal equations and orthogonality	Dot products, sums through $\mathbf{1}^T\mathbf{a}$, Euclidean norm, orthogonality, and the least-squares normal-equation bridge	Ch02 Sec. 2.4; Ch05 Secs. 5.1, 5.3, 5.5, 5.8; Ch12 Secs. 12.9, 12.11, 12.12
Design matrix dimensions	Matrix notation, tall matrices, transposes, identity matrices, and symmetry	Ch07 Secs. 7.1, 7.3, 7.4
Fitted values as $\mathbf{X}\widehat{\boldsymbol{\beta}}$	Matrix-vector multiplication as row inner products and as a linear combination of columns	Ch03 Sec. 3.5; Ch05 Secs. 5.1, 5.3; Ch07 Secs. 7.1, 7.6.1, 7.6.2
Full column rank requirement	Linear independence, bases, unique linear-combination intuition, and full-rank conditions	Ch06 Secs. 6.3, 6.5; Ch10 Sec. 10.5
Software implementation	Python/NumPy arrays, shapes, transposes, and matrix multiplication with `@`	Ch07 Secs. 7.1, 7.4, 7.7

9.21 Practice and Workflow

Focused Study Checklist

Use this as a final check after fitting and interpreting a multiple linear regression model.

1. Can you identify the response, predictors, and their units?
2. Can you write both the scalar model and the design-matrix form?
3. Can you interpret each coefficient as an adjusted effect with other predictors held fixed?
4. Can you use the global F-test for the question "is at least one predictor useful?"
5. Can you use coefficient tests for the question "which predictors contribute after adjustment?"
6. Can you report RSE, R^2, and adjusted R^2 without relying on any one of them alone?
7. Can you use residual plots and EDA to check whether the fitted model is defensible?
8. Can you distinguish a confidence interval for a mean response from a prediction interval for a single future response?

Chapter 10

Truth + Error, Estimator Behavior, and Residual Geometry

10.1 The Big Picture

The previous chapter treated multiple linear regression as a practical modeling tool. We used the design matrix, the coefficient vector, and the least squares fit to answer applied questions about predictors, model fit, and prediction. This chapter steps back into theory for one reason:

> We want to understand what the least squares estimates and residuals are doing, so that later diagnostics and hypothesis tests are more than button-clicking.

The model is still

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{10.1}$$

The difference is that we now treat this equation as a *truth plus error* representation. The term $\mathbf{X}\boldsymbol{\beta}$ is the systematic part of the response, and $\boldsymbol{\epsilon}$ is the random part that the model does not explain.

Notation bridge. Chapters 7, 9, 11, and 12 use $\mathbf{y}$ for the observed response vector from a realized data set. This chapter temporarily switches to $\mathbf{Y}$ because the full response vector is being treated as random once the design matrix $\mathbf{X}$ is fixed.

The central questions are:

1. If $\boldsymbol{\beta}$ is the true coefficient vector, how does $\widehat{\boldsymbol{\beta}}$ behave?
2. Is $\widehat{\boldsymbol{\beta}}$ centered on the true value $\boldsymbol{\beta}$?
3. How variable is $\widehat{\boldsymbol{\beta}}$?
4. How do the residuals $\mathbf{e} = \mathbf{Y} - \widehat{\mathbf{Y}}$ relate to the true errors $\boldsymbol{\epsilon}$?
5. Why do residual plots tell us whether our modeling assumptions are plausible?

This chapter is the bridge from regression geometry to regression diagnostics. It explains why residuals are useful, but also why they are not the same thing as the true errors.

Math Foundations Connection. This chapter uses vector notation, transposes, matrix-vector multiplication, identity matrices, orthogonality, rank, column-space/residual-space geometry, least-squares normal equations, and the $\mathbf{A}^T\mathbf{A}$ Hessian. These connections are covered in the Math Foundations textbook, Chapters 2, 5, 7, 10, 11, 12, and 13.

10.2 The Regression Model with Fixed Design Matrix

We start with the multiple linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{10.2}$$

where

$$\mathbf{Y} \in \mathbb{R}^n, \qquad \mathbf{X} \in \mathbb{R}^{n\times(p+1)}, \qquad \boldsymbol{\beta} \in \mathbb{R}^{p+1}, \qquad \boldsymbol{\epsilon} \in \mathbb{R}^n. \tag{10.3}$$

The design matrix $\mathbf{X}$ includes an intercept column of ones and p predictor columns.

In this chapter, we use a fixed-design view:

> Treat $\mathbf{X}$ as fixed, and let $\mathbf{Y}$ vary randomly because $\boldsymbol{\epsilon}$ varies randomly.

That is the answer to the question, "Where does the randomness come from?" It comes from the noise vector $\boldsymbol{\epsilon}$. Once $\mathbf{X}$ is fixed, the only random object on the right side of (10.2) is $\boldsymbol{\epsilon}$, so the only random object on the left side is $\mathbf{Y}$.

Conditional view. If the predictors are themselves random in the data-generating process, the results in this chapter are usually read conditionally on the observed $\mathbf{X}$. In words: after we observe the predictor values, we analyze the randomness in the response around the regression surface.

10.3 The Big Three Regression Assumptions

Chapter 5 introduced a three-step regression assumption ladder. We now restate that same ladder in matrix form. The stronger assumptions support stronger inferential claims.

Assumption 1: Mean-zero errors

The first assumption is

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}. \tag{10.4}$$

Equivalently, for each observation,

$$\mathbb{E}[\epsilon_i \mid \mathbf{X}] = 0. \tag{10.5}$$

This says the model is centered correctly. After accounting for the predictors in $\mathbf{X}$, the remaining error should not systematically lean positive or negative.

Under this assumption,

$$\mathbb{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}. \tag{10.6}$$

Thus $\mathbf{X}\boldsymbol{\beta}$ is the conditional mean of the response.

Assumption 2: Constant variance and uncorrelated errors

The second assumption is

$$\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n. \tag{10.7}$$

This compact matrix statement contains two ideas:

1. Each error has the same variance: $\mathrm{Var}(\epsilon_i \mid \mathbf{X}) = \sigma^2$.
2. Different errors are uncorrelated: $\mathrm{Cov}(\epsilon_i, \epsilon_j \mid \mathbf{X}) = 0$ for $i \neq j$.

The first part is called homoskedasticity. The second part rules out error correlation. Violations of this assumption are why residual plots, time-order plots, and variance checks matter.

Assumption 3: Gaussian errors

The third assumption is

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n). \tag{10.8}$$

This says the full error vector is multivariate normal with mean zero and covariance $\sigma^2 \mathbf{I}_n$. This is stronger than the first two assumptions. It supports exact finite-sample t- and F-based inference for the usual linear model quantities.

Practical interpretation. The first assumption gives correct centering. The second assumption gives the usual variance formulas. The third assumption gives exact distributional statements. In practice, diagnostics help us decide how believable these assumptions are.

10.4 Recall: Least Squares Estimator

This section restates the Chapter 7 least-squares formula in the fixed-design notation of this chapter. The formula is not new; the new question is how it behaves when $\mathbf{Y}$ is random.

The least squares estimator minimizes the squared residual length:

$$\widehat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b} \in \mathbb{R}^{p+1}} \left\| \mathbf{Y} - \mathbf{X}\mathbf{b} \right\|^2 . \tag{10.9}$$

Assuming $\mathbf{X}$ has full column rank, the normal equations give

$$\mathbf{X}^T \mathbf{X} \widehat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{Y}, \tag{10.10}$$

and therefore

$$\boxed{\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}.} \tag{10.11}$$

The fitted response vector is

$$\widehat{\mathbf{Y}} = \mathbf{X} \widehat{\boldsymbol{\beta}}. \tag{10.12}$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \widehat{\mathbf{Y}}. \tag{10.13}$$

10.5 Dimension Check

The formulas in this chapter are much easier to trust if the dimensions work. Let $q = p + 1$, the number of columns in $\mathbf{X}$. Then:

Object	Meaning	Dimension
$\mathbf{Y}$	response vector	$n \times 1$
$\mathbf{X}$	design matrix	$n \times q$
$\boldsymbol{\beta}$	true coefficient vector	$q \times 1$
$\widehat{\boldsymbol{\beta}}$	estimated coefficient vector	$q \times 1$
$\boldsymbol{\epsilon}$	true error vector	$n \times 1$
$\widehat{\mathbf{Y}}$	fitted response vector	$n \times 1$
$\mathbf{e}$	residual vector	$n \times 1$
$\mathbf{H}$	hat matrix	$n \times n$
$\mathbf{I}_n - \mathbf{H}$	residual-maker matrix	$n \times n$

The matrix

$$\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \tag{10.14}$$

has dimension $q \times n$. It maps the response vector $\mathbf{Y} \in \mathbb{R}^n$ into the coefficient estimate $\widehat{\boldsymbol{\beta}} \in \mathbb{R}^q$:

$$\widehat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}. \tag{10.15}$$

10.6 Truth + Error Representation for $\widehat{\boldsymbol{\beta}}$

Start with the least squares estimator:

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}. \tag{10.16}$$

Substitute the regression model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\begin{aligned}
\widehat{\boldsymbol{\beta}} &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) && (10.17)\\
&= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon} && (10.18)\\
&= \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}. && (10.19)
\end{aligned}$$

Therefore,

$$\boxed{\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}.} \tag{10.20}$$

This is the truth plus error representation for the coefficient estimator. It says:

> The estimated coefficient vector equals the true coefficient vector plus a transformed version of the random noise vector.

Each coefficient estimate has its own associated error term. That error is not a new source of randomness; it is produced by the same $\boldsymbol{\epsilon}$ that created the random response values.

10.7 Unbiasedness of $\widehat{\boldsymbol{\beta}}$

Using (10.20), take conditional expectations given $\mathbf{X}$:

$$\mathbb{E}[\widehat{\boldsymbol{\beta}} \mid \mathbf{X}] = \mathbb{E}\left[\boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon} \mid \mathbf{X}\right] \tag{10.21}$$

$$= \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}]. \tag{10.22}$$

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\widehat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}.} \tag{10.23}$$

So $\widehat{\boldsymbol{\beta}}$ is unbiased under the mean-zero error assumption.

What unbiased means here. Unbiased does not mean that $\widehat{\boldsymbol{\beta}}$ equals $\boldsymbol{\beta}$ in a particular dataset. It means that if we repeatedly observed new response vectors from the same design matrix and same true model, the average of the coefficient estimates would center on the true coefficient vector.

10.8 Variance of $\widehat{\boldsymbol{\beta}}$

The truth plus error representation also makes the variance calculation direct. Define

$$\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T. \tag{10.24}$$

Then

$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \mathbf{A}\boldsymbol{\epsilon}. \tag{10.25}$$

Since $\boldsymbol{\beta}$ is fixed, it has no variance. Therefore,

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \mathrm{Var}(\mathbf{A}\boldsymbol{\epsilon} \mid \mathbf{X}). \tag{10.26}$$

The matrix variance rule is

$$\mathrm{Var}(\mathbf{A}\mathbf{Z}) = \mathbf{A}\,\mathrm{Var}(\mathbf{Z})\mathbf{A}^T. \tag{10.27}$$

Using $\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2\mathbf{I}_n$,

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \mathbf{A}\,\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X})\mathbf{A}^T \tag{10.28}$$

$$= \mathbf{A}(\sigma^2\mathbf{I}_n)\mathbf{A}^T \tag{10.29}$$

$$= \sigma^2\mathbf{A}\mathbf{A}^T. \tag{10.30}$$

Now substitute $\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$:

$$\mathbf{A}\mathbf{A}^T = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\left[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\right]^T \tag{10.31}$$

$$= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\left[(\mathbf{X}^T\mathbf{X})^{-1}\right]^T. \tag{10.32}$$

Because $\mathbf{X}^T\mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{A}\mathbf{A}^T = (\mathbf{X}^T\mathbf{X})^{-1}. \tag{10.33}$$

So

$$\boxed{\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.} \tag{10.34}$$

What this means. The coefficient uncertainty depends on two things:

1. the error variance σ^2, and
2. the geometry of the design matrix through $(\mathbf{X}^T\mathbf{X})^{-1}$.

If the errors are noisy, coefficient estimates are noisy. If the predictor columns are poorly arranged, nearly dependent, or badly scaled, then $(\mathbf{X}^T\mathbf{X})^{-1}$ can contain large entries, making coefficient estimates unstable.

10.9 Standard Errors of the Coefficients

The diagonal entries of $\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X})$ give the variances of the individual coefficient estimates. Let

$$\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1}. \tag{10.35}$$

Then

$$\mathrm{Var}(\widehat{\beta}_j \mid \mathbf{X}) = \sigma^2 C_{jj}. \tag{10.36}$$

The standard error of $\widehat{\beta}_j$ is

$$\mathrm{SE}(\widehat{\beta}_j) = \sigma\sqrt{C_{jj}}. \tag{10.37}$$

Because σ is unknown, we estimate it using the residuals:

$$\widehat{\sigma}^2 = \frac{\mathbf{e}^T\mathbf{e}}{n-p-1} = \frac{\mathrm{RSS}}{n-p-1}. \tag{10.38}$$

Thus the estimated standard error is

$$\boxed{\widehat{\mathrm{SE}}(\widehat{\beta}_j) = \widehat{\sigma}\sqrt{C_{jj}}.} \tag{10.39}$$

This is the quantity reported in regression output tables. It is the denominator of the usual t-statistic.

10.10 Distribution of $\widehat{\boldsymbol{\beta}}$

If we also assume Gaussian errors,

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_n), \tag{10.40}$$

then $\widehat{\boldsymbol{\beta}}$ is a linear transformation of a multivariate normal random vector. Linear transformations of multivariate normal vectors are multivariate normal. Therefore,

$$\boxed{\widehat{\boldsymbol{\beta}} \mid \mathbf{X} \sim N\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\right).} \tag{10.41}$$

For an individual coefficient,

$$\widehat{\beta}_j \mid \mathbf{X} \sim N\left(\beta_j, \sigma^2 C_{jj}\right). \tag{10.42}$$

After estimating σ, the usual test statistic is

$$t_j = \frac{\widehat{\beta}_j - \beta_{j,0}}{\widehat{\mathrm{SE}}(\widehat{\beta}_j)}, \tag{10.43}$$

where $\beta_{j,0}$ is the hypothesized value, usually 0. Under the Gaussian model and the null hypothesis,

$$t_j \sim t_{n-p-1}. \tag{10.44}$$

Connection to inference. This is why regression output can include coefficient tests and confidence intervals. The theory starts with the truth plus error representation, then uses the error assumptions to describe the sampling behavior of $\widehat{\boldsymbol{\beta}}$.

10.11 Takeaways for Coefficients

The coefficient estimator has three central properties under the standard linear model assumptions:

1. **Truth plus error:**
$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}. \tag{10.45}$$

2. **Unbiasedness under mean-zero errors:**
$$\mathbb{E}[\widehat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}. \tag{10.46}$$

3. **Variance under homoskedastic uncorrelated errors:**
$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}. \tag{10.47}$$

If the Gaussian assumption is added, then $\widehat{\boldsymbol{\beta}}$ has a multivariate normal distribution.

10.12 The Hat Matrix Revisited

The fitted values are

$$\begin{aligned}\widehat{\mathbf{Y}} &= \mathbf{X}\widehat{\boldsymbol{\beta}} &(10.48)\\ &= \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}. &(10.49)\end{aligned}$$

Define the hat matrix
$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.} \tag{10.50}$$

Then
$$\widehat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}. \tag{10.51}$$

The matrix $\mathbf{H}$ puts the hat on $\mathbf{Y}$, which is why it is called the hat matrix.

The hat matrix is an orthogonal projection matrix onto $\mathrm{Col}(\mathbf{X})$. It has two key properties:

1. Symmetry:
$$\mathbf{H}^T = \mathbf{H}. \tag{10.52}$$

2. Idempotence:
$$\mathbf{H}^2 = \mathbf{H}. \tag{10.53}$$

Symmetry and idempotence are the algebraic signatures of an orthogonal projection.

10.13 The Residual Vector

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \widehat{\mathbf{Y}}. \tag{10.54}$$

Using $\widehat{\mathbf{Y}} = \mathbf{HY}$,

$$\mathbf{e} = \mathbf{Y} - \mathbf{HY} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}. \tag{10.55}$$

The matrix

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H} \tag{10.56}$$

is often called the residual-maker matrix because

$$\mathbf{e} = \mathbf{MY}. \tag{10.57}$$

Now substitute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \tag{10.58}$$

$$= (\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.59}$$

Because $\mathbf{H}$ projects onto $\mathrm{Col}(\mathbf{X})$, it leaves every vector in $\mathrm{Col}(\mathbf{X})$ unchanged. Since $\mathbf{X}\boldsymbol{\beta} \in \mathrm{Col}(\mathbf{X})$,

$$\mathbf{HX}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}. \tag{10.60}$$

Therefore,

$$(\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}, \tag{10.61}$$

and

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \tag{10.62}$$

This is the truth plus error representation for residuals.

Important interpretation. Residuals are not the true errors. The true errors are $\boldsymbol{\epsilon}$. The residuals are projected errors:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.63}$$

Least squares removes from $\boldsymbol{\epsilon}$ the component that lies in $\mathrm{Col}(\mathbf{X})$. What remains is the component orthogonal to the fitted-value space.

10.14 Why $\mathbf{I}_n - \mathbf{H}$ Is Also an Orthogonal Projection

If $\mathbf{H}$ is an orthogonal projection, then it is symmetric and idempotent. Let

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}. \tag{10.64}$$

We show that $\mathbf{M}$ is also an orthogonal projection.

First, symmetry:

$$\mathbf{M}^T = (\mathbf{I}_n - \mathbf{H})^T \tag{10.65}$$

$$= \mathbf{I}_n^T - \mathbf{H}^T \tag{10.66}$$

$$= \mathbf{I}_n - \mathbf{H} \tag{10.67}$$

$$= \mathbf{M}. \tag{10.68}$$

Second, idempotence:

$$\mathbf{M}^2 = (\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H}) \tag{10.69}$$
$$= \mathbf{I}_n - 2\mathbf{H} + \mathbf{H}^2 \tag{10.70}$$
$$= \mathbf{I}_n - 2\mathbf{H} + \mathbf{H} \tag{10.71}$$
$$= \mathbf{I}_n - \mathbf{H} \tag{10.72}$$
$$= \mathbf{M}. \tag{10.73}$$

Therefore $\mathbf{I}_n - \mathbf{H}$ is also an orthogonal projection matrix.

Geometrically, $\mathbf{H}$ projects $\mathbf{Y}$ onto $\mathrm{Col}(\mathbf{X})$, while $\mathbf{I}_n - \mathbf{H}$ projects $\mathbf{Y}$ onto the orthogonal complement of $\mathrm{Col}(\mathbf{X})$. That orthogonal complement is where residuals live.

10.15 Expectation of the Residual Vector

Using (10.62),

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.74}$$

Take conditional expectations:

$$\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = (\mathbf{I}_n - \mathbf{H})\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}]. \tag{10.75}$$

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = \mathbf{0}.} \tag{10.76}$$

This is why we expect residual plots to be centered around zero when the mean structure is correctly specified.

10.16 Variance of the Residual Vector

Again use

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.77}$$

By the matrix variance rule,

$$\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = (\mathbf{I}_n - \mathbf{H})\,\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X})(\mathbf{I}_n - \mathbf{H})^T. \tag{10.78}$$

If $\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2\mathbf{I}_n$, then

$$\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^T \tag{10.79}$$
$$= \sigma^2(\mathbf{I}_n - \mathbf{H})^2 \tag{10.80}$$
$$= \sigma^2(\mathbf{I}_n - \mathbf{H}). \tag{10.81}$$

Thus

$$\boxed{\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}).} \tag{10.82}$$

This formula has an important consequence. The ith residual has variance

$$\mathrm{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}), \tag{10.83}$$

where h_{ii} is the ith diagonal entry of the hat matrix. The off-diagonal entries give residual covariances:

$$\mathrm{Cov}(e_i, e_j \mid \mathbf{X}) = -\sigma^2 h_{ij}, \qquad i \neq j. \tag{10.84}$$

Key diagnostic warning. Even if the true errors are uncorrelated and have equal variance, the raw residuals generally do not have equal variance and are generally correlated. This is why leverage-adjusted residuals and studentized residuals are useful in diagnostics.

10.17 Distribution of the Residual Vector

If the Gaussian assumption holds,

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n), \tag{10.85}$$

then $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}$ is a linear transformation of a multivariate normal vector. Therefore,

$$\boxed{\mathbf{e} \mid \mathbf{X} \sim N\left(\mathbf{0}, \sigma^2(\mathbf{I}_n - \mathbf{H})\right).} \tag{10.86}$$

The covariance matrix is singular because $\mathbf{e}$ is constrained to lie in the residual subspace, which has dimension $n - p - 1$, not dimension n. The residual vector cannot point anywhere it wants in $\mathbb{R}^n$; it must be orthogonal to every column of $\mathbf{X}$.

10.18 Residual Degrees of Freedom

The residual degrees of freedom are the dimensions left over after fitting the model. Algebraically,

$$\mathrm{df}_{\mathrm{res}} = \mathrm{tr}(\mathbf{I}_n - \mathbf{H}). \tag{10.87}$$

Because $\mathrm{tr}(\mathbf{I}_n) = n$ and $\mathrm{tr}(\mathbf{H}) = p + 1$ when $\mathbf{X}$ has full column rank,

$$\boxed{\mathrm{df}_{\mathrm{res}} = n - p - 1.} \tag{10.88}$$

This also explains the denominator in the residual variance estimate:

$$\widehat{\sigma}^2 = \frac{\mathbf{e}^T\mathbf{e}}{n - p - 1}. \tag{10.89}$$

The numerator $\mathbf{e}^T\mathbf{e}$ is the residual sum of squares. The denominator is the number of leftover residual dimensions.

10.19 Worked Mini Example: A Three-Point Simple Regression

Consider three observations with one predictor and an intercept:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}. \tag{10.90}$$

Here $n = 3$, $p = 1$, and $p + 1 = 2$. The fitted values live in a two-dimensional plane inside $\mathbb{R}^3$, and the residuals live in the one-dimensional orthogonal complement.

Compute

$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \tag{10.91}$$

and

$$(\mathbf{X}^T\mathbf{X})^{-1} = \frac{1}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}. \tag{10.92}$$

Thus the coefficient variance matrix is

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \frac{\sigma^2}{6}\begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix}. \tag{10.93}$$

So

$$\mathrm{Var}(\widehat{\beta}_0) = \frac{5}{6}\sigma^2, \qquad \mathrm{Var}(\widehat{\beta}_1) = \frac{1}{2}\sigma^2. \tag{10.94}$$

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \begin{bmatrix} \frac{5}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{5}{6} \end{bmatrix}. \tag{10.95}$$

The residual-maker matrix is

$$\mathbf{I}_3 - \mathbf{H} = \begin{bmatrix} \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \end{bmatrix}. \tag{10.96}$$

Notice that

$$\mathrm{tr}(\mathbf{H}) = 2, \qquad \mathrm{tr}(\mathbf{I}_3 - \mathbf{H}) = 1. \tag{10.97}$$

This matches the geometry: two fitted-value degrees of freedom and one residual degree of freedom.

10.20 What Residual Plots Are Really Checking

The residual representation

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon} \tag{10.98}$$

explains why residual analysis works. If the true errors have mean zero, constant variance, and approximate normality, then the residuals should show compatible behavior.

Common diagnostic checks include:

Diagnostic plot	What we hope to see	Possible problem if violated
Residuals vs fitted values	random cloud around zero	nonlinearity or missing structure
Residuals vs predictor	random cloud around zero	predictor transformation needed
Scale-location plot	similar vertical spread	heteroskedasticity
Q-Q plot	points close to a line	non-normal error behavior
Residuals vs time/order	no runs or waves	correlated errors
Residuals vs leverage	no extreme influential points	high leverage or outliers

A good residual plot does not prove that the model is true. It only suggests that the most visible violations are not obvious. A bad residual plot is more informative: it gives evidence that something in the modeling assumptions or model form may need attention.

10.21 Good Residual Behavior

A residual plot with no visible pattern is usually a good sign. A healthy residual-vs-fitted plot should look like a random scatter of points around the horizontal line at zero. The points do not need to be perfectly symmetric, but they should not show a systematic curve, trend, funnel, or clustering pattern.

A random-looking residual cloud suggests:

1. the conditional mean may be reasonably specified,
2. the variance may be roughly stable over the fitted range, and
3. there may not be an obvious missing nonlinear term.

This is why residuals are the natural next object after coefficients. Coefficients tell us what the fitted model says. Residuals tell us what the fitted model failed to explain.

10.22 Bad Residual Behavior

Bad residual behavior usually points to one of a few issues:

1. **Curvature:** residuals form a U-shape or other smooth pattern. This suggests the mean function is not linear in the current predictors.
2. **Funnel shape:** residual spread increases or decreases with fitted values. This suggests nonconstant variance.
3. **Runs over time:** residuals stay positive or negative for long stretches in time order. This suggests correlated errors.
4. **Extreme residuals:** a few residuals are much larger than the rest. This suggests outliers or observations not explained by the model.
5. **High leverage:** some rows of $\mathbf{X}$ are far from the rest. These observations may strongly influence the fitted model.

Residual analysis is not just a plot-making exercise. It is a theory-backed health check of the assumptions that allowed us to interpret $\widehat{\boldsymbol{\beta}}$, standard errors, t-statistics, and prediction intervals.

10.23 Residuals, Leverage, and Studentization

The diagonal hat values

$$h_{ii} \tag{10.99}$$

measure leverage. An observation has high leverage when its predictor pattern is unusual relative to the rest of the design matrix.

Because

$$\mathrm{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}), \tag{10.100}$$

raw residuals do not all have the same variance. A leverage-adjusted residual divides by its estimated standard deviation:

$$r_i = \frac{e_i}{\widehat{\sigma}\sqrt{1 - h_{ii}}}. \tag{10.101}$$

This is often called an internally studentized residual. It gives a more comparable scale for judging which residuals are unusually large.

Why leverage matters. A point with a large residual but low leverage may be an outlier in the response direction. A point with high leverage may control the fitted hyperplane even if its residual is not large. A point with both high leverage and a large residual is especially important to inspect.

Chapter 12 turns these objects into an applied influence workflow. The diagonal values h_{ii}, leverage-adjusted residuals, and Cook's distance help identify observations that deserve investigation. High influence is not an automatic deletion order; it is a signal to check data quality, model form, and sensitivity of the conclusion.

10.24 A Clean Interpretation of Diagnostics

The following table connects each theoretical assumption to the residual behavior we expect to see.

Assumption about $\boldsymbol{\epsilon}$	Residual implication	Diagnostic clue
$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = 0$	residuals centered near zero	no systematic trend
$\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}$	stable spread, no strong correlation	no funnel or tracking
Gaussian errors	residuals roughly normal after adjustment	Q-Q plot roughly linear
Correct linear mean structure	no missing pattern in residuals	no curve or clusters
Full column rank	unique coefficients	no exact multicollinearity

10.25 Why This Matters for Modeling Decisions

The coefficient theory and residual theory combine into a modeling workflow:

1. Fit a model.
2. Interpret $\widehat{\boldsymbol{\beta}}$, standard errors, and model fit.
3. Check residual behavior.
4. If residuals behave badly, consider model repair.

Model repair might include transforming predictors, transforming the response, adding interactions, adding nonlinear terms, using weighted least squares, using robust standard errors, or moving to a different model class. The next chapters build toward these tools.

The key point is that residual diagnostics are not separate from the math. They come directly from

$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon} \tag{10.102}$$

and

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.103}$$

Coding companion reference. Package-specific Python/R commands for extracting fitted values, residuals, leverage, studentized residuals, diagnostic plots, and coefficient covariance matrices have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the theoretical residual and estimator behavior.

10.26 Common Mistakes

Mistake 1: Treating residuals as the true errors

The residuals are not $\boldsymbol{\epsilon}$. They are

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{10.104}$$

They are the part of the error vector left after projecting away the fitted-value component.

Mistake 2: Forgetting the fixed-design condition

The cleanest formulas are conditional on $\mathbf{X}$. If the predictors are random, we usually interpret the results after conditioning on the observed design matrix.

Mistake 3: Thinking unbiased means correct in this dataset

Unbiased means centered correctly over repeated samples. It does not mean the estimate is close in one observed dataset.

Mistake 4: Ignoring multicollinearity in the variance formula

The variance formula

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} \tag{10.105}$$

shows why nearly dependent predictor columns can cause large standard errors.

Mistake 5: Using residual plots as proof

A clean residual plot is not proof that assumptions hold. It is evidence that obvious violations are not visible. A bad residual plot is stronger evidence because it points to a concrete failure mode.

10.27 Operational Briefing Checklist

When briefing a regression model, be ready to answer the following questions:

1. What is the response variable, and what are the predictors?
2. What is the design matrix dimension $n \times (p + 1)$?

3. Is $\mathbf{X}$ full column rank, or is there an identifiability concern?
4. What assumptions are needed for the coefficient standard errors?
5. What assumptions are needed for exact t-tests and confidence intervals?
6. What do the residual plots show?
7. Are residuals centered near zero?
8. Is residual spread roughly stable?
9. Are there patterns suggesting nonlinearity, correlated errors, or missing variables?
10. Are there high leverage points or unusually large studentized residuals?

10.28 Chapter Summary

The core equations of this chapter are:

$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}, \tag{10.106}$$

$$\mathbb{E}[\widehat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}, \tag{10.107}$$

$$\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}, \tag{10.108}$$

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}, \tag{10.109}$$

$$\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}). \tag{10.110}$$

The coefficient estimator is the truth plus a transformed error. Under mean-zero errors, it is unbiased. Under constant-variance uncorrelated errors, its variance is governed by $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$. Under Gaussian errors, it has a multivariate normal distribution.

The residual vector is the true error vector after projection onto the residual subspace. This explains why residuals can be used to diagnose model assumptions. Residuals should be centered around zero, have roughly stable spread, and show no systematic patterns if the model assumptions are reasonable. Chapter 12 turns these facts into a practical diagnostic and repair workflow.

10.29 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 10 use	Math Foundations connection	MF textbook location
Observation-space vectors	Vector notation in $\mathbb{R}^n$, coordinates, and subvectors	Ch02 Secs. 2.1, 2.2, 2.6; Ch06 Sec. 6.5
Regression matrix expressions	Matrix sizes, transposes, identity matrices, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.3, 7.4, 7.6.1, 7.6.2
Residual orthogonality	Dot products, Euclidean norms, and the condition $\mathbf{a}^T\mathbf{b} = 0$ for orthogonality	Ch02 Sec. 2.6; Ch05 Secs. 5.1, 5.3, 5.5, 5.8; Ch12 Sec. 12.12
Least-squares normal-equation bridge	Stationary points, least-squares minimization, the normal equations, and the positive-semidefinite $\mathbf{A}^T\mathbf{A}$ argument	Ch05 Sec. 5.5; Ch10 Sec. 10.6; Ch12 Secs. 12.8, 12.9, 12.11, 12.12; Ch13 Sec. 13.4
Full-rank requirement for unique coefficients	Left inverses, linearly independent columns, rank, and full column rank	Ch06 Sec. 6.3; Ch10 Secs. 10.1, 10.3, 10.5; Ch11 Sec. 11.4
Computational matrix operations	Python/NumPy arrays, shapes, transposes, and matrix multiplication	Ch07 Secs. 7.1, 7.4, 7.7

10.30 Practice and Workflow

Focused Study Checklist

Use this as a final check after studying the truth-plus-error representation.

1. Can you write $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and explain why $\mathbf{Y}$ is random under fixed design?
2. Can you state the coefficient decomposition

$$\widehat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\epsilon}?$$

3. Can you identify which assumption gives unbiasedness and which assumptions support the usual variance and distribution results?
4. Can you write the residual relation $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}$?
5. Can you explain why residuals are useful diagnostics but not the same as the true errors?
6. Can you connect a residual pattern to a likely assumption failure?

Chapter 11

Linear Models Beyond Straight Lines: Transformations, Categorical Predictors, and Interactions

11.1 The Big Picture

The previous chapters built multiple linear regression from two viewpoints:

1. Algebraically, the fitted values are produced by a design matrix and a coefficient vector.
2. Geometrically, the fitted response vector is the projection of $\mathbf{y}$ onto the column space of the design matrix.

This chapter makes an important expansion. Earlier chapters used "linear" mainly through the geometric picture of lines, planes, and hyperplanes in the displayed predictors. That picture was useful, but incomplete:

> A linear model does not require the response to be a straight-line function of the original predictor values. It only requires the model to be linear in the unknown coefficients.

That distinction is easy to miss. The model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \tag{11.1}$$

is curved as a function of x, but it is still a linear model in the parameters $\beta_0, \beta_1, \beta_2$. Once we create the transformed predictor column x^2, ordinary least squares can be used exactly as before. The same idea lets us use category indicators and interaction columns while keeping the model linear in the coefficients.

The central question is therefore not, "Is the graph a straight line?" The better question is:

> Can the model be written as a design matrix times a coefficient vector, plus error?

If the answer is yes, then the linear regression machinery still applies.

We also return to the sample-level notation of Chapters 7 and 9: $\mathbf{y}$ is the observed response vector from the data set being fit.

Math Foundations Connection. This chapter uses the Math Foundations ideas of vectors, sets, linear combinations, bases, matrix-vector multiplication, submatrices, linear independence, null spaces, and full-column-rank inverse-like formulas. Transformed regression, categorical regression, and interaction modeling are all ways of building columns for the design matrix. Those columns form candidate directions in $\mathbb{R}^n$, and least squares chooses the best linear combination of those directions. See the Math Foundations textbook, Chapters 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and 12.

11.2 What "Linear" Means Here

A model is linear in parameters when it can be written as

$$m(x;\boldsymbol{\theta}) = \theta_0 g_0(x) + \theta_1 g_1(x) + \cdots + \theta_q g_q(x), \tag{11.2}$$

where the functions $g_0, g_1, \ldots, g_q$ are known functions of the input values, and the unknowns are the coefficients $\theta_0, \theta_1, \ldots, \theta_q$. Usually $g_0(x) = 1$, which gives the intercept column.

The key is that the unknown parameters appear only as coefficients multiplying known columns. They are not multiplied by each other, raised to powers, placed inside nonlinear functions, or used as exponents.

For observations $i = 1, \ldots, n$, define

$$z_{ij} = g_j(x_i), \qquad j = 0, 1, \ldots, q. \tag{11.3}$$

Then the model becomes

$$Y_i = \theta_0 z_{i0} + \theta_1 z_{i1} + \cdots + \theta_q z_{iq} + \epsilon_i. \tag{11.4}$$

Stacking the observations gives

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \tag{11.5}$$

where

$$\mathbf{Z} = \begin{bmatrix} z_{10} & z_{11} & \cdots & z_{1q} \\ z_{20} & z_{21} & \cdots & z_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n0} & z_{n1} & \cdots & z_{nq} \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_q \end{bmatrix}. \tag{11.6}$$

This is the same matrix form as ordinary multiple linear regression. The only difference is that the design matrix may contain transformed versions of the original variables, category indicators, and interaction columns.

Original predictor space versus feature space

It is helpful to distinguish two spaces:

1. **Original predictor space:** the variables as measured, such as x, x_1, x_2, temperature, distance, or time.
2. **Feature space:** the columns actually used in the design matrix, such as x, x^2, $\log x$, $\sin x$, $x_1 x_2$, or category indicators.

Linear regression is linear in the columns of the design matrix. Those columns can be created from nonlinear transformations of the original predictors.

11.3 The Least Squares Problem with Transformed Columns

Once the transformed design matrix $\mathbf{Z}$ is built, the least squares objective is

$$\min_{\mathbf{b}\in\mathbb{R}^{q+1}} \|\mathbf{y} - \mathbf{Z}\mathbf{b}\|^2 . \tag{11.7}$$

The normal equations are

$$\mathbf{Z}^T\mathbf{Z}\widehat{\boldsymbol{\theta}} = \mathbf{Z}^T\mathbf{y}. \tag{11.8}$$

If $\mathbf{Z}$ has full column rank, then

$$\boxed{\widehat{\boldsymbol{\theta}} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{y}.} \tag{11.9}$$

The fitted values are

$$\widehat{\mathbf{y}} = \mathbf{Z}\widehat{\boldsymbol{\theta}}. \tag{11.10}$$

The hat matrix for the transformed model is

$$\mathbf{H}_{\mathbf{Z}} = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T, \tag{11.11}$$

and therefore

$$\widehat{\mathbf{y}} = \mathbf{H}_{\mathbf{Z}}\mathbf{y}, \qquad \mathbf{e} = (\mathbf{I}_n - \mathbf{H}_{\mathbf{Z}})\mathbf{y}. \tag{11.12}$$

Thus the geometry from Chapter 7 and the residual theory from Chapter 10 still apply. The projection subspace has changed because the design matrix columns changed, but the projection logic is the same.

11.4 A Dimension Check

Let $q + 1$ be the number of columns in the transformed design matrix $\mathbf{Z}$. Then:

Object	Meaning	Dimension
$\mathbf{y}$	response vector	$n \times 1$
$\mathbf{Z}$	transformed design matrix	$n \times (q+1)$
$\boldsymbol{\theta}$	coefficient vector for transformed columns	$(q+1) \times 1$
$\widehat{\boldsymbol{\theta}}$	least squares estimate	$(q+1) \times 1$
$\widehat{\mathbf{y}}$	fitted response vector	$n \times 1$
$\mathbf{e}$	residual vector	$n \times 1$
$\mathbf{H}_{\mathbf{Z}}$	hat matrix for transformed model	$n \times n$

The model is valid as a least squares linear model if the columns of $\mathbf{Z}$ are well-defined and have enough independent information to identify $\boldsymbol{\theta}$. If $\mathbf{Z}^T\mathbf{Z}$ is singular, then the coefficient vector is not uniquely identified.

11.5 Mayer's Moon Motion Problem

The motivating example in the source slides is the Moon motion problem. The Moon is tidally locked to Earth, but not perfectly. Its apparent motion includes libration in longitude, latitude, and diurnal effects. Because of this wobble, observers can see more than exactly half of the Moon's near side over time.

Tobias Mayer collected measurements to describe this apparent wobbling motion, called libration. The important statistical point is that the response did not need to be modeled as a straight line in the original physical angle. Instead, Mayer's equation used transformed predictors:

$$Y = \beta + \alpha x_1 + \gamma x_2 + \epsilon, \tag{11.13}$$

where

$$Y = \text{arc/libration measurement}, \qquad x_1 = \sin(\text{angle}), \qquad x_2 = \cos(\text{angle}). \tag{11.14}$$

For observation i, write

$$Y_i = \beta + \alpha \sin(\theta_i) + \gamma \cos(\phi_i) + \epsilon_i. \tag{11.15}$$

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & \sin(\theta_1) & \cos(\phi_1) \\ 1 & \sin(\theta_2) & \cos(\phi_2) \\ \vdots & \vdots & \vdots \\ 1 & \sin(\theta_n) & \cos(\phi_n) \end{bmatrix}, \qquad \boldsymbol{\theta} = \begin{bmatrix} \beta \\ \alpha \\ \gamma \end{bmatrix}. \tag{11.16}$$

Then the model is simply

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \tag{11.17}$$

This is a linear model because β, α, and γ enter linearly. The sine and cosine transformations are part of the known design matrix, not unknown parameters.

Why the plot may not look linear. If we plot the response against the raw angles, the shape may look curved or oscillatory. But once the columns $\sin(\theta)$ and $\cos(\phi)$ are created, the fitted response vector is a linear combination of those columns and the intercept column. In the transformed feature space, the fit is a plane.

11.6 Basis Functions

A basis function is a known function used to build a column of the design matrix. In the notation from (11.2), the functions

$$g_0(x), g_1(x), \ldots, g_q(x) \tag{11.18}$$

are basis functions.

A model using basis functions has the form

$$f(x) = \theta_0 g_0(x) + \theta_1 g_1(x) + \cdots + \theta_q g_q(x). \tag{11.19}$$

This is a linear combination of known functions. The coefficients determine how much of each basis direction is used.

Common basis choices include:

Basis type	Example columns	Useful for
Polynomial	$1, x, x^2, x^3$	smooth curvature
Interaction	$1, x_1, x_2, x_1x_2$	combined predictor effects
Trigonometric	$1, \sin x, \cos x$	cyclic or angular behavior
Logarithmic	$1, \log x$	diminishing returns and scale effects
Indicator/step	$I(x < c), I(x \geq c)$	piecewise constant effects
Spline	piecewise polynomial basis columns	flexible smooth curves

The design choice is not automatic. The analyst chooses basis functions based on domain knowledge, EDA, residual behavior, predictive performance, and interpretability.

11.7 Polynomial Regression

A polynomial regression model of degree d is

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \epsilon_i. \tag{11.20}$$

Although this model is nonlinear in x, it is linear in the coefficients $\beta_0, \beta_1, \ldots, \beta_d$.

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_2 & x_2^2 & \cdots & x_2^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{bmatrix}. \tag{11.21}$$

This is a Vandermonde-style design matrix. The Math Foundations textbook uses a square Vandermonde matrix to fit an interpolating polynomial through n points; see Ch08 Sec. 8.7. In data analysis, we usually do something more modest: choose a small degree d, keep $d + 1 \ll n$, and solve a least squares problem instead of forcing the curve through every point.

Interpolation versus regression

Polynomial interpolation and polynomial regression can look similar algebraically, but their goals differ.

Task	Matrix shape	Goal
Interpolation	usually square $n \times n$	pass exactly through all points
Regression	usually tall $n \times (d + 1)$	find a stable trend with residual error

Interpolation can be useful in numerical methods, but in noisy data analysis it often overfits. Regression accepts residuals because the data include noise and because the goal is usually inference, prediction, or explanation rather than exact passage through every observed point.

Degree choice

A degree-one polynomial is the ordinary straight-line model. A degree-two polynomial allows one bend. A degree-three polynomial allows more shape. But high-degree polynomials can behave badly, especially near the edges of the observed data range.

A practical rule:

> Use the lowest-degree polynomial that addresses the visible pattern and is defensible for the problem.

Residual plots and test error help decide whether the added flexibility is helping or merely fitting noise.

11.8 Categorical Predictors as Indicator Columns

A categorical predictor cannot be used as an arbitrary numerical label unless the numbers have quantitative meaning. If a curriculum is coded as OR = 1, Physics = 2, Applied Math = 3, and so on, the number 3 is not "larger" than the number 1 in the same way that a height or budget is larger. The category must first be encoded as design-matrix columns.

For a binary category, define an indicator variable

$$X_i = \begin{cases} 1, & \text{if observation } i \text{ is in category } C, \\ 0, & \text{otherwise.} \end{cases} \tag{11.22}$$

Using the source-slide example, suppose Y is IQ for an NPS student and $X_i = 1$ if the student is in the OR curriculum. The model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \tag{11.23}$$

has two fitted group means:

$$\widehat{Y} \mid X = 0 = \widehat{\beta}_0, \tag{11.24}$$

$$\widehat{Y} \mid X = 1 = \widehat{\beta}_0 + \widehat{\beta}_1. \tag{11.25}$$

Thus $\widehat{\beta}_0$ is the fitted mean for the reference category, and $\widehat{\beta}_1$ is the contrast from the reference category to the coded category. In this example, $\widehat{\beta}_1$ is the fitted OR adjustment from the non-OR baseline; as the slides joke, we hope it is positive.

Changing the reference category changes the intercept and the sign of the contrast, but it does not change the fitted values. The reference category is an interpretation choice, not a different least-squares method.

11.9 Multi-Level Categorical Predictors

If a categorical predictor has k levels and the model includes an intercept, use $k - 1$ indicator columns. Pick one level as the reference category and create indicators for the remaining levels.

For example, let curriculum have levels

$$\{\text{OR},\ \text{Physics},\ \text{Applied Math},\ \text{Engineering},\ \text{Other}\}. \tag{11.26}$$

If Other is the reference category, define

$$X_{i1} = 1\{\text{student } i \text{ is in OR}\}, \tag{11.27}$$
$$X_{i2} = 1\{\text{student } i \text{ is in Physics}\}, \tag{11.28}$$
$$X_{i3} = 1\{\text{student } i \text{ is in Applied Math}\}, \tag{11.29}$$
$$X_{i4} = 1\{\text{student } i \text{ is in Engineering}\}. \tag{11.30}$$

Then

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i3} + \beta_4 X_{i4} + \epsilon_i. \tag{11.31}$$

The fitted reference mean is $\widehat{\beta}_0$. The fitted OR mean is $\widehat{\beta}_0 + \widehat{\beta}_1$, the fitted Physics mean is $\widehat{\beta}_0 + \widehat{\beta}_2$, and so on.

Why not use all k indicators? With an intercept and all five curriculum indicators, every row satisfies

$$X_{i,\text{OR}} + X_{i,\text{Physics}} + X_{i,\text{Applied Math}} + X_{i,\text{Engineering}} + X_{i,\text{Other}} = 1. \tag{11.32}$$

That sum is exactly the intercept column. The design matrix is therefore rank deficient, so the coefficient vector is not uniquely identified. This is the dummy-variable trap. The fix is to drop one level as the reference category or to remove the intercept and use all group indicators with a different interpretation.

11.10 Mixed Numeric and Categorical Predictors

Numeric and categorical predictors can appear in the same design matrix. For example,

$$Y_i = \beta_0 + \beta_1 \texttt{hours}_i + \beta_2 \texttt{OR}_i + \epsilon_i \tag{11.33}$$

uses a numeric predictor and a binary curriculum indicator. If $\texttt{OR}_i = 0$, the fitted line is

$$\widehat{Y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 \texttt{hours}_i. \tag{11.34}$$

If $\texttt{OR}_i = 1$, the fitted line is

$$\widehat{Y}_i = (\widehat{\beta}_0 + \widehat{\beta}_2) + \widehat{\beta}_1 \texttt{hours}_i. \tag{11.35}$$

The two groups share the same slope but have different intercepts. That is a modeling assumption. If the slope should differ by group, we need an interaction.

11.11 Interactions as Transformed Columns

An interaction means that the effect of one predictor depends on another predictor. It is not merely correlation between predictors. In the Advertising example, Newspaper and Radio may be correlated because marketing budgets are allocated together, but an interaction asks a different question: does the marginal effect of Radio change when Newspaper spending changes?

With two numeric predictors, the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i \tag{11.36}$$

is linear in the coefficients. The interaction column is

$$z_{i3} = x_{i1} x_{i2}, \tag{11.37}$$

and the design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}x_{12} \\ 1 & x_{21} & x_{22} & x_{21}x_{22} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}x_{n2} \end{bmatrix}. \tag{11.38}$$

The model can be rewritten as

$$Y_i = \beta_0 + (\beta_1 + \beta_3 x_{i2})\, x_{i1} + \beta_2 x_{i2} + \epsilon_i. \tag{11.39}$$

Thus the slope with respect to x_1 depends on x_2.

Numeric-by-categorical interactions

Suppose Y is health-insurance charges, x is BMI, and $S_i = 1$ if person i is a smoker. The model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 S_i + \beta_3 x_i S_i + \epsilon_i \tag{11.40}$$

allows smokers and nonsmokers to have different intercepts and different BMI slopes:

$$\widehat{Y}_i \mid S_i = 0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_i, \tag{11.41}$$

$$\widehat{Y}_i \mid S_i = 1 = (\widehat{\beta}_0 + \widehat{\beta}_2) + (\widehat{\beta}_1 + \widehat{\beta}_3) x_i. \tag{11.42}$$

The interaction coefficient $\widehat{\beta}_3$ is the difference between the smoker and nonsmoker BMI slopes.

Factor crossings. When two categorical predictors interact, the model allows each combination of levels to have its own adjustment relative to a reference cell. This is useful when the categories combine in a way that is not additive. The cost is more columns, more degrees of freedom, and more care in interpretation.

Hierarchy principle. If a model includes an interaction such as $x_1 x_2$, it is usually good practice to include the corresponding main effects x_1 and x_2. This makes interpretation more stable and avoids treating the combined effect as meaningful while excluding its component directions. Exceptions require a clear domain reason.

Coding companion reference. Package-specific commands for factors, reference levels, dummy-variable inspection, and formula syntax belong in the Coding and Software Cookbook companion. The main text focuses on the design-matrix logic and coefficient interpretation.

11.12 Transforming the Response

So far, the transformations have been applied to predictors. Sometimes the response is transformed instead. Examples include

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{11.43}$$

or

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i. \tag{11.44}$$

A response transformation changes the scale on which the model is linear. For example, if

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{11.45}$$

then the model says that the conditional mean structure is simpler on a log scale than on the original scale. This can help when residual spread grows with the fitted values or when effects are multiplicative rather than additive.

Important caution. A transformed-response model is not the same as an untransformed-response model. If we fit $\log(Y)$, then predictions are naturally produced on the log scale. Transforming back with $\exp(\cdot)$ gives a prediction on the original scale, but uncertainty and bias require care because

$$\mathbb{E}[\exp(W)] \neq \exp(\mathbb{E}[W]) \tag{11.46}$$

in general.

11.13 Classifying Models: Linear or Not Linear?

The quickest check is to ask whether the model can be written as $\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ for a design matrix $\mathbf{Z}$ that can be computed from the observed data without knowing the parameters.

Model expression	Linear model?	Why
$\beta_0 + \beta_1 x$	Yes	coefficients multiply known columns
$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$	Yes	polynomial columns are known
$\beta_0 x^{\beta_1}$	No	parameter appears as exponent
$\beta_0 + \beta_1 x_1 + \beta_0 \beta_1 x_1^2$	No	parameters are multiplied together
$1/(\beta_0 + \beta_1 x)$	No	parameters are inside reciprocal transformation
$\beta_1 \sin x + \beta_2 \cos x + \beta_3 \sin x \cos x$	Yes	transformed columns are known
$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$	Yes	interaction column is known

The logistic regression mean function

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \tag{11.47}$$

is not a linear regression model for $p(x)$. It is, however, linear in the logit scale:

$$\log\left(\frac{p(x)}{1 - p(x)}\right) = \beta_0 + \beta_1 x. \tag{11.48}$$

That is why logistic regression belongs to the generalized linear model family rather than ordinary least squares linear regression.

11.14 Projection Geometry Still Works

A transformed design matrix changes the subspace, not the projection rule. In ordinary MLR, fitted values live in $\mathrm{Col}(\mathbf{X})$. In this chapter, fitted values live in

$$\mathrm{Col}(\mathbf{Z}). \tag{11.49}$$

Least squares chooses

$$\widehat{\mathbf{y}} \in \mathrm{Col}(\mathbf{Z}) \tag{11.50}$$

so that

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} \tag{11.51}$$

is orthogonal to every column of $\mathbf{Z}$. Therefore,

$$\mathbf{Z}^T\mathbf{e} = \mathbf{0}. \tag{11.52}$$

This is the transformed-model version of the normal equations.

Residual interpretation. If a residual plot shows curvature after fitting a straight-line model, we can add a transformed column that represents the missing shape. If the new transformed model is appropriate, the residual pattern should weaken or disappear. This does not prove the model is true, but it provides evidence that the new basis is capturing structure the old basis missed.

11.15 Truth Plus Error with Transformed Designs

All of the estimator behavior from the previous chapter carries over with $\mathbf{X}$ replaced by $\mathbf{Z}$. For the theoretical expectation and variance calculations, treat the response vector as random. If

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \tag{11.53}$$

then

$$\begin{aligned}
\widehat{\boldsymbol{\theta}} &= (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{Y} && (11.54)\\
&= (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T(\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}) && (11.55)\\
&= \boldsymbol{\theta} + (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\boldsymbol{\epsilon}. && (11.56)
\end{aligned}$$

Thus,

$$\boxed{\widehat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\boldsymbol{\epsilon}.} \tag{11.57}$$

For the realized data set, we compute the coefficient estimate using the observed response vector:

$$\widehat{\boldsymbol{\theta}} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{y}. \tag{11.58}$$

The truth-plus-error expression above is used to study the estimator's behavior; it is not a computational formula because $\boldsymbol{\theta}$ and $\boldsymbol{\epsilon}$ are unknown.

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{Z}] = \mathbf{0}$, then

$$\mathbb{E}[\widehat{\boldsymbol{\theta}} \mid \mathbf{Z}] = \boldsymbol{\theta}. \tag{11.59}$$

If $\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{Z}) = \sigma^2 \mathbf{I}_n$, then

$$\mathrm{Var}(\widehat{\boldsymbol{\theta}} \mid \mathbf{Z}) = \sigma^2 (\mathbf{Z}^T \mathbf{Z})^{-1}. \tag{11.60}$$

This matters because transformed columns do not exempt us from regression assumptions. They only change the systematic part of the model.

11.16 Model Comparison for Transformations

When comparing candidate transformations, avoid relying on a single number or a single test in isolation. Useful evidence includes:

1. residual plots,
2. test MSE or cross-validation error,
3. adjusted R^2, AIC, or BIC,
4. nested-model F-tests when the models are nested,
5. coefficient interpretability,
6. domain plausibility.

Nested transformed models

Suppose a reduced model has residual sum of squares RSS_R and degrees of freedom df_R, and a full model has residual sum of squares RSS_F and degrees of freedom df_F. If the reduced model is nested inside the full model, the usual comparison statistic is

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \tag{11.61}$$

For example, the straight-line model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i \tag{11.62}$$

is nested inside the quadratic model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \tag{11.63}$$

The nested test asks whether the added x^2 column explains enough additional variation to justify the extra parameter.

Prediction-focused comparison

For prediction, a transformation that improves training RSS may not improve performance on new data. Use a train/test split or cross-validation when the modeling goal is predictive accuracy. This connects directly to the bias-variance ideas from earlier chapters: more basis columns increase flexibility, which can reduce bias but increase variance.

11.17 Practical Transformation Patterns

Polynomial terms

Use polynomial terms when residual plots suggest smooth curvature. Example:

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon. \tag{11.64}$$

Interpretation is about the whole curve, not just one coefficient in isolation.

Log predictor

Use $\log x$ when the effect of x appears to show diminishing returns or when x spans orders of magnitude:

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon. \tag{11.65}$$

This requires $x > 0$.

Log response

Use $\log Y$ when the response is positive and variability grows with the mean:

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon. \tag{11.66}$$

This requires $Y > 0$.

Square-root response

Use $\sqrt{Y}$ for nonnegative responses, especially count-like responses where variance grows with the mean:

$$\sqrt{Y} = \beta_0 + \beta_1 x + \epsilon. \tag{11.67}$$

This requires $Y \geq 0$.

Trigonometric terms

Use sine and cosine terms for cyclic predictors such as time of day, heading, angle, season, or orbital position:

$$Y = \beta_0 + \beta_1 \sin(t) + \beta_2 \cos(t) + \epsilon. \tag{11.68}$$

For a period P, use

$$\sin\left(\frac{2\pi t}{P}\right), \qquad \cos\left(\frac{2\pi t}{P}\right). \tag{11.69}$$

Interactions

Use interactions when the effect of one predictor plausibly depends on another:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \tag{11.70}$$

11.18 Numerical Stability and Identifiability

Transformed columns can create numerical problems. For example, if x is large, then x^2, x^3, and x^4 may have very different scales. This can make $\mathbf{Z}^T\mathbf{Z}$ ill-conditioned.

Centering before powers

A common fix is to center before creating powers:

$$u_i = x_i - \bar{x}, \tag{11.71}$$

and then fit

$$Y_i = \beta_0 + \beta_1 u_i + \beta_2 u_i^2 + \epsilon_i. \tag{11.72}$$

Centering often improves interpretability and numerical stability.

Orthogonal polynomial columns

Another fix is to use orthogonal polynomial basis columns. This is a computational version of the Gram-Schmidt idea from Math Foundations: transform a set of related columns into a better-conditioned set of orthogonal directions; see the Math Foundations textbook, Ch09 Secs. 9.1, 9.3, and 9.11. The fitted values can represent the same polynomial space, but the coefficient estimation is more stable.

Full column rank

The transformed design matrix must still have full column rank for the usual inverse formula to apply. Problems occur when columns are exact linear combinations of each other. Examples include:

1. including duplicate columns,
2. creating indicators that sum to the intercept column without dropping a reference category,
3. including high-degree terms with too few distinct predictor values,
4. creating transformations that are constant in the observed data.

The regression problem is not saved by calling a column "transformed." If the resulting columns are linearly dependent, the model is still not identified.

11.19 Interpretation After Transformations

Transformations improve flexibility, but they change interpretation.

Polynomial coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon, \tag{11.73}$$

the coefficient β_1 is not simply the global slope. The instantaneous slope of the fitted mean is

$$\frac{d}{dx}\left(\beta_0 + \beta_1 x + \beta_2 x^2\right) = \beta_1 + 2\beta_2 x. \tag{11.74}$$

Thus the effect of changing x depends on where you are on the curve.

Categorical coefficient interpretation

For an indicator variable, the coefficient is a contrast from the reference category. In a model with one indicator,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \tag{11.75}$$

β_0 is the fitted reference mean and β_1 is the fitted adjustment for the coded group. In a model with $k-1$ indicators, each contrast is interpreted relative to the omitted reference level.

Interaction coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon, \tag{11.76}$$

the slope with respect to x_1 is

$$\frac{\partial}{\partial x_1}\mathbb{E}[Y \mid x_1, x_2] = \beta_1 + \beta_3 x_2. \tag{11.77}$$

The coefficient β_3 tells us how the slope in x_1 changes as x_2 increases.

Log predictor interpretation

In

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon, \tag{11.78}$$

small percentage changes in x correspond approximately to additive changes in Y. The exact change from x to cx is

$$\Delta\widehat{Y} = \beta_1 \log(c). \tag{11.79}$$

Log response interpretation

In

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon, \tag{11.80}$$

a one-unit increase in x changes the fitted log-response by β_1. On the original scale, this corresponds approximately to multiplying the response by e^{β_1}, before accounting for retransformation issues.

11.20 Worked Example: Building a Polynomial Design Matrix

Suppose we observe four data points (x_i, y_i) and want to fit a quadratic model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \tag{11.81}$$

Let

$$x = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \qquad y = \begin{bmatrix} 2 \\ 1 \\ 2 \\ 5 \end{bmatrix}. \tag{11.82}$$

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}. \tag{11.83}$$

The model is

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{11.84}$$

Here $\boldsymbol{\beta}$ is the coefficient vector for this quadratic transformed design; it plays the same role as $\boldsymbol{\theta}$ above. The least squares estimate is

$$\widehat{\boldsymbol{\beta}} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{y}. \tag{11.85}$$

The fitted curve is then

$$\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x + \widehat{\beta}_2 x^2. \tag{11.86}$$

Although the graph is a parabola, the computation is ordinary least squares.

Coding companion reference. Package-specific Python/R commands for polynomial terms, interactions, response transformations, and prediction grids have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the linear-in-parameters logic and interpretation guidance.

11.21 Common Mistakes

Mistake 1: Thinking curved means not linear

A polynomial curve can be a linear model. Linearity refers to the unknown coefficients, not the visual shape of the fitted curve in the original predictor.

Mistake 2: Transforming after the split incorrectly

For prediction workflows, transformations that learn quantities from the data, such as centering and scaling, should be learned on the training set and then applied to the test set. Otherwise, information leaks from test data into the training process.

Mistake 3: Ignoring domain restrictions

Log transformations require positive values. Square-root transformations require nonnegative values. A transformation that silently drops rows can change the analysis population.

Mistake 4: Overinterpreting individual polynomial coefficients

In polynomial models, the individual coefficients depend heavily on the chosen basis and scaling. Interpret the fitted curve and its derivative, not just isolated coefficient signs.

Mistake 5: Adding flexibility without checking test performance

More columns usually improve training fit. That does not guarantee better prediction on new observations. Use residual plots and validation.

Mistake 6: Creating rank-deficient transformed designs

A transformed design matrix can still be singular. Check for duplicate columns, redundant indicators, too many polynomial terms, and variables with too few distinct values.

Mistake 7: Treating category codes as numeric measurements

A category code is not a quantity just because it is stored as a number. Encode categories with indicator variables or another defensible contrast system.

Mistake 8: Using all category indicators plus an intercept

With an intercept, use $k-1$ indicators for a k-level categorical predictor. Otherwise the indicator columns sum to the intercept column and the coefficients are not uniquely identified.

11.22 Operational Briefing Checklist

When briefing a transformed linear model, answer the following:

1. What is the response variable?
2. What original predictors were measured?
3. What transformed, categorical, or interaction columns were created?
4. Can the model be written as $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$?
5. Why are these transformations operationally or scientifically plausible?
6. Does the transformed design matrix have full column rank?
7. What residual pattern motivated the transformation?
8. Did residual behavior improve after the transformation?

9. Was model comparison based on residuals, test MSE, adjusted R^2, AIC/BIC, nested tests, or a combination?
10. What is the reference category for each categorical predictor?
11. If interactions are included, what effect modification is being claimed?
12. How does the transformation or encoding change interpretation for the customer or commander?

11.23 Chapter Summary

The core idea is:

$$\boxed{\text{linear model} = \text{linear in unknown coefficients, not necessarily linear in original predictors.}} \tag{11.87}$$

If we can create a transformed design matrix $\mathbf{Z}$ such that

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \tag{11.88}$$

then, if $\mathbf{Z}$ has full column rank, ordinary least squares uses the inverse form:

$$\widehat{\boldsymbol{\theta}} = (\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{y}. \tag{11.89}$$

If $\mathbf{Z}^T\mathbf{Z}$ is singular, the fitted values may still be defined, but the coefficient vector is not uniquely identified. The fitted values are still a projection:

$$\widehat{\mathbf{y}} = \mathbf{H}_{\mathbf{Z}}\mathbf{y}. \tag{11.90}$$

The residuals are still orthogonal to the model columns:

$$\mathbf{Z}^T\mathbf{e} = \mathbf{0}. \tag{11.91}$$

Transformations let us fit more realistic structure while keeping the interpretability and inference tools of linear regression. Polynomial terms, sine/cosine terms, log terms, indicator columns, and interactions are all ways to build richer design matrices. For a k-level categorical predictor with an intercept, use $k-1$ indicators and interpret coefficients relative to the reference category. The cost is that interpretation, rank, and overfitting must be handled carefully.

11.24 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 11 use	Math Foundations connection	MF textbook location
Transformed columns as candidate directions	Vector notation in $\mathbb{R}^n$, scalar-vector multiplication, and linear combinations	Ch02 Sec. 2.1; Ch03 Secs. 3.4, 3.5
Basis and spanning viewpoint	Bases, linear independence, and unique linear-combination representations	Ch06 Secs. 6.1, 6.3, 6.5
Design-matrix mechanics	Matrix notation, submatrices, transposes, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.4, 7.6.1, 7.6.2
Categorical indicators and reference levels	Set membership, subsets, and variables as data vectors	Ch02 Sec. 2.6; Ch04 Secs. 4.1, 4.4
Dummy-variable trap	Linear independence, rank, full column rank, and nonzero null spaces	Ch06 Sec. 6.3; Ch10 Secs. 10.3, 10.5; Ch11 Sec. 11.4
Solving transformed systems	QR decomposition, triangular systems, and solving linear systems	Ch08 Sec. 8.1; Ch09 Sec. 9.9
Vandermonde / polynomial design	Vandermonde matrices for polynomial systems and interpolation	Ch08 Sec. 8.7
Orthogonal polynomial intuition	Gram-Schmidt orthogonalization and orthonormal directions	Ch05 Secs. 5.5, 5.8; Ch08 Sec. 8.7; Ch09 Secs. 9.1, 9.3, 9.11
Full column rank requirement	Left inverses, rank, and full-column-rank conditions	Ch10 Secs. 10.1, 10.3, 10.5
Computational matrix operations	Python/NumPy arrays, shapes, transposes, and matrix multiplication	Ch07 Secs. 7.1, 7.4, 7.7

11.25 Practice and Workflow

Focused Study Checklist

Use this as a final check after proposing a transformed linear model.

1. Can you state the difference between original predictors and created features?
2. Can you write the transformed or engineered design matrix $\mathbf{Z}$?
3. Can you express the model as $\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$?
4. Can you explain why the model is linear in the coefficients even if it is curved in the original variables?
5. Can you identify reference categories and interpret categorical contrasts?
6. Can you check whether transformed, indicator, or interaction columns create rank or identifiability problems?
7. Can you compare residual behavior before and after the transformation?

8. Can you interpret the transformed model in language that remains useful to the customer?

Chapter 12

Residual Analysis and Model Repair

12.1 The Big Picture

A fitted model is not the end of the analysis. It is the beginning of the model check.

The previous chapters built multiple linear regression from the design-matrix point of view:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \qquad \widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}}. \tag{12.1}$$

The residual vector $\mathbf{e}$ is the part of the observed response vector that the fitted model did not explain. Residual analysis asks:

> Does the leftover pattern look like random noise, or does it look like structure the model failed to capture?

This chapter is about using residuals as a diagnostic tool and then deciding how to repair a model when the diagnostics look bad. The three practical repair moves emphasized in the source slides are:

1. transform the predictors and/or the response,
2. add interaction terms,
3. admit that the current linear-model frame is not adequate and move to a different modeling strategy.

The most important habit is to avoid treating a single hypothesis test, a single R^2, or a single residual plot as the whole story. Residual analysis is evidence. Test MSE, cross-validation, domain knowledge, and interpretability are also evidence.

As in Chapters 7, 9, and 11, we use $\mathbf{y}$ for the observed response vector from the fitted data set. Chapter 10 used $\mathbf{Y}$ when the response vector was being treated as random. Here the emphasis is back on the realized fit and its residuals.

Math Foundations Connection. Residual analysis is geometric, but the supporting Math Foundations ideas are more basic: norms, orthogonality, matrix-vector multiplication, rank, column-space/residual-space geometry, and least-squares minimization. These ideas are covered in the Math Foundations textbook, Chapters 5, 7, 10, 11, and 12. The projection language used in this chapter is a Data Analysis synthesis built from those underlying ideas. Model repair changes the design columns and therefore changes which patterns can be represented by the fitted part of the model.

12.2 Recall from Chapters 7 and 10: The Residual Vector

Chapters 7 and 10 already developed the linear-algebra machinery behind residuals. We recall only the pieces needed for diagnostics.

For a full-rank design matrix $\mathbf{X} \in \mathbb{R}^{n\times(p+1)}$, ordinary least squares gives

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}, \qquad \widehat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \qquad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T. \tag{12.2}$$

The residual vector is therefore

$$\boxed{\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}.} \tag{12.3}$$

If the linear truth-plus-error model is appropriate, then

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{12.4}$$

Since $\mathbf{HX} = \mathbf{X}$, the residuals can be written as

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \tag{12.5}$$

This identity is the key diagnostic reminder: residuals are not the true errors. They are the parts of the errors left over after projecting onto the fitted model space.

The normal equations also give

$$\mathbf{X}^T\mathbf{e} = \mathbf{0}. \tag{12.6}$$

Thus residuals are orthogonal to every column of the design matrix. If the model includes an intercept column $\mathbf{1}$, then

$$\mathbf{1}^T\mathbf{e} = \sum_{i=1}^{n} e_i = 0. \tag{12.7}$$

This is an algebraic fact, not a diagnostic victory. The residuals can sum to zero and still show curvature, changing spread, time dependence, or outliers. The modeling assumption we care about is closer to

$$\mathbb{E}[\epsilon_i \mid \mathbf{x}_i] = 0, \tag{12.8}$$

meaning that the unobserved errors are centered correctly at each predictor setting. Residual plots give practical evidence for or against that claim.

12.3 What Residuals Are Trying to Tell Us

The source slides summarize the residual-checking logic as follows:

Residual behavior	Suggests about the errors	Common diagnostic view
Mean-zero pattern for each x	$\mathbb{E}[\epsilon \mid X = x] = 0$ may be plausible	residuals versus fitted values or predictors
Approximately constant spread	$\mathrm{Var}(\epsilon \mid X = x) = \sigma^2$ may be plausible	residuals versus fitted values; scale-location plot
Approximately normal shape	$\epsilon \sim N(0, \sigma^2)$ may be plausible	histogram or normal Q-Q plot
No sequential pattern	uncorrelated errors may be plausible	residuals versus time/order

The phrase "suggests" is important. Residual plots do not prove assumptions. They help us identify visible violations.

12.4 Residuals Under the Big Three Regression Assumptions

Residual diagnostics connect directly to the three-step assumption ladder from earlier regression chapters. This section gives the diagnostic consequences, rather than re-deriving the full residual theory.

12.4.1 Assumption 1: Mean-zero errors

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$, then the model is centered correctly and residuals should not show systematic mean structure. A residual plot with no visible pattern around zero supports this assumption. A curved residual pattern suggests that the mean model is missing structure, such as a transformed predictor, a polynomial term, or an interaction.

12.4.2 Assumption 2: Constant variance and uncorrelated errors

If $\mathrm{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$, then the usual least-squares standard-error formulas are supported. Chapter 10 showed that residuals then satisfy

$$\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}), \qquad \mathrm{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}). \tag{12.9}$$

The important practical point is that raw residuals are affected by leverage. Changing spread in residual plots suggests heteroskedasticity, while waves or runs over time/order suggest correlated errors.

12.4.3 Assumption 3: Gaussian errors

If $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, then the usual finite-sample t- and F-based inference is exact under the model. Normal Q–Q plots and residual histograms are used to look for strong skewness, heavy tails, or outliers. For prediction, mild nonnormality may be less damaging than nonlinearity, heteroskedasticity, or dependence.

12.5 Residual Scaling: Raw, Standardized, and Studentized

The raw residual is

$$e_i = y_i - \widehat{y}_i. \tag{12.10}$$

Raw residuals are measured in the units of the response, which makes them easy to interpret operationally.

The residual standard error is

$$RSE = \widehat{\sigma} = \sqrt{\frac{RSS}{n-p-1}}, \qquad RSS = \sum_{i=1}^{n} e_i^2. \tag{12.11}$$

A common standardized residual is

$$r_i = \frac{e_i}{\widehat{\sigma}\sqrt{1-h_{ii}}}. \tag{12.12}$$

This accounts for the fact that residual variance depends on leverage and is often called an internally studentized residual. Externally studentized residuals use a leave-one-out estimate of σ; the idea is the same but the scaling is slightly more protective against the observation being tested. Large standardized or studentized residuals are potential outliers. A rough screening rule is that values beyond ± 2 deserve attention and values beyond ± 3 deserve a closer look.

Do not automate judgment. A large residual is a flag, not an automatic deletion order. Ask whether the observation is a data-entry error, an unusual but valid case, evidence of missing predictors, or evidence that the model class is too limited.

12.6 Core Diagnostic Plots

12.6.1 Residuals versus fitted values

This is the primary diagnostic plot for linear regression. The horizontal axis is $\widehat{y}_i$, and the vertical axis is e_i. A good residual-versus-fitted plot looks like an unstructured cloud centered around zero.

Common patterns include:

Pattern	Likely issue	Candidate repair
Curved shape	missing nonlinear mean structure	polynomial term, log term, spline, or other transformed predictor
Fan shape	nonconstant variance	transform response, weighted least squares, robust standard errors
Separate bands by group	missing categorical effect or interaction	add group indicator or interaction
Large isolated points	outliers or data errors	inspect data, sensitivity analysis, robust method

12.6.2 Residuals versus predictors

A residual-versus-fitted plot may hide which predictor is causing the problem. Plotting residuals against individual predictors can reveal a missed nonlinear effect in one variable.

For example, if residuals plotted against x_1 show a curve, then a candidate repair is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \epsilon. \tag{12.13}$$

If residuals plotted against x_2 show increasing spread, then a transformation or weighting strategy may be needed.

12.6.3 Normal Q-Q plot

A normal Q-Q plot compares standardized residual quantiles to normal quantiles. If the residuals are approximately normal, the points should lie roughly along a straight line.

Common departures include:

1. heavy tails: ends bend away from the line,
2. skewness: one tail departs more than the other,
3. isolated outliers: one or a few points far from the line.

12.6.4 Scale-location plot

A scale-location plot often uses

$$\sqrt{|r_i|} \tag{12.14}$$

against fitted values. It is designed to make changing variance easier to see. A rising trend suggests that the residual spread increases with the fitted value.

12.6.5 Residuals versus time or collection order

If the observations have a natural order, plot residuals against time, mission phase, batch number, route order, or collection order. A visible wave, trend, or clustering pattern can indicate correlated errors or missing time-varying predictors.

12.7 A Practical Diagnostic Hierarchy

Different diagnostics answer different questions. A useful order is:

1. **Test MSE or cross-validation error** for prediction performance.
2. **Residual plots** for missed structure, nonconstant variance, and outliers.
3. **Residuals versus individual predictors** to identify which variable may need a transformation or interaction.
4. **Formal tests** such as the Breusch–Pagan test as supporting evidence, not as the sole decision rule.

5. **t- and F-tests** for inference after the modeling assumptions are plausible enough for the claim being made.

Use the hierarchy to avoid overreacting to one number. A p-value can flag a problem, but a plot usually shows the shape of the problem and suggests the repair.

12.8 Problem 1: Nonlinearity

A linear model assumes that the conditional mean is linear in the columns of the design matrix:

$$\mathbb{E}[Y \mid \mathbf{x}] = \mathbf{x}^T \boldsymbol{\beta}. \tag{12.15}$$

If the true mean function is curved and the design matrix has only straight-line columns, residuals will show structure.

Diagnostic sign

A residual plot with a curve is the classic sign. For example, a straight-line fit to a parabola will often leave residuals that are negative-positive-negative across the predictor range.

Repairs

Candidate repairs include:

1. adding polynomial terms such as x^2 or x^3,
2. using transformations such as $\log x$, $\sqrt{x}$, or $1/x$,
3. using sine/cosine terms for cyclic predictors,
4. using splines or generalized additive models when a simple transformation is not enough,
5. moving to a different model class if the response type requires it.

Model repair as new columns

If the original model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \tag{12.16}$$

then adding a quadratic predictor changes the design matrix to

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}. \tag{12.17}$$

The repaired model is

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \tag{12.18}$$

So model repair is still a projection problem; we changed the subspace.

12.9 Problem 2: Nonconstant Variance

The homoskedastic model assumes

$$\mathrm{Var}(\epsilon_i \mid \mathbf{x}_i) = \sigma^2 \tag{12.19}$$

for every observation. If residual spread changes with fitted values or predictor values, then the constant-variance assumption is questionable.

Diagnostic sign

A common pattern is a funnel shape: residuals are tightly clustered for small fitted values and widely spread for large fitted values, or vice versa.

Why it matters

Nonconstant variance can make standard errors, confidence intervals, and hypothesis tests unreliable. It may not destroy point predictions, but it can damage uncertainty quantification.

Repairs

Candidate repairs include:

1. transform the response, especially $\log Y$ or $\sqrt{Y}$,
2. use weighted least squares when the variance pattern is understood,
3. use heteroskedasticity-robust standard errors for inference,
4. use a model family designed for the response type, such as logistic or Poisson regression.

A log response model has the form

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{12.20}$$

which often helps when variability grows multiplicatively with the mean.

A square-root response model has the form

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i, \tag{12.21}$$

which can help for nonnegative count-like responses.

Breusch–Pagan test as supporting evidence

The Breusch–Pagan test is a formal check for evidence that error variance is not constant. Operationally, it asks whether squared residuals show systematic structure as a function of the predictors or fitted values. A small p-value is evidence against constant variance, but it does not identify the correct repair. Residual plots remain the primary diagnostic tool because they reveal the shape: funneling, curvature, group-specific spread, or a few unusual points.

Do not use the BP test in a vacuum. If the plot and test disagree, investigate why: the model may be nonlinear, the variance pattern may be subtle, or one influential observation may be driving the result.

12.10 Problem 3: Nonnormal Residuals

The Gaussian noise model assumes

$$\epsilon_i \sim N(0, \sigma^2). \tag{12.22}$$

A histogram or Q-Q plot of residuals can reveal skewness, heavy tails, or outliers.

Why it matters

Normality is most important for exact small-sample inference. If the sample is large, coefficient estimates may still behave well under weaker conditions, but heavy tails and outliers can still destabilize the fit.

Repairs

Candidate repairs include:

1. transform the response,
2. inspect and correct data-entry issues,
3. use robust regression or bootstrap uncertainty intervals,
4. use a model family more appropriate to the response distribution.

12.11 Problem 4: Leverage and Influence

An outlier is unusual in the response direction. A high-leverage point is unusual in predictor space. An influential point is one that can noticeably change the fitted model. These are related but not identical.

The leverage of observation i is the ith diagonal entry of the hat matrix:

$$h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i. \tag{12.23}$$

Here $\mathbf{x}_i^T$ is the full design row, including the intercept and any transformed, categorical, or interaction columns. High leverage means the design row is far from the center of the other design rows. The model has fewer nearby observations to stabilize the fit around that point.

Because $\text{trace}(\mathbf{H}) = p + 1$ under full column rank, the average leverage is

$$\bar{h} = \frac{p+1}{n}. \tag{12.24}$$

Here $p+1$ means the number of fitted coefficients, equivalently the number of columns in the current full-rank design matrix. Rules such as $h_{ii} > 2\bar{h}$ or $h_{ii} > 3\bar{h}$ are screening heuristics, not deletion rules.

Cook's distance combines residual size and leverage:

$$D_i = \frac{e_i^2}{(p+1)\widehat{\sigma}^2} \frac{h_{ii}}{(1-h_{ii})^2}. \tag{12.25}$$

Large Cook's distance is context-dependent screening evidence that deleting observation i would noticeably change the fitted model, not a deletion rule. The most concerning observations usually have both high leverage and a large adjusted residual.

Keep, remove, or repair?

Influential is not the same as bad. Use EDA and the problem context before deciding what to do.

Situation	Reasonable response
Data-entry error, such as age 400 instead of 40	Correct if the truth is recoverable; otherwise remove or mark missing with documentation.
Valid extreme case, such as a very tall basketball player or a mansion-sized home	Usually keep; report that the model is sensitive to an operationally real case.
Influential pattern at the edge of a curved relationship	Repair the model, often with a transformation or interaction, rather than deleting the endpoints.
Observation outside the intended population	Exclude only under a pre-stated, reproducible population rule.

A defensible analysis often reports a sensitivity check: fit with and without the point, explain the difference, and justify the final choice.

12.12 Problem 5: Correlated Errors

Least squares formulas for standard errors often assume uncorrelated errors:

$$\operatorname{Cov}(\epsilon_i, \epsilon_j \mid \mathbf{X}) = 0 \qquad \text{for } i \neq j. \tag{12.26}$$

This assumption is especially vulnerable when observations are collected over time, across space, within units, or by repeated measurements on the same platform or person.

Diagnostic sign

Residuals plotted over time or collection order may show runs, waves, or clusters. Adjacent residuals may have similar signs or magnitudes.

Repairs

Candidate repairs include:

1. include time, batch, route, or group predictors,
2. include lagged predictors or lagged responses when appropriate,
3. use generalized least squares or time-series models,
4. use clustered or block-robust standard errors,
5. redesign the data collection plan when possible.

12.13 Problem 6: Missing Interactions

An additive model assumes that the effect of one predictor does not depend on another predictor. Interactions repair that limitation.

The additive two-predictor model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \tag{12.27}$$

The interaction model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \tag{12.28}$$

Equivalently,

$$Y = \beta_0 + (\beta_1 + \beta_3 x_2) x_1 + \beta_2 x_2 + \epsilon. \tag{12.29}$$

The slope in x_1 is now allowed to depend on x_2.

Diagnostic signs

Possible signs of a missing interaction include:

1. residuals have different patterns in different groups,
2. one predictor appears helpful only at certain levels of another predictor,
3. residuals plotted by color or symbol for a second variable reveal structure,
4. domain knowledge says the variables should combine synergistically.

12.14 A Model Repair Menu

When residuals look bad, the repair should be tied to the defect.

Residual symptom	Likely interpretation	First repair candidates
Curved residual trend	wrong mean shape	transform predictor, polynomial, spline
Fan-shaped spread	heteroskedasticity	transform response, WLS, robust SEs
Heavy tails in Q-Q plot	outliers or nonnormal errors	inspect data, robust method, transformation
Runs over time/order	correlated errors	time terms, lags, GLS, clustered SEs
Different patterns by group	missing categorical effect or interaction	add indicators or interactions
Large influential point	unusual observation dominates fit	audit, sensitivity analysis, robust method
No repair makes sense	model class mismatch	change model family or data strategy

12.15 Transformation Repairs

The source slides compare several model-repair candidates: a linear fit, a third-degree polynomial transformation, a square-root transformation, and a logarithmic-linear transformation. The general lesson is that there is rarely a purely mechanical answer to "which one wins."

12.15.1 Linear fit

The simplest model is

$$Y = \beta_0 + \beta_1 x + \epsilon. \tag{12.30}$$

Use this as the baseline. It is interpretable and low-variance, but it may leave curved residual patterns.

12.15.2 Polynomial predictor transformation

A third-degree polynomial model is

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon. \tag{12.31}$$

This can capture curvature while remaining linear in the coefficients. The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{bmatrix}. \tag{12.32}$$

The risk is overfitting and unstable behavior near the edges.

12.15.3 Square-root transformation

A square-root predictor model might be

$$Y = \beta_0 + \beta_1 \sqrt{x} + \epsilon, \tag{12.33}$$

with $x \geq 0$. A square-root response model might be

$$\sqrt{Y} = \beta_0 + \beta_1 x + \epsilon, \tag{12.34}$$

with $Y \geq 0$. These are different models and must be interpreted differently.

12.15.4 Logarithmic-linear transformation

A logarithmic predictor model is

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon, \tag{12.35}$$

with $x > 0$. This is useful for diminishing returns.

A log-response model is

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon, \tag{12.36}$$

with $Y > 0$. This is useful for multiplicative behavior and variance that grows with the mean.

Scale warning. Do not compare RMSE from a model fitted on Y to RMSE from a model fitted on $\log(Y)$ without putting predictions on the same scale. Model comparison must be done on a common operational scale.

12.16 Which One Wins?

The source slides emphasize that model repair includes some qualitative judgment. A defensible answer combines several checks:

1. Do residual plots look less patterned?
2. Does test MSE or cross-validated error improve on the operational prediction scale?
3. Is the transformation plausible for the domain?
4. Does the repair preserve interpretability?
5. Did the repair create new problems, such as extrapolation risk or rank deficiency?
6. Are nested-model tests, adjusted R^2, AIC, or BIC consistent with the conclusion?

Nested-model comparison

If a reduced model is nested inside a full model, use the familiar comparison statistic:

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \tag{12.37}$$

For example, the linear model

$$Y = \beta_0 + \beta_1 x + \epsilon \tag{12.38}$$

is nested inside the cubic model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon. \tag{12.39}$$

The test asks whether the added polynomial columns explain enough additional variation to justify the added degrees of freedom.

Prediction-focused comparison

For prediction, validation should dominate training fit. The model with the smallest training RSS is often the most flexible model, not necessarily the best generalizing model. Use a train/test split or cross-validation to estimate test error.

Inference-focused comparison

For inference, residual repair should support assumptions needed for standard errors, intervals, and tests. A model with slightly worse prediction but clearer interpretation and better-behaved residuals may be the better briefing model.

12.17 Admit Defeat: When Repair Means Changing the Model Class

Sometimes bad residuals are not a minor defect. They are evidence that ordinary least squares is the wrong tool.

Examples:

Situation	Why OLS may struggle	Better direction
Binary response	predictions can fall outside $[0, 1]$	logistic regression
Count response	variance often grows with mean	Poisson or negative binomial model
Positive skewed response	additive normal errors may be implausible	log transform or Gamma GLM
Repeated measures	errors are correlated within unit	mixed model or clustered SEs
Time series	residuals may be autocorrelated	time-series regression or GLS
Strong nonlinear boundary	linear mean is structurally wrong	nonlinear model, tree, GAM, or ML method

Admitting defeat is not failure. It is a modeling conclusion: the residuals are telling us the current model family is missing something important.

12.18 Worked Example: Residual Pattern to Repair Decision

Suppose a linear fit gives the residual pattern

$$e_i \approx \text{curved function of } x_i. \tag{12.40}$$

A reasonable repair sequence is:

1. Fit the baseline model:
$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \tag{12.41}$$
2. Plot residuals versus x and fitted values.
3. Add a transformation motivated by the pattern:
$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \tag{12.42}$$
4. Refit the model and replot residuals.
5. Compare the two models using residual behavior, test MSE, and a nested-model F-test if appropriate.
6. Brief the result in terms of the original problem, not just the equation.

A clean result would be: the repaired model reduces the visible residual curve, improves validation error on the response scale, and has a defensible interpretation.

12.19 Residual Analysis Workflow

A reusable workflow is:

1. Fit a simple, interpretable baseline model.
2. Compute fitted values and residuals.
3. Check residuals versus fitted values.
4. Check residuals versus important predictors.
5. Check residual distribution with a histogram and Q-Q plot.
6. Check residuals over time/order/group if applicable.
7. Check leverage and influence for unusual observations.
8. Propose one repair at a time.
9. Refit and repeat diagnostics.
10. Compare models on a common scale using validation or other appropriate criteria.
11. Document the decision.

The last step is critical. If an analyst tries several transformations and only reports the one that looks good, the analysis is less defensible. A good report explains what was tried and why the final choice was made.

Coding companion reference. Package-specific Python/R commands for residual plots, Q-Q plots, influence measures, transformations, interactions, and validation RMSE have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the diagnostic logic and model-repair workflow.

12.20 Common Mistakes

Mistake 1: Treating residuals as the true errors

Residuals estimate error behavior, but they are not the unobserved errors. They are constrained by the fitted model and satisfy $\mathbf{X}^T\mathbf{e} = \mathbf{0}$.

Mistake 2: Celebrating mean-zero residuals

If the model has an intercept, the residuals sum to zero automatically. The meaningful question is whether residuals have no systematic pattern across predictor values.

Mistake 3: Comparing transformed and untransformed RMSE directly

RMSE on Y and RMSE on $\log(Y)$ are not on the same scale. Compare models on the operational scale that matters.

Mistake 4: Deleting outliers without explanation

Outlier removal must be repeatable and defensible. If the point is valid and operationally relevant, deleting it may hide exactly the case the model needs to understand.

Mistake 5: Adding every possible transformation

More columns reduce training RSS but can increase test error. Add transformations because the residuals, domain knowledge, and validation evidence justify them.

Mistake 6: Ignoring data quality

Sometimes residuals look bad because the data are bad: wrong units, miscoded categories, impossible values, repeated rows, or data-entry errors. Model repair does not replace data cleaning.

Mistake 7: Taking one hypothesis test in a vacuum

A small p-value for one transformation does not automatically make it the best model. Consider residual behavior, prediction performance, assumptions, and operational meaning together.

Mistake 8: Letting the Breusch–Pagan test replace the plot

The BP test can support a heteroskedasticity concern, but it does not show the shape of the problem or tell you how to repair it. Use the plot, the test, and the modeling context together.

Mistake 9: Deleting influential points automatically

A high-leverage or high-Cook's-distance observation is a flag for investigation. It may be a data error, a valid extreme case, or evidence that the model form is too simple.

12.21 Operational Briefing Checklist

When briefing residual analysis and model repair, answer:

1. What was the baseline model?
2. What residual pattern indicated a problem?
3. Which assumption did that pattern threaten?
4. What repair options were considered?
5. Which repair was selected, and why?
6. Did residual behavior improve after the repair?
7. Did test MSE or cross-validation error improve on the response scale?
8. Did the repair change interpretation?

9. Were outliers, high-leverage observations, or influential observations investigated?
10. Did BP or another formal test support the visual diagnostic evidence, and was it interpreted cautiously?
11. Were any observations removed? If yes, what was the defensible rule?
12. Are remaining limitations clear to the customer or commander?

12.22 Chapter Summary

The central residual-analysis loop is:

$$\boxed{\text{fit} \longrightarrow \text{diagnose residuals} \longrightarrow \text{repair model} \longrightarrow \text{re-check.}} \tag{12.43}$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \widehat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}. \tag{12.44}$$

If the model is correctly specified, then

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \tag{12.45}$$

Residual plots help us evaluate whether the leftover variation behaves like random noise or contains missed structure.

A good residual plot is not proof that the model is true. A bad residual plot is useful evidence that the model needs work. Formal tests such as Breusch–Pagan can provide supporting evidence, but they do not replace plots or judgment. Repairs include transformed predictors, transformed responses, interaction terms, different model families, robust methods, and sometimes better data. Leverage and Cook's distance help identify observations that may drive the fit and require investigation.

The guiding principle is:

> Residuals are the model's after-action report. Read them before trusting the fit.

12.23 Math Foundations Source Map

This source map records the Math Foundations support used in this chapter and where those ideas appear in the completed Math Foundations textbook.

Chapter 12 use	Math Foundations connection	MF textbook location
Residual vectors in observation space	Vector notation in $\mathbb{R}^n$, coordinates, and subvectors	Ch02 Secs. 2.1, 2.2, 2.6; Ch03 Sec. 3.1; Ch06 Sec. 6.5
Orthogonality condition $\mathbf{X}^T\mathbf{e} = \mathbf{0}$	Dot products, Euclidean norms, and the zero-dot-product condition for orthogonality	Ch05 Secs. 5.1, 5.3, 5.5, 5.8
Hat-matrix computations as linear transformations	Matrix notation, identity matrices, transposes, and matrix-vector multiplication	Ch07 Secs. 7.1, 7.3, 7.4, 7.6.1, 7.6.2; Ch12 Sec. 12.12
Leverage and influence diagnostics	Hat-matrix diagonal entries, trace, Euclidean residual length, and projection geometry	Ch05 Secs. 5.5, 5.8; Ch07 Secs. 7.6.1, 7.6.2; Ch11 Sec. 11.2
Column-space / residual-space viewpoint	Column rank, column space, and full-rank conditions	Ch02 Sec. 2.6; Ch05 Sec. 5.8; Ch10 Sec. 10.5; Ch11 Sec. 11.2
Least-squares repair criterion	Stationary points, least-squares minimization, the normal equations, and the positive-semidefinite $\mathbf{A}^T\mathbf{A}$ argument	Ch05 Sec. 5.5; Ch10 Sec. 10.6; Ch12 Secs. 12.8, 12.9, 12.11; Ch13 Sec. 13.4
Identifiability after repair	Left inverses, linearly independent columns, rank, and full column rank	Ch06 Sec. 6.3; Ch10 Secs. 10.1, 10.3, 10.5; Ch11 Sec. 11.4
Computational matrix operations	Python/NumPy arrays, shapes, transposes, and matrix multiplication	Ch07 Secs. 7.1, 7.4, 7.7

12.24 Practice and Workflow

Focused Study Checklist

Use this as a final check after diagnosing and repairing a fitted model.

1. Can you name the residual plot or diagnostic view you used?
2. Can you describe the observed residual pattern without overstating it?
3. Can you connect the pattern to the threatened assumption or modeling weakness?
4. Can you choose a repair: transform predictors, transform the response, add interactions, audit data, use a robust method, or change model family?
5. Can you refit and compare residual behavior, validation error, and interpretability?
6. Can you explain whether the repair changed the meaning of the coefficients or predictions?
7. Can you state the remaining limitations honestly in a briefing-style conclusion?